

# 3GPP TS 29.572 V19.6.0 (2026-03)

*Technical Specification*

## **3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; 5G System; Location Management Services; Stage 3 (Release 19)**



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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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- x the first digit:
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- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

- shall** indicates a mandatory requirement to do something
- shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

- should** indicates a recommendation to do something
- should not** indicates a recommendation not to do something
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- need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

- can** indicates that something is possible
- cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

- will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

---

## 1 Scope

The present document specifies the stage 3 protocol and data model for the Nlmf Service Based Interface. It provides stage 3 protocol definitions and message flows, and specifies the API for each service offered by the LMF.

The 5G System stage 2 architecture and procedures are specified in 3GPP TS 23.501 [2] and 3GPP TS 23.502 [3].

The Technical Realization of the Service Based Architecture and the Principles and Guidelines for Services Definition are specified in 3GPP TS 29.500 [4] and 3GPP TS 29.501 [5].

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## 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".
- [3] 3GPP TS 23.502: "Procedures for the 5G System; Stage 2".
- [4] 3GPP TS 29.500: "5G System; Technical Realization of Service Based Architecture; Stage 3".
- [5] 3GPP TS 29.501: "5G System; Principles and Guidelines for Services Definition; Stage 3".
- [6] IETF RFC 4776: "Dynamic Host Configuration Protocol (DHCPv4 and DHCPv6) Option for Civic Addresses Configuration Information".
- [7] IETF RFC 5139: "Revised Civic Location Format for Presence Information Data Format Location Object (PIDF-LO)".
- [8] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".
- [9] 3GPP TS 33.501: "Security architecture and procedures for 5G system".
- [10] IETF RFC 6749: "The OAuth 2.0 Authorization Framework".
- [11] 3GPP TS 29.510: "Network Function Repository Services; Stage 3".
- [12] IETF RFC 9113: "HTTP/2".
- [13] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

- [14] OpenAPI Initiative, "OpenAPI Specification Version 3.0.0", <https://spec.openapis.org/oas/v3.0.0>.
- [15] IETF RFC 9457: "Problem Details for HTTP APIs".
- [16] 3GPP TR 21.900: "Technical Specification Group working methods".
- [17] 3GPP TS 22.071: "Location Services (LCS); Service description; Stage 1".
- [18] 3GPP TS 29.002: "Mobile Application Part (MAP) specification".
- [19] 3GPP TS 23.273: "5G System (5GS) Location Services (LCS); Stage 2".
- [20] 3GPP TS 24.080: "Mobile radio interface layer 3 Supplementary services specification; Formats and coding".
- [21] 3GPP TS 37.355: "LTE Positioning Protocol (LPP)".
- [22] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".
- [23] 3GPP TS 29.518: "Access and Mobility Management Services".
- [24] 3GPP TS 29.171: "Location Services (LCS); LCS Application Protocol (LCS-AP) between the Mobile Management Entity (MME) and Evolved Serving Mobile Location Centre (E-SMLC); SLs interface".
- [25] IETF RFC 4119: "A Presence-based GEOPRIV Location Object Format".
- [26] 3GPP TS 33.256: "Security aspects of Uncrewed Aerial Systems (UAS)".
- [27] 3GPP TS 29.515: "5G System; Gateway Mobile Location Services Stage 3".
- [28] Void
- [29] Void
- [30] 3GPP TS 38.355: "NR; Sidelink Positioning Protocol (SLPP); Protocol Specification".
- [31] 3GPP TS 38.455: "NR Positioning Protocol A (NRPPa)".
- [32] IETF RFC 6902: "JavaScript Object Notation (JSON) Patch".
- [33] IETF RFC 2387: "The MIME Multipart/Related Content-type".
- [34] 3GPP TS 29.503: "Unified Data Management Services; Stage 3".
- [35] IETF RFC 2045: "Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies".

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## 3 Definitions and abbreviations

### 3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

|       |                                             |
|-------|---------------------------------------------|
| LDR   | Location Deferred Request                   |
| LIR   | Location Immediate Request                  |
| LMF   | Location Management Function                |
| LPHAP | Low Power and High Accuracy Positioning     |
| MBSR  | Mobile Base Station Relay                   |
| MWAB  | Mobile gNB with Wireless Access Backhauling |

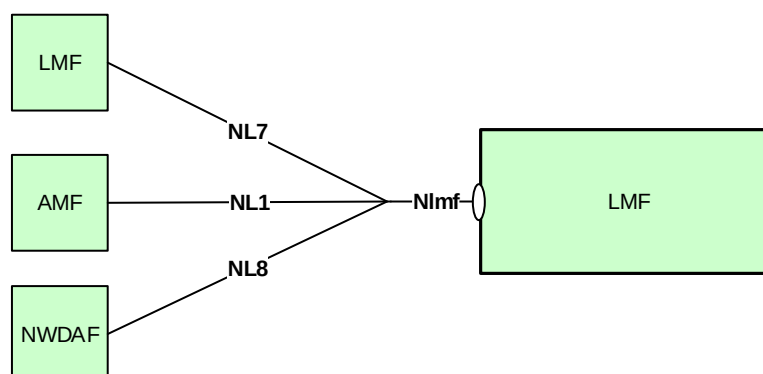
## 4 Overview

The Location Management Function (LMF) is the network entity in the 5G Core Network (5GC) supporting the following functionality:

- Supports location determination for a UE.
- Obtains downlink location measurements or a location estimate from the UE.
- Obtains uplink location measurements from the NG RAN.
- Obtains non-UE associated assistance data from the NG RAN.
- Provides broadcast assistance data to UEs and forwards associated ciphering keys to an AMF.
- Obtains ML model(s) from the NWDAF.
- Provide input data for ML model training or ML model performance monitoring for LMF-based AI/ML Positioning.

Other functions of an LMF are listed in clause 4.3.8 of 3GPP TS 23.273 [19].

Figure 4-1 provides the reference model (in service-based interface representation and in reference point representation), with focus on the LMF:



**Figure 4-1: Reference model – LMF**

## 5 Services Offered by the LMF

### 5.1 Introduction

The LMF offers to other NFs the following services:

- Nlmf\_Location
- Nlmf\_Broadcast
- Nlmf\_DataExposure

Table 5.1-1 summarizes the corresponding APIs defined for this specification.

**Table 5.1-1: API Descriptions**

| Service Name      | Clause | Description               | OpenAPI Specification File     | apiName           | Annex |
|-------------------|--------|---------------------------|--------------------------------|-------------------|-------|
| Nlmf_Location     | 6.1    | LMF Location Service      | TS29572_Nlmf_Location.yaml     | nlmf-loc          | A.2   |
| Nlmf_Broadcast    | 6.2    | LMF Broadcast Service     | TS29572_Nlmf_Broadcast.yaml    | nlmf-broadcast    | A.3   |
| Nlmf_DataExposure | 6.3    | LMF Data Exposure Service | TS29572_Nlmf_DataExposure.yaml | nlmf-dataexposure | A.4   |

### 5.2 Nlmf\_Location Service

#### 5.2.1 Service Description

The Nlmf\_Location service enables an NF to request location determination (current geodetic and optionally local and/or civic location) for a target UE or to request periodic or triggered location for a target UE.

#### 5.2.2 Service Operations

##### 5.2.2.1 Introduction

The service operations defined for the Nlmf\_Location service are as follows:

- DetermineLocation: It provides UE location information to the consumer NF.
- EventNotify: It notifies the consumer NF of an event for periodic or triggered location for a target UE, or cumulative event report for the location reporting over user plane from the UE.
- CancelLocation: It enables a consumer NF to cancel an ongoing periodic or triggered location for a target UE.
- LocationContextTransfer: It enables a consumer NF to transfer location context information for periodic or triggered location of a target UE to a new LMF.
- MeasurementData: It enables a consumer NF to request the PRU location measurements from PRU serving LMFs.
- UPLSubscribe: It enables a consumer NF to subscribe the status of a secure LCS-UP connection for a target UE.
- UPLUnSubscribe: It enables the consumer NF to unsubscribe the status of a secure LCS-UP connection for a target UE.
- UPNotify: It notifies the consumer NF of the status or modification of a secure LCS-UP connection for a target UE.

- UPConfig: It enables a consumer NF to set up, modify or terminate a secure LCS-UP connection for a target UE.

## 5.2.2.2 DetermineLocation

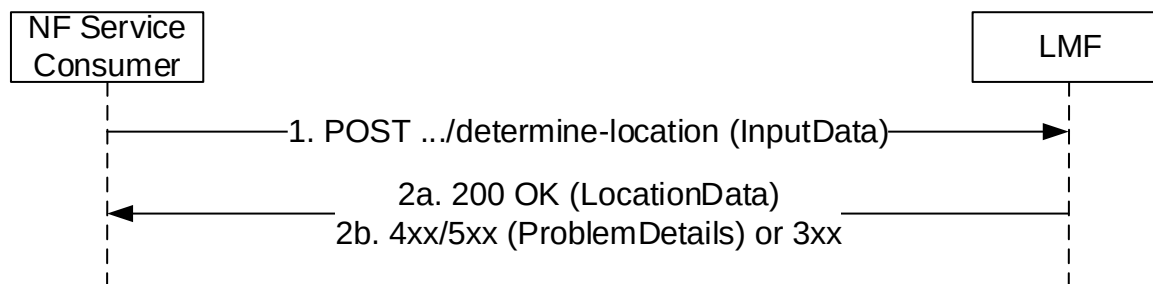
### 5.2.2.2.1 General

The following procedures are defined, using the "DetermineLocation" service operation:

- Retrieve UE Location
- Retrieve UE Location for 5G-MO-LR
- Retrieve UE Location for Ranging and Sidelink Positioning

#### 5.2.2.2.2 Retrieve UE Location

This procedure allows a consumer NF to request the location information (geodetic location and, optionally, local and/or civic location) for a target UE or to activate periodic or triggered deferred location for a target UE.



**Figure 5.2.2.2.2-1: DetermineLocation Request**

1. The NF Service Consumer shall send an HTTP POST request to the resource URI associated with the "determine-location" custom operation. The input parameters for the request (external client type, LCS correlation identifier, serving cell identifier, location QoS, mapped location QoS applicable to EPS, supported GAD shapes, LDR Type, H-GMLC address, LDR Reference, UE connectivity state per access type, TNAP identifier, TWAP identifier, scheduled location time, LpHapType, UE User Plane Positioning Capability, reporting indication, MBSR Info, Additional UE Info, integrity requirements, requested ranging\_SL location results, related UEs, SUPI and/or GPSI, Ranging and sidelink positioning capabilities supported by the target UE, MWAB support indication, etc) may be included in the HTTP POST request body;

If UE geographical area identified by the country, area within a country or international area needs to be determined, the NF Service Consumer shall include UE geographical area determination indication for PLMN selection verification in the request;

If UE Unaware Positioning is required, the NF Service Consumer shall include UE unaware indication in the request;

If UE LCS Capability is received in the request indicating LPP is not supported by the UE, the LMF shall not send LPP messages to the UE in subsequent positioning procedures.

During 5GC-MT-LR multiple location procedure for regulatory location service, the AMF shall also include the indication of acceptance for intermediate response and the maximum response time, the GMLC callback address and the LIR reference number, if received from the GMLC. The AMF may overwrite the received maximum response time (e.g. to avoid HTTP service request timeout) when passing it to the LMF.

For deferred periodic or triggered 5GC-MT-LR procedures, if the NF Service Consumer requests the location reporting over user plane, the NF Service Consumer shall include the endpoint address of the location reporting over user plane, the cumulative event report timer, or the maximum number of location reports over user plane.

- 2a. On success, "200 OK" shall be returned. The response body shall contain the parameters related to the determined position of the UE if any (geodetic position, local location, civic location, positioning methods, LOS/NLOS measurement indication, integrity result, ...);

If the NF service consumer has requested to determine UE country, area within a country or international area, the LMF shall also include `ueAreaInd`.

If the indication of acceptance for intermediate response was received in the request, the LMF shall perform positioning procedures and determines multiple location estimates within the maximum response time. The LMF shall include the FINAL location in the content of this response message. If any intermediate location(s) are determined, the LMF shall send intermediate location reporting event notification(s) to the GMLC (see clause 5.2.2.3.3).

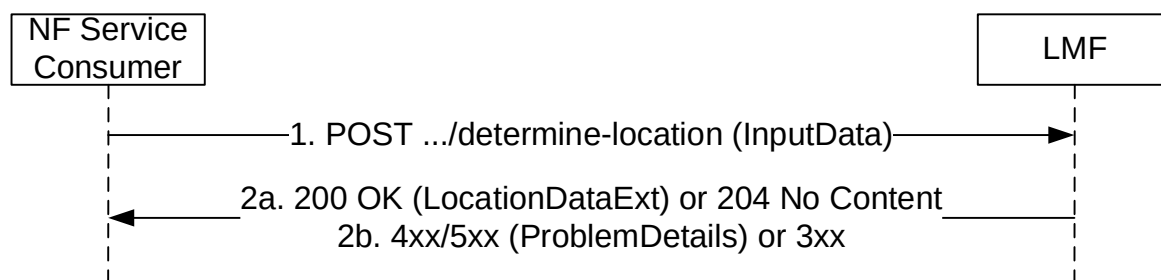
During deferred periodic or triggered 5GC-MT-LR procedures, if the mapped location QoS applicable to EPS was received in the request and if the access type allowed for the UE for event reporting includes "E-UTRAN connected to EPC", the LMF shall forward the mapped location QoS applicable to EPS to the UE.

During deferred periodic 5GC-MT-LR procedure, if the LMF has determined to generate Periodic Location Estimates based on NRPPa Periodic Measurement, the LMF shall include the `nrppaPeriodicInd` IE with the value true in the response.

- 2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.4.2.2-2 shall be returned. For a 4xx/5xx response, the message body should contain a `ProblemDetails` structure with the "cause" attribute set to one of the application error listed in Table 6.1.4.2.2-2.

#### 5.2.2.2.3 Retrieve UE Location for 5G-MO-LR

This procedure allows a consumer NF (i.e. an AMF) to request the location information or location assistance data for a target UE which initiates MO-LR procedure (see 3GPP TS 23.273 [19]).



**Figure 5.2.2.2.3-1: DetermineLocation Request for 5G-MO-LR**

The same requirements in clause 5.2.2.2.2 shall be applied with following modifications:

1. Same as step 1 of figure 5.2.2.2.2-1, the request body shall include the following additional information:
  - The indication received from UE indicating whether a location estimate or location assistance data is required.
  - The LPP messages if received in MO-LR Request from UE
  - UE's subscribed assistance data for 5GC-MO-LR if received from UDM.
- 2a. Same as step 2a of figure 5.2.2.2.2-1 if a consumer NF requests the location information for a target UE. If a NF consumer requests location assistance data for a target UE and LMF has successfully delivered location assistance data to the UE, 204 No Content shall be returned.
- 2b. Same as step 2b of figure 5.2.2.2.2-1.

This procedure is also used to retrieve UE Location for Ranging and Sidelink Positioning.

### 5.2.2.3 EventNotify

#### 5.2.2.3.1 General

The following procedures are defined, using the "EventNotify" service operation:

- Periodic or Triggered Event Notification

#### 5.2.2.3.2 Periodic or Triggered Event Notification

This procedure notifies the NF Service Consumer (i.e. GMLC) about event information related to periodic or triggered location of a target UE. The notification is delivered to:

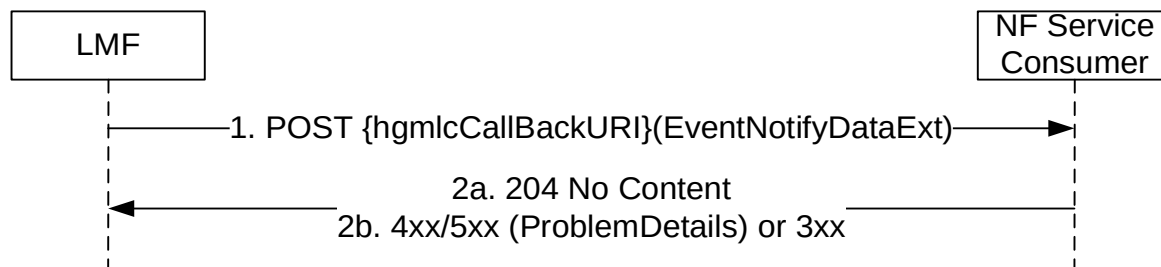
- the callback URI of an H-GMLC received (from an AMF) during an earlier DetermineLocation service operation if still available and if the LMF is configured for direct access to the H-GMLC;
- the callback URI of an H-GMLC received (from another LMF) during an earlier LocationContextTransfer service operation if still available and if the LMF is configured for direct access to the H-GMLC;
- the callback URI of an H-GMLC received (from the target UE) in a supplementary services event report if the LMF is configured for direct access to the H-GMLC;

otherwise (if not available),

- the callback URI of a V-GMLC registered in the NRF, if the V-GMLC registered to the NRF with notification endpoints for periodic or triggered event notifications; or

otherwise (if not available),

- the URI of a V-GMLC locally provisioned in the LMF.



**Figure 5.2.2.3.2-1: EventNotify Request**

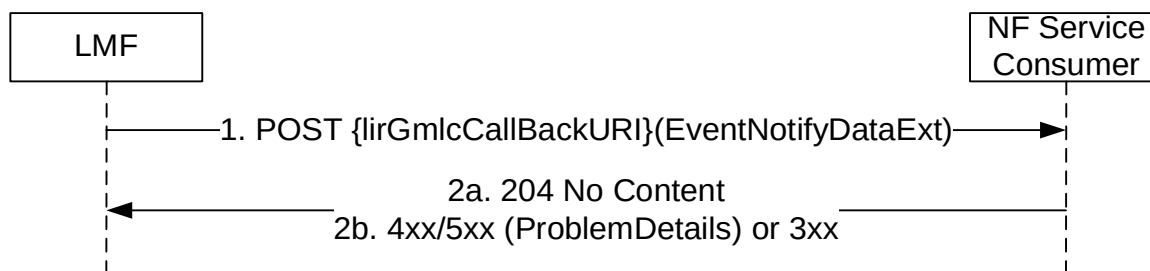
1. The LMF shall send a POST request to the GMLC callback URI determined as described above. The request body shall include a notification correlation ID (LDR reference), the UE identification (SUPI and if available GPSI), the type of event and may include a geodetic location, local location, civic location, position methods used, and other available parameters related to the position if any (e.g. Velocity, Altitude etc.), H-GMLC callback URI (if the NF consumer is a V-GMLC), serving LMF identification, LOS/NLOS measurement indication, integrity result, and the statistics on the location reporting over user plane.
- 2a. On success, "204 No content" shall be returned by the NF Service Consumer.
- 2b. On failure or redirection, one of the appropriate HTTP status code listed in Table 6.1.5.1.3.1-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

#### 5.2.2.3.3 Intermediate location reporting Event Notification

This procedure notifies the NF Service Consumer (i.e. GMLC) about event information related to intermediate location of a target UE. The notification is delivered to:



- the callback URI of the GMLC for multiple location request received (from an AMF) during an earlier DetermineLocation service operation (see clause 5.2.2.2.2).



**Figure 5.2.2.3.3-1: Intermediate location reporting Event Notification Request**

1. The LMF shall send a POST request to the GMLC callback URI determined as described above. The request body shall include the associated LIR reference number, the UE identification (SUPI and if available GPSI), the type of event ("INTERMEDIATE\_EVENT") and the intermediate location of the UE (geodetic location, local location, civic location, position methods used), and other available parameters related to the position if any (e.g. Velocity, Altitude etc.).
- 2a. On success, "204 No content" shall be returned by the NF Service Consumer.
- 2b. On failure or redirection, one of the appropriate HTTP status code listed in Table 6.1.5.1.3.1-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

## 5.2.2.4 CancelLocation

### 5.2.2.4.1 General

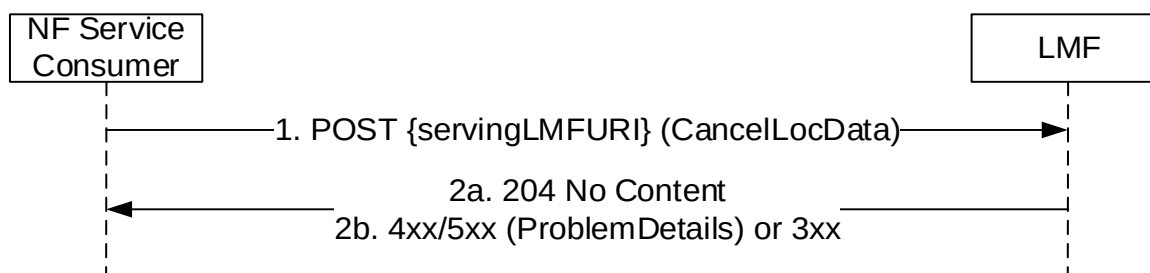
The following procedures are defined, using the "CancelLocation" service operation:

- Cancel Periodic or Triggered Location
- Cancel 5GC-MT-LR / 5GC-MO-LR procedure during UE mobility from 5GS to EPS with N26 interface

#### 5.2.2.4.2 Cancel Periodic, Triggered Location or 5G-MT-LR / 5G-MO-LR

This procedure allows a consumer NF to cancel periodic or triggered location for a target UE, or allows a consumer NF to cancel 5G-MT-LR/5G-MO-LR location during UE mobility from 5GS to EPS with N26 interface.

The cancellation is delivered to a resource URI on the serving LMF identified by the serving LMF identification provided to the consumer NF (i.e. AMF) by a V-GMLC or H-GMLC (see 3GPP TS 23.273 [19]).



**Figure 5.2.2.4.2-1: CancelLocation Request**

1. The NF Service Consumer shall send an HTTP POST request to the resource URI of "cancel-location" custom operation on the serving LMF.

If used to cancel periodic location or triggered location, the request body shall include a notification correlation ID (LDR reference) and an H-GMLC callback URI.

If used to cancel 5G-MT-LR / 5G-MO-LR location during UE mobility from 5GC to EPS with N26 interface, the request body shall include the LCS Correlation ID.

2a. On success, "204 No content" shall be returned by the LMF.

2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.4.3.2-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.4.3.2-2.

## 5.2.2.5 LocationContextTransfer

### 5.2.2.5.1 General

The following procedures are defined, using the "LocationContextTransfer" service operation:

- Transfer Location Context

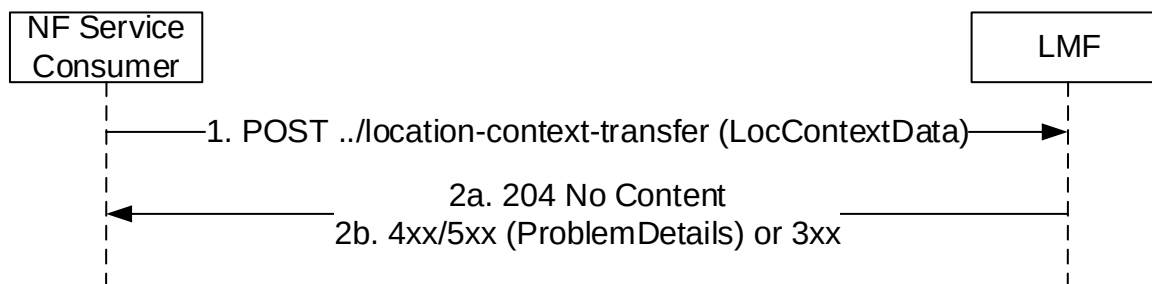
#### 5.2.2.5.2 Transfer Location Context

This procedure allows a NF service consumer (e.g. the old LMF) to transfer location context information for periodic or triggered location for a target UE (see clause 6.4 and clause 6.7.2 of 3GPP TS 23.273 [19]). The NF service consumer discovers the service URI of the new LMF by performing a discovery via NRF using:

- the identification of the LMF received (from an AMF) during an earlier Namf\_Communication\_N1MessageNotify service operation to the consumer NF;

otherwise (if not available),

- the identification of an LMF locally provisioned in the consumer NF.



**Figure 5.2.2.5.2-1: LocationContextTransfer Request**

1. The NF Service Consumer shall send an HTTP POST request to the Custom operation URI ("/location-context-transfer") on the Service URI discovered as described above.

The request body shall include an AMF identity, Deferred location type, Deferred location parameters, Notification Target Address (H-GMLC callback URI), Notification Correlation ID (LDR reference), an embedded event report message and may include an event reporting status, UE location information, scheduled location time and LOS/NLOS measurement indication, or UE user plane positioning capabilities.

If an N1 message is received from the UE with Control Plane CIoT 5GS Optimisation, the request body shall include an indication of Control Plane CIoT 5GS Optimisation.

If the location context information for periodic or triggered location for a target UE includes the endpoint address of the location reporting over user plane, the NF Service Consumer may include the cumulative event report timer, or the maximum number of location reports over user plane.

If the location context(s) being transferred are associated with the LCS-UPP connection between the UE and the source LMF, the source LMF shall include the `lcsUppExistInd` IE with the value `true` for the associated location context(s) and, for UE which supports multiple LCS-UPP connections, also include the source LMF routing information (i.e. the source LMF Identification). If the event report message from the UE is not available for the location context(s) being transferred, the NF Service consumer shall set the event class of the `eventReportMessage` IE to the value `"DUMMY"` in the `LocContextData` object and the target LMF shall ignore the received `eventReportMessage` IE with event class set to the value `"DUMMY"`.

2a. On success, `"204 No content"` shall be returned by the LMF.

2b. On failure or redirection, one of the HTTP status codes listed in Table 6.1.4.4.2-2 shall be returned. For a `4xx/5xx` response, the message body should contain a `ProblemDetails` structure with the `"cause"` attribute set to one of the application error listed in Table 6.1.4.4.2-2.

### 5.2.2.5.3 Multiple Location Contexts Transfer

If there are multiple LCS sessions for the UE with LCS UP connection, and if the source and target LMF support the MCTX feature, the source LMF shall send the additional LCS contexts of the UE to the target LMF in the `addLocContextList` (see clause 6.18.3 of 3GPP TS 23.273 [19]).

The requirement in clause 5.2.2.5.2 shall be applied with following addition:

Same as step1 of clause 5.2.2.5.2, the HTTP content of the request shall additionally contain:

- `addLocationContext` attribute.

### 5.2.2.6 MeasurementData

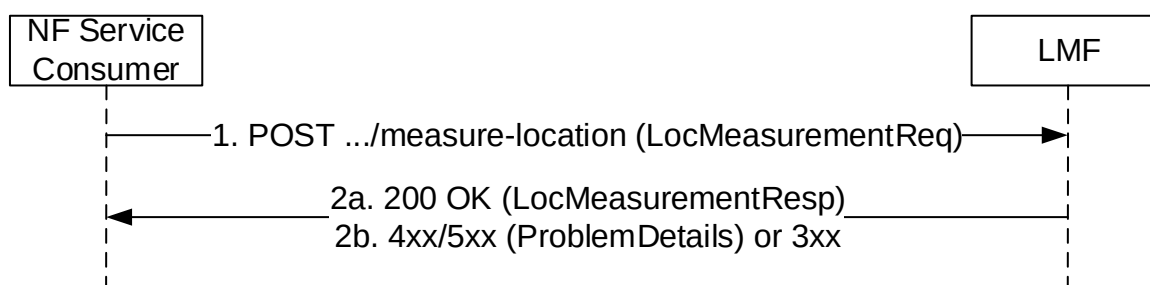
#### 5.2.2.6.1 General

The following procedures are defined, using the `"MeasurementData"` service operation:

- Location Measurements

#### 5.2.2.6.2 Location Measurements

This procedure allows a consumer NF (e.g. LMF) to request PRU location measurements from PRU serving LMF.



**Figure 5.2.2.6.2-1: LocationMeasure Request**

1. The NF Service Consumer shall send an HTTP POST request to the resource URI associated with the `"measure-location"` custom operation. The request body shall include target UE cell ID and may include the Pre-calculated location of target UE and time windows.
- 2a. On success, `"200 OK"` shall be returned. The response body shall contain the parameters related to the location measurements of the PRU and the known location of the associated PRU.

- 2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.4.5.2-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.4.5.2-2.

### 5.2.2.7 UPLSubscribe

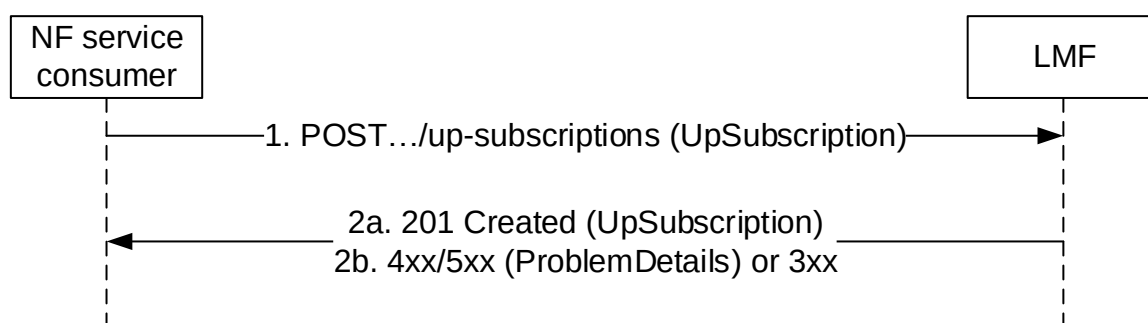
#### 5.2.2.7.1 General

The following procedures are defined, using the "UPLSubscribe" service operation:

- Subscribe to Notification of LCS-UP connection status

#### 5.2.2.7.2 Subscribe to Notification of LCS-UP connection status

This procedure allows a consumer NF to subscribe the status of a secure LCS-UP connection for a target UE.



**Figure 5.2.2.7.2-1: NF service consumer subscribes to notifications**

1. The NF service consumer shall send a POST request to the resource URI representing the "up-subscriptions" collection resource.

The callback URI of the NF service consumer shall be included in the request content to receive notifications of LCS-UP connection status from the LMF.

- 2a. On success, the LMF responds with "201 Created". The response shall contain the data related to the created subscription, and the HTTP Location header shall contain the URI of the created subscription.
- 2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body containing a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.2.3.1-3.

### 5.2.2.8 UPNotify

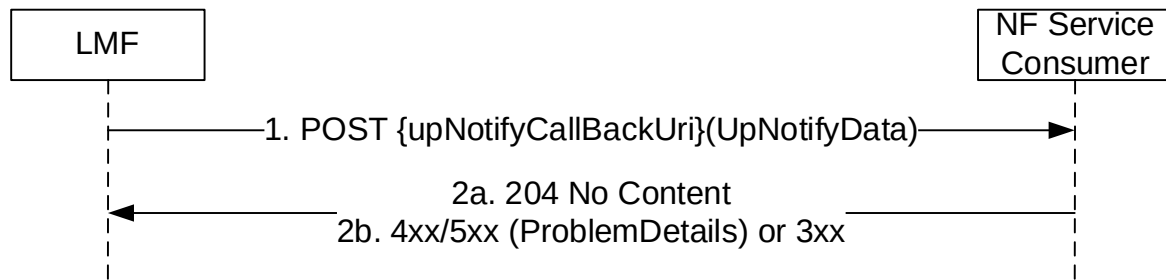
#### 5.2.2.8.1 General

The following procedures are defined, using the "UPNotify" service operation:

- Notification of LCS-UP connection

#### 5.2.2.8.2 Notification of LCS-UP connection

This procedure notifies the NF Service Consumer (i.e. AMF) about status or modification of a secure LCS-UP connection for a target UE.



**Figure 5.2.2.8.2-1: UPNotify**

1. The LMF shall send a POST request to the upNotifyCallbackUri. The request body shall include a notification correlation ID, LCS-UP connection status and may include a target LMF identifier.
- 2a. On success, "204 No content" shall be returned by the NF Service Consumer.
- 2b. On failure or redirection, one of the appropriate HTTP status code listed in Table 6.1.5.2.3.1-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

## 5.2.2.9 UPConfig

### 5.2.2.9.1 General

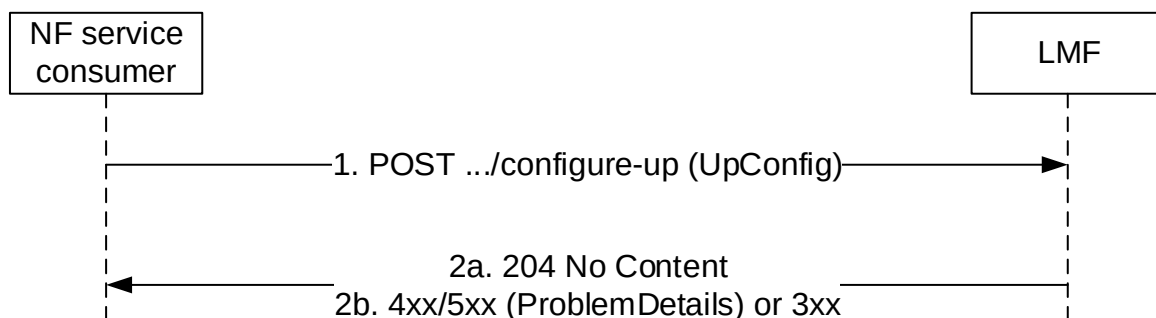
The following procedures are defined, using the "UPConfig" service operation:

- Configure LCS-UP connection

#### 5.2.2.9.2 Configure LCS-UP connection

This procedure allows a consumer NF (i.e. AMF) to set up, modify or terminate a secure LCS-UP connection for a target UE and to get the notification of LCS-UPP connection status.

Additionally, if AMF change happens during UE mobility, the target AMF shall also invoke the service operation to inform the serving LMF, which has established LCS-UPP connection with the UE, that the serving AMF has changed.



**Figure 5.2.2.9.2-1: LCS-UP connection**

1. The NF service consumer shall send an HTTP POST request to the resource URI associated with the "configure-up" custom operation to set up, modify or terminate LCS-UP connection. The request content shall include a UP notify callback URI, a notification correlation ID, the UE identity, and, if available, the UE user plane positioning capabilities.
- 2a. On success, "204 No Content" shall be returned.
- 2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.4.7.2-2 shall be returned. For a 4xx/5xx response, the message body may contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.1.4.7.2-2.

### 5.2.2.10 UPUUnSubscribe

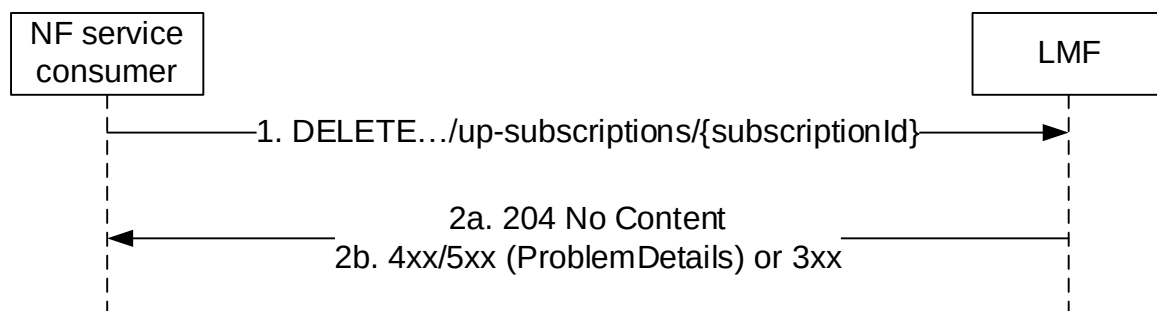
#### 5.2.2.10.1 General

The following procedure is defined using the "UPUnSubscribe" service operation:

- Unsubscribe to notification of LCS-UP connection status

#### 5.2.2.10.2 Unsubscribe to notification of LCS-UP connection status

This procedure allows the consumer NF to unsubscribe the status of a secure LCS-UP connection for a target UE.



**Figure 5.2.2.10.2-1: NF service consumer unsubscribes to notifications**

1. The NF service consumer shall send a DELETE request to the resource identified by the URI previously received during subscription creation.
- 2a. On success, the LMF responds with "204 No Content".
- 2b. On failure or redirection, one of the HTTP status code listed in Table 6.1.3.3.3.1-3 shall be returned. For a 4xx/5xx response, the message body containing a ProblemDetails structure with the "cause" attribute set to one of the application error listed in Table 6.1.3.3.3.1-3.

## 5.3 Nlmf\_Broadcast Service

### 5.3.1 Service Description

The Nlmf\_Broadcast service enables an NF to obtain ciphering keys and associated parameters applicable to location assistance data that is broadcast to subscribed UEs in ciphered form.

### 5.3.2 Service Operations

#### 5.3.2.1 Introduction

The service operations defined for the Nlmf\_Broadcast service are as follows:

- CipheringKeyData: It provides the ciphering key information to the consumer NF.

#### 5.3.2.2 CipheringKeyData

##### 5.3.2.2.1 General

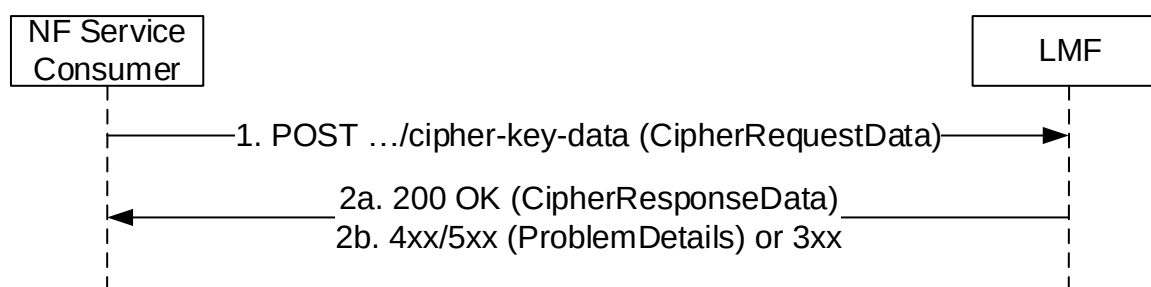
The following procedures are defined, using the "CipheringKeyData" service operation:

- Request Ciphering Key Information
- Provide Ciphering Key Information

NOTE: The Request Ciphering Key procedure is included in order to provide a valid context in OpenAPI version 3 for the Provide Ciphering Key Information procedure. The Request Ciphering Key procedure is not used for support of ciphering key transfer in 3GPP TS 23.273 [19] and hence need not be supported by an NF Service Consumer or by an LMF.

#### 5.3.2.2.2 Request Ciphering Key Information

This procedure allows a consumer NF to request ciphering key information.



**Figure 5.3.2.2.2-1: CipheringKeyData Request**

1. The NF Service Consumer shall send an HTTP POST request to the resource URI associated with the "cipher-key-data" custom operation. The request body shall include a notification callback URI.
- 2a. On success, "200 OK" shall be returned. The response body shall indicate whether the LMF has ciphering key data. If the LMF has ciphering key data, the Provide Ciphering Key Information procedure is used to provide the ciphering key data to the NF Service Consumer.
- 2b. On failure or redirection, one of the HTTP status codes listed in Table 6.2.4.4.2-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in Table 6.2.7.3-1.

#### 5.3.2.2.3 Provide Ciphering Key Information

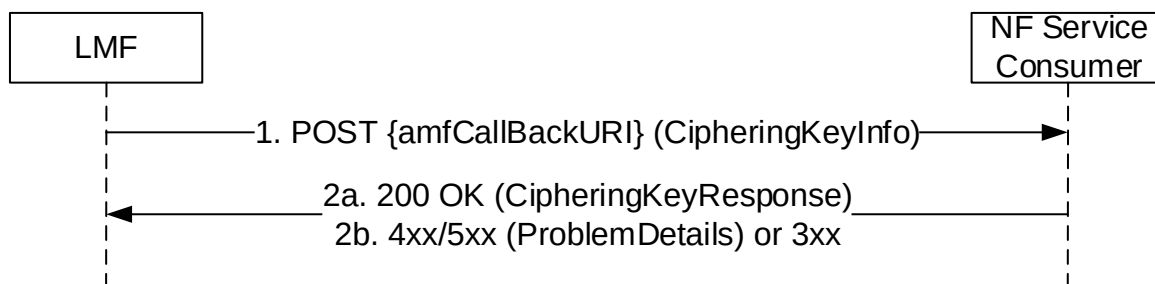
This procedure notifies the NF Service Consumer (i.e. AMF) about updated ciphering key information applicable to broadcast of location assistance data in ciphered form to subscribed UEs. The notification is delivered to:

- the callback URI of an AMF received during an earlier CipheringKeyData request service operation if still available; or
- a callback URI registered in the NRF, if the AMF registered to the NRF with notification endpoints for ciphering key data notifications;

Otherwise (if not available),

- an AMF callback URI locally provisioned in the LMF.

The procedure is invoked by issuing a POST request to the callback URI of the NF Service Consumer. See figure 5.3.2.2.3-1.



**Figure 5.3.2.2.3-1: CipheringKeyData Notify**

1. The LMF shall send an HTTP POST request to the callback URI for the NF service consumer determined as described above. The request body shall include one or more ciphering keys and for each ciphering key may include a ciphering key value, ciphering key identifier, validity period and set of applicable types of broadcast assistance data.
- 2a. On success or partial success, "200 OK" shall be returned. The response body shall indicate which ciphering key information was successfully stored by the NF service consumer.
- 2b. On failure or redirection to store any ciphering key information, one of the HTTP status codes listed in table 6.2.5.1.3.1-2 shall be returned. For a 4xx/5xx response, the message body shall contain a ProblemDetails structure with the "cause" attribute set to one of the application errors listed in table 6.2.5.1.3.1-2.

## 5.4 Nlmf\_DataExposure Service

### 5.4.1 Service Description

The Nlmf\_DataExposure service enables an NF to subscribe to input data for requested location measurement data for ML training-based positioning.

### 5.4.2 Service Operation

#### 5.4.2.1 Introduction

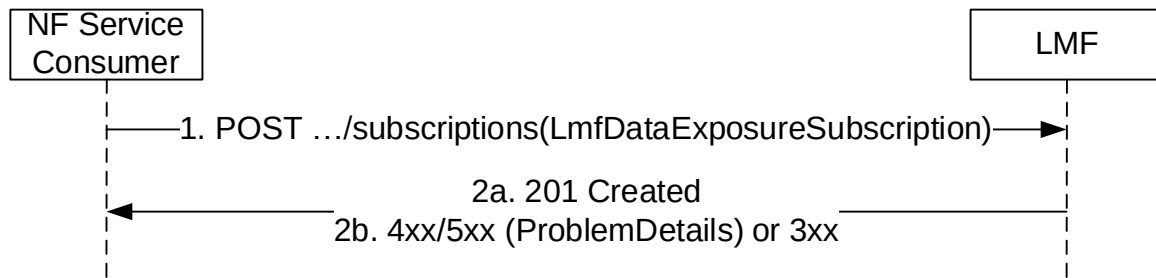
The service operations defined for Nlmf\_DataExposure service are defined as follows:

- Lmf Data Exposure Subscription: It allows the consumer NF to subscribe to the input data used for ML model training or performance for LMF based AI/ML positioning.
- Lmf Data Exposure Modification: It allows the NF consumer to modify the subscription used for ML model training or performance monitoring for LMF based AI/ML positioning.
- Lmf Data Exposure Notification: It allows the consumer NF to get notification about the input data used for ML model training or performance monitoring for LMF based AI/ML positioning.
- Lmf Data Exposure Unsubscription: It allows the consumer NF to cancel subscription to input data used for ML model training or performance monitoring for LMF based AI/ML positioning.

#### 5.4.2.2 LMF Data Exposure Subscription

This procedure allows NF Service Consumer to subscribe for notifications of sampling data used for ML model training or ML model performance monitoring for LMF-based AI/ML Positioning by using HTTP POST method with the URI of individual subscription resource.





**Figure 5.4.2.2-1: Subscription of NF service consumer to LMF data exposure**

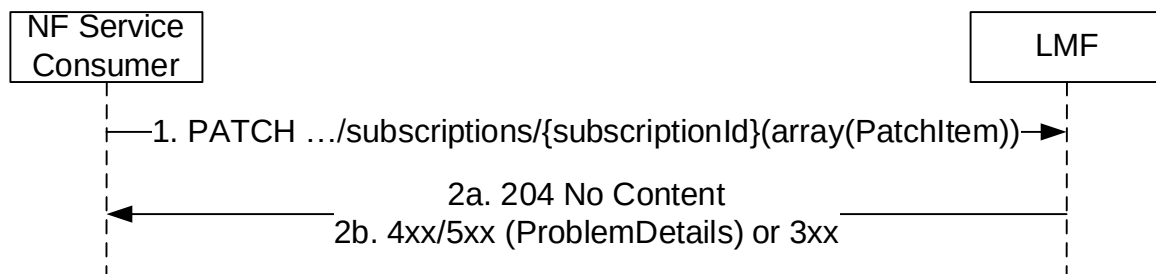
1. The NF Service Consumer shall send an HTTP POST request to the resource URI associated with the subscription collection. The input parameters for the request shall include notification target address, notification correlation ID and AoI. The request may include other optional parameters for data collection (see clause 6.3.6.2.2).

If the subscription is for performance monitoring of LMF based AI/ML positioning, the request shall also include the ML model identifier to be monitored.

- 2a. On success, "201 Created" shall be returned. The response body shall include a HTTP Location header including the subscription ID together with the status code 201 indicating the requested resource is created in the response message.
  - If the LmfDataExposureSubscription contains "recurTime" IE, then LMF shall generate a report at the end of every time interval provided in the "recurTime" IE.
  - If the LmfDataExposureSubscription contains "recurTime" IE and "numOfSamples" IE, then LMF shall generate a report when the requested number of samples are met or at the end of every time interval provided in the "recurTime" IE.
  - When the NF Service Consumer is LMF, if the LmfDataExposureSubscription contains "pruIndication" IE and set to true, then LMF shall report the PRU locations and the measurement data from the NG-RAN for the PRUs only.
- 2b. On failure or redirection, one of the HTTP status codes listed in Table 6.3.3.2.3.1-3 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

### 5.4.2.3 LMF Data Exposure Modification

This procedure allows NF Service Consumer to modify an existing subscription by using HTTP PATCH method with the URI of individual subscription resource.

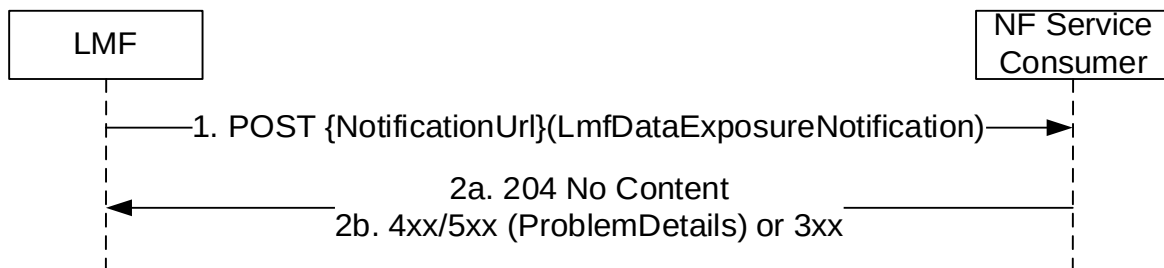


**Figure 5.4.2.3-1: Modification of subscription for LMF data exposure**

1. The NF Service Consumer shall send a PATCH request to modify a subscription resource in the LMF.
- 2a. On success, "204 No Content" shall be returned.
- 2b. On failure or redirection, one of the HTTP status codes listed in Table 6.3.3.3.3.1-3 shall be returned. For a 4xx/5xx response, the message body contains a ProblemDetails indicating appropriate additional error information.

#### 5.4.2.4 LMF Data Exposure Notification

This procedure allows the LMF to send a notification to the NF consumer including the sampling data used for ML model training or ML model performance monitoring for LMF-based AI/ML Positioning.



**Figure 5.4.2.4-1: Notification for LMF data exposure**

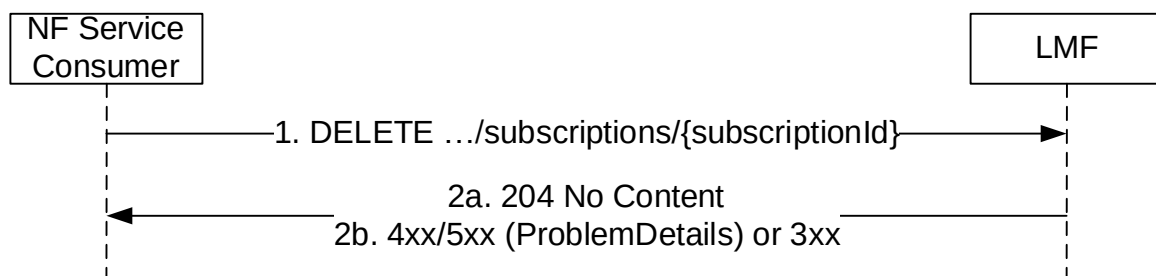
1. LMF shall send a POST request to the NF Service Consumer with the information on Notification target address. The request body shall include notification correlation ID, a list of report(s) to be notified to the NF Service Consumer. If the subscription is for performance monitoring of LMF based AI/ML positioning, the request shall also include the ML model identifier to be monitored and the location of the PRU/UE estimated by the LMF using LMF based AI/ML positioning with the indicated ML model.

The LMF may provide a cause code in the report to indicate additional information to the NF consumer, e.g. when the requested number of sampling data cannot be met within the Area of Interest during the request time window.

- 2a. On success, "204 No content" shall be returned by the NF Service Consumer.
- 2b. On failure or redirection, one of the appropriate HTTP status codes listed in Table 6.3.5.2.3-2 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

#### 5.4.2.5 LMF Data Exposure Unsubscription

This procedure allows a consumer NF to delete the notification about the input data used for ML model training or ML model performance monitoring for LMF-based AI/ML Positioning.



**Figure 5.4.2.5-1: Unsubscribe ServiceOperation for LMF data exposure**

1. The NF consumer shall send a Delete request to LMF to cancel the subscription to input data used for ML model training or model performance monitoring for LMF based AI/ML positioning.
- 2a. On success, "204 No content" shall be returned by the NF Service Consumer.
- 2b. On failure or redirection, one of the appropriate HTTP status codes listed in Table 6.3.3.3.2-3 shall be returned. For a 4xx/5xx response, the message body should contain a ProblemDetails structure indicating appropriate additional error information.

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## 6 API Definitions

### 6.1 Nlmf\_Location Service API

#### 6.1.1 API URI

The Nlmf\_Location service shall use the Nlmf\_Location API.

The API URI of the Nlmf\_Location API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "nlmf-loc".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.1.3.

#### 6.1.2 Usage of HTTP

##### 6.1.2.1 General

HTTP/2, as defined in IETF RFC 9113 [12], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

HTTP messages and bodies for the Nlmf\_Location service shall comply with the OpenAPI [14] specification contained in Annex A.

##### 6.1.2.2 HTTP Standard Headers

###### 6.1.2.2.1 General

###### 6.1.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [13], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of 3GPP TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [15]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

Multipart messages shall also be supported (see clause 6.1.2.4) using the content type "multipart/related", comprising:

- one JSON body part with the "application/json" content type; and
- one or more binary body parts with 3gpp vendor specific content subtypes.

The 3gpp vendor specific content subtypes defined in Table 6.1.2.2.2-1 shall be supported.

**Table 6.1.2.2-1: 3GPP vendor specific content subtypes**

| content subtype                                                                                                                                                             | Description                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| vnd.3gpp.lpp                                                                                                                                                                | Binary encoded content, encoding LTE Positioning Protocol (LPP) IEs, as specified in 3GPP TS 37.355 [21]. |
| NOTE: Using 3GPP vendor content subtypes allows to describe the nature of the opaque content (e.g. LPP information) without having to rely on metadata in the JSON content. |                                                                                                           |

See clause 6.1.2.4 for the binary contents supported in the binary body part of multipart messages.

### 6.1.2.3 HTTP custom headers

#### 6.1.2.3.1 General

The following HTTP custom headers shall be supported:

- 3gpp-Sbi-Message-Priority: See 3GPP TS 29.500 [4], clause 5.2.3.2.2.

This API does not define any new HTTP custom headers.

### 6.1.2.4 HTTP multipart messages

HTTP multipart messages shall be supported, to transfer opaque LPP Information, in the following service operations (and HTTP messages):

- DetermineLocation Request (POST);

HTTP multipart messages shall include one JSON body part and one or more binary body parts comprising:

- one LPP payload (see clause 6.1.6.4).

The JSON body part shall be the "root" body part of the multipart message. It shall be encoded as the first body part of the multipart message. The "Start" parameter does not need to be included.

The multipart message shall include a "type" parameter (see IETF RFC 2387 [33]) specifying the media type of the root body part, i.e. "application/json".

NOTE: The "root" body part (or "root" object) is the first body part the application processes when receiving a multipart/related message, see IETF RFC 2387 [33]. The default root is the first body within the multipart/related message. The "Start" parameter indicates the root body part, e.g. when this is not the first body part in the message.

For each binary body part in a HTTP multipart message, the binary body part shall include a Content-ID header (see IETF RFC 2045 [35]), and the JSON body part shall include an attribute, defined with the RefToBinaryData type, that contains the value of the Content-ID header field of the referenced binary body part.

## 6.1.3 Resources

### 6.1.3.1 Overview

The structure of the Resource URIs of the Nlmf\_Location service is shown in figure 6.1.3.1-1.

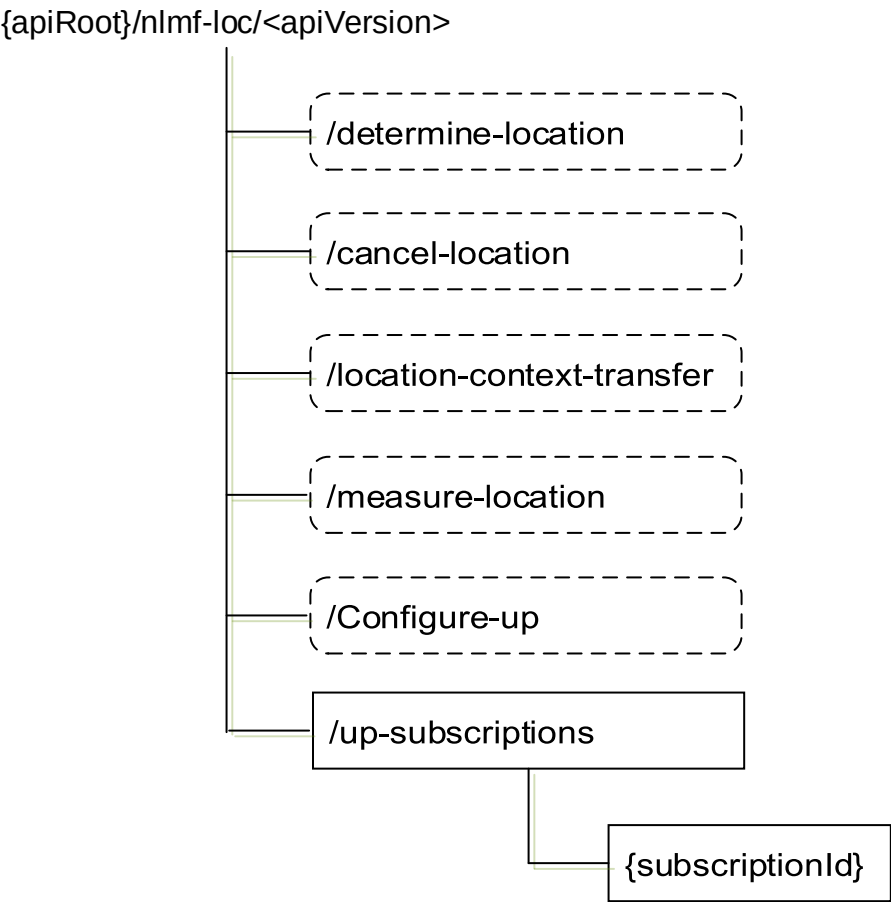


Figure 6.1.3.1-1: Resource URI structure of the Nlmf\_Location API

Table 6.1.3.1-1 provides an overview of the resources and applicable HTTP methods.

Table 6.1.3.1-1: Resources and methods overview

| Resource name                 | Resource URI                       | HTTP method or custom operation | Description                                            |
|-------------------------------|------------------------------------|---------------------------------|--------------------------------------------------------|
| up-subscriptions (Collection) | /up-subscriptions                  | POST                            | Create a subscription                                  |
| up-subscription (Document)    | /up-subscriptions/{subscriptionId} | DELETE                          | Delete the subscription identified by {subscriptionId} |

6.1.3.2 Resource: up-subscriptions (Collection)

6.1.3.2.1 Description

This resource represents a collection of subscriptions to notifications.

6.1.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nlmf-loc/<apiVersion>/up-subscriptions**

This resource shall support the resource URI variables defined in table 6.1.3.2.2-1.

**Table 6.1.3.2.2-1: Resource URI variables for this resource**

| Name       | Data type | Definition       |
|------------|-----------|------------------|
| apiRoot    | string    | See clause 6.1.1 |
| apiVersion | string    | See clause 6.1.1 |

### 6.1.3.2.3 Resource Standard Methods

#### 6.1.3.2.3.1 POST

This method creates a new subscription. This method shall support the URI query parameters specified in table 6.1.3.2.3.1-1.

**Table 6.1.3.4.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|------|-----------|---|-------------|-------------|
| n/a  |           |   |             |             |

This method shall support the request data structures specified in table 6.1.3.2.3.1-2 and the response data structures and response codes specified in table 6.1.3.2.3.1-3.

**Table 6.1.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type      | P | Cardinality | Description                                     |
|----------------|---|-------------|-------------------------------------------------|
| UpSubscription | M | 1           | Input parameters to the "UPSubscribe" operation |

**Table 6.1.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                           | P | Cardinality | Response codes         | Description                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UpSubscription                                                                                                                                                                                                                                      | M | 1           | 201 Created            | This case represents the successful creation of a subscription.<br><br>Upon success, the HTTP response shall include a "Location" HTTP header that contains the resource URI of the created resource. |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 307 Temporary Redirect | Temporary redirection.                                                                                                                                                                                |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 308 Permanent Redirect | Permanent redirection.                                                                                                                                                                                |
| NOTE: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                       |

**Table 6.1.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name     | Data type | P | Cardinality | Description                                                                                                                         |
|----------|-----------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Location | string    | M | 1           | Contains the URI of the newly created resource, according to the structure: {apiRoot}/nlmf-loc/v1/up-subscriptions/{subscriptionId} |

**Table 6.1.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

**Table 6.1.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

### 6.1.3.3 Resource: up-subscription (Document)

#### 6.1.3.3.1 Description

This resource represents an individual subscription.

#### 6.1.3.3.2 Resource Definition

Resource URI: {apiRoot}/nlmf-loc/v1/up-subscriptions/{subscriptionId}

This resource shall support the resource URI variables defined in table 6.1.3.5.2-1.

**Table 6.1.3.3.2-1: Resource URI variables for this resource**

| Name           | Data type | Definition                         |
|----------------|-----------|------------------------------------|
| apiRoot        | string    | See clause 6.1.1                   |
| subscriptionId | string    | Represents a specific subscription |

#### 6.1.3.3.3 Resource Standard Methods

##### 6.1.3.3.3.1 DELETE

This method terminates an existing subscription. This method shall support the URI query parameters specified in table 6.1.3.3.3.1-1.

**Table 6.1.3.3.3.1-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description |
|------|-----------|---|-------------|-------------|
| n/a  |           |   |             |             |

This method shall support the request data structures specified in table 6.1.3.3.3.1-2 and the response data structures and response codes specified in table 6.1.3.3.3.1-3.

**Table 6.1.3.3.3.1-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|-----------|---|-------------|-------------|
| n/a       |   |             |             |

**Table 6.1.3.3.1-3: Data structures supported by the DELETE Response Body on this resource**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|------------------------|
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content         |                        |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection. |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection. |
| NOTE: The mandatory HTTP error status codes for the DELETE method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                        |

**Table 6.1.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

**Table 6.1.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

## 6.1.4 Custom Operations without associated resources

### 6.1.4.1 Overview

The URI structure for Custom Operations without associated resources is included as part of the Figure 6.1.3.1-1

**Table 6.1.4.1-1: Custom operations without associated resources**

| Operation Name            | Custom operation URI       | Mapped HTTP method | Description (Service Operation)                   |
|---------------------------|----------------------------|--------------------|---------------------------------------------------|
| determine-location        | /determine-location        | POST               | Determine Location                                |
| cancel-location           | /cancel-location           | POST               | Cancel Location                                   |
| location-context-transfer | /location-context-transfer | POST               | Transfer Location Context                         |
| measure-location          | /measure-location          | POST               | Location Measure                                  |
| configure-up              | /configure-up              | POST               | Create, Modify or Terminate the LCS-UP connection |

NOTE: The Custom operation URI above are deviating from the URI Path Segment Naming Conventions defined in clause 5.1.3.2 of 3GPP TS 29.501 [5], but they are not changed to maintain backwards compatibility.

### 6.1.4.2 Operation: determine-location

#### 6.1.4.2.1 Description

This clause will describe the custom operation and what it is used for, and the custom operation's URI.



### 6.1.4.2.2 Operation Definition

This operation shall support the response data structures and response codes specified in tables 6.1.4.2.2-1 and 6.1.4.2.2-2.

**Table 6.1.4.2.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description                                            |
|-----------|---|-------------|--------------------------------------------------------|
| InputData | M | 1           | Input parameters to the "Determine Location" operation |

**Table 6.1.4.2.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LocationDataExt                                                                                                                                                                                                                                       | M | 1           | 200 OK                    | This case represents the successful retrieval of the location of the UE or successful activation of periodic or triggered location in the UE.<br><br>Upon success, a response body is returned containing the different parameters of the location data with one or more corresponding UEs if obtained, such as:<br><ul style="list-style-type: none"> <li>- Geographic Area</li> <li>- Civic Location</li> <li>- Local location,</li> <li>- Positioning methods</li> </ul> |
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content            | This case represents the successful delivery of location assistance data to the UE, during MO-LR requesting for location assistance data for the UE.                                                                                                                                                                                                                                                                                                                        |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect    | Temporary redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect    | Permanent redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden             | The "cause" attribute may be used to indicate the following application errors:<br><ul style="list-style-type: none"> <li>- POSITIONING_DENIED</li> <li>- UNSPECIFIED</li> <li>- UNSUPPORTED_BY_UE</li> <li>- PAGING_NOT_ALLOWED</li> </ul> See table 6.1.7.3-1 for the description of these errors.                                                                                                                                                                        |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 500 Internal Server Error | The "cause" attribute may be used to indicate the following application error:<br><ul style="list-style-type: none"> <li>- POSITIONING_FAILED</li> </ul> See table 6.1.7.3-1 for the description of these errors.                                                                                                                                                                                                                                                           |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 501 Not Implemented       | The "cause" attribute may be used to indicate one of the following application errors:<br><ul style="list-style-type: none"> <li>- UNSUPPORTED_EVENT_TYPE</li> </ul>                                                                                                                                                                                                                                                                                                        |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 504 Gateway Timeout       | The "cause" attribute may be used to indicate the following application error:<br><ul style="list-style-type: none"> <li>- UNREACHABLE_USER</li> </ul> See table 6.1.7.3-1 for the description of this error.                                                                                                                                                                                                                                                               |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Table 6.1.4.2.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

**Table 6.1.4.2.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

### 6.1.4.3 Operation: cancel-location

#### 6.1.4.3.1 Description

This clause describes the custom operation and what it is used for.

#### 6.1.4.3.2 Operation Definition

This operation shall support the request and response data structures and response codes specified in table 6.1.4.3.2-1 and table 6.1.4.3.2-2.

**Table 6.1.4.3.2-1: Data structures supported by the POST Request Body on this resource**

| Data type     | P | Cardinality | Description                              |
|---------------|---|-------------|------------------------------------------|
| CancelLocData | M | 1           | The information used to cancel location. |

**Table 6.1.4.3.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content         | This case represents successful cancellation of location.                                                                                                                                |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                       |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                       |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors:<br>- UNSPECIFIED<br>- LOCATION_SESSION_UNKNOWN<br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                          |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                          |

**Table 6.1.4.3.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

**Table 6.1.4.3.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

#### 6.1.4.4 Operation: location-context-transfer

##### 6.1.4.4.1 Description

This clause will describe the custom operation and what it is used for.

##### 6.1.4.4.2 Operation Definition

This operation shall support the request and response data structures and response codes specified in table 6.1.4.4.2-1 and table 6.1.4.4.2-2.

**Table 6.1.4.4.2-1: Data structures supported by the POST Request Body on this resource**

| Data type      | P | Cardinality | Description                                                   |
|----------------|---|-------------|---------------------------------------------------------------|
| LocContextData | M | 1           | Input parameters to the "Location Context Transfer" operation |

**Table 6.1.4.4.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content         | This case represents successful transfer of the location context.                                                                                                                                                                                                                                                           |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                                          |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                                          |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors:<br><ul style="list-style-type: none"> <li>- UNSPECIFIED</li> <li>- LOCATION_TRANSFER_NOT_SUPPORTED</li> <li>- INSUFFICIENT_RESOURCES</li> <li>- EVENT_REPORT_UNRECOGNIZED</li> </ul> See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                                                                                             |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                                                                                                                                                             |

**Table 6.1.4.4.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

**Table 6.1.4.4.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

## 6.1.4.5 Operation: measure-location

### 6.1.4.5.1 Description

This clause will describe the custom operation and what it is used for, and the custom operation's URI.

### 6.1.4.5.2 Operation Definition

This operation shall support the response data structures and response codes specified in tables 6.1.4.5.2-1 and 6.1.4.5.2-2.

**Table 6.1.4.5.2-1: Data structures supported by the POST Request Body on this resource**

| Data type         | P | Cardinality | Description                                         |
|-------------------|---|-------------|-----------------------------------------------------|
| LocMeasurementReq | M | 1           | Input parameters to the "MeasurementData" operation |

**Table 6.1.4.5.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                           | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LocMeasurementResp                                                                                                                                                                                                                                  | M | 1           | 200 OK                 | This case represents the successful retrieval of the location measurements of the PRU.                                                                                                                                                                                                                                                                         |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 307 Temporary Redirect | Temporary redirection. The response shall include a Location header field containing a different URI, or the same URI if a request is redirected to the same target resource via a different SCP. In the former case, the URI shall be an alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 308 Permanent Redirect | Permanent redirection. The response shall include a Location header field containing a different URI, or the same URI if a request is redirected to the same target resource via a different SCP. In the former case, the URI shall be an alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. |
| ProblemDetails                                                                                                                                                                                                                                      | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors: <ul style="list-style-type: none"> <li>- UNSPECIFIED</li> <li>- LOCATION_MEASUREMENT_UNKNOWN</li> </ul> See table 6.1.7.3-1 for the description of these errors.                                                                                                               |
| NOTE: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                                                                                                                                |

**Table 6.1.4.5.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

**Table 6.1.4.5.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

## 6.1.4.6 Void

### 6.1.4.7 Operation: configure-up

#### 6.1.4.7.1 Description

This clause will describe the custom operation and what it is used for, and the custom operation's URI.

#### 6.1.4.7.2 Operation Definition

This operation shall support the response data structures and response codes specified in tables 6.1.4.7.2-1 and 6.1.4.7.2-2.

**Table 6.1.4.7.2-1: Data structures supported by the POST Request Body on this resource**

| Data type | P | Cardinality | Description                                  |
|-----------|---|-------------|----------------------------------------------|
| UpConfig  | M | 1           | Input parameters to the "UPConfig" operation |

**Table 6.1.4.7.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                           | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                                 |   |             | 204 No Content         | This case represents the successful set up, modify or terminate LCS-UP connection.                                                                                                                                                                                                                                                                             |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 307 Temporary Redirect | Temporary redirection. The response shall include a Location header field containing a different URI, or the same URI if a request is redirected to the same target resource via a different SCP. In the former case, the URI shall be an alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. |
| RedirectResponse                                                                                                                                                                                                                                    | O | 0..1        | 308 Permanent Redirect | Permanent redirection. The response shall include a Location header field containing a different URI, or the same URI if a request is redirected to the same target resource via a different SCP. In the former case, the URI shall be an alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. |
| ProblemDetails                                                                                                                                                                                                                                      | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors:<br>- UNSPECIFIED<br><br>See table 6.1.7.3-1 for the description of these errors.                                                                                                                                                                                               |
| NOTE: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                                                                                                                                |

**Table 6.1.4.7.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

**Table 6.1.4.7.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. Or the same URI, if a request is redirected to the same target resource via a different SCP. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                            |

## 6.1.5 Notifications

This clause specifies the notifications provided by the Nlmf\_Location service.

**Table 6.1.5.1-1: Notifications overview**

| Notification | Callback URI          | HTTP method or custom operation | Description (service operation) |
|--------------|-----------------------|---------------------------------|---------------------------------|
| EventNotify  | {hgmlcCallbackURI}    | POST                            |                                 |
| UPNotify     | {upNotifyCallbackUri} | POST                            | UPNotify service operation      |

### 6.1.5.1 EventNotify

#### 6.1.5.1.1 Description

The EventNotify operation is used to notify the occurrence of periodic or triggered location event for a target UE to a consumer NF (e.g. GMLC).

#### 6.1.5.1.2 Notification Definition

Callback URI: {hgmlcCallbackURI}

See clause 5.2.2.2.2 for the description of how the LMF obtains the Callback URI of the NF Service Consumer (e.g. GMLC).

#### 6.1.5.1.3 Notification Standard Methods

##### 6.1.5.1.3.1 POST

This method sends a Location event notify to the NF Service Consumer.

This method shall support the request and response data structures and response codes specified in table 6.1.5.1.3.1-1 and table 6.1.5.1.3.1-2.

**Table 6.1.5.1.3.1-1: Data structures supported by the POST Request Body**

| Data type          | P | Cardinality | Description                                               |
|--------------------|---|-------------|-----------------------------------------------------------|
| EventNotifyDataExt | M | 1           | Input parameters to the "Location Event Notify" operation |

**Table 6.1.5.1.3.1-2: Data structures supported by the POST Response Body**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content         | This case represents successful notification of the event.                                                                                                                                   |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                           |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                           |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors:<br>- UNSPECIFIED<br>- LOCATION_SESSION_UNKNOWN<br><br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                              |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                              |

**Table 6.1.5.1.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

**Table 6.1.5.1.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

## 6.1.5.2 UPNotify

### 6.1.5.2.1 Description

The UPNotify operation is used to notify the status or modification of a secure LCS-UP connection for a target UE to a consumer NF (e.g. AMF).

### 6.1.5.2.2 Notification Definition

Callback URI: {upNotifyCallBackUri}

See clause 5.2.2.7.2 for the description of how the LMF obtains the Callback URI of the NF Service Consumer (e.g. AMF).



## 6.1.5.2.3 Notification Standard Methods

## 6.1.5.2.3.1 POST

This method sends a notification of LCS-UP connection status to the NF service consumer.

This method shall support the request and response data structures and response codes specified in table 6.1.5.2.3.1-1 and table 6.1.5.2.3.1-2.

**Table 6.1.5.2.3.1-1: Data structures supported by the POST Request Body**

| Data type    | P | Cardinality | Description                                  |
|--------------|---|-------------|----------------------------------------------|
| UpNotifyData | M | 1           | Input parameters to the "UPNotify" operation |

**Table 6.1.5.2.3.1-2: Data structures supported by the POST Response Body**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                                   |   |             | 204 No Content         | This case represents successful notification of the event.                                                                                                                                   |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                           |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                           |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application errors:<br>- UNSPECIFIED<br>- LOCATION_SESSION_UNKNOWN<br><br>See table 6.1.7.3-1 for the description of this error. |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                              |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                              |

**Table 6.1.5.2.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

**Table 6.1.5.2.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

## 6.1.6 Data Model

### 6.1.6.1 General

This clause specifies the application data model supported by the API.

Table 6.1.6.1-1 specifies the data types defined for the Nlmf\_Location service based interface protocol.

Table 6.1.6.1-1: Nlmf\_Location specific Data Types

| Data type                                    | Clause defined | Description                                                                           |
|----------------------------------------------|----------------|---------------------------------------------------------------------------------------|
| InputData                                    | 6.1.6.2.2      | Information within Determine Location Request                                         |
| LocationData                                 | 6.1.6.2.3      | Information within Determine Location Response                                        |
| GeographicalCoordinates                      | 6.1.6.2.4      | Geographical coordinates                                                              |
| GeographicArea                               | 6.1.6.2.5      | Geographic area specified by different shape                                          |
| Point                                        | 6.1.6.2.6      | Ellipsoid Point                                                                       |
| PointUncertaintyCircle                       | 6.1.6.2.7      | Ellipsoid point with uncertainty circle                                               |
| PointUncertaintyEllipse                      | 6.1.6.2.8      | Ellipsoid point with uncertainty ellipse                                              |
| Polygon                                      | 6.1.6.2.9      | Polygon                                                                               |
| PointAltitude                                | 6.1.6.2.10     | Ellipsoid point with altitude                                                         |
| PointAltitudeUncertainty                     | 6.1.6.2.11     | Ellipsoid point with altitude and uncertainty ellipsoid                               |
| EllipsoidArc                                 | 6.1.6.2.12     | Ellipsoid Arc                                                                         |
| LocationQoS                                  | 6.1.6.2.13     | QoS of Location request                                                               |
| CivicAddress                                 | 6.1.6.2.14     | Indicates a Civic address                                                             |
| PositioningMethodAndUsage                    | 6.1.6.2.15     | Indicates the usage of a positioning method                                           |
| GnssPositioningMethodAndUsage                | 6.1.6.2.16     | Indicates the usage of a Global Navigation Satellite System (GNSS) positioning method |
| VelocityEstimate                             | 6.1.6.2.17     | Velocity estimate                                                                     |
| HorizontalVelocity                           | 6.1.6.2.18     | Horizontal velocity                                                                   |
| HorizontalWithVerticalVelocity               | 6.1.6.2.19     | Horizontal and vertical velocity                                                      |
| HorizontalVelocityWithUncertainty            | 6.1.6.2.20     | Horizontal velocity with speed uncertainty                                            |
| HorizontalWithVerticalVelocityAndUncertainty | 6.1.6.2.21     | Horizontal and vertical velocity with speed uncertainty                               |
| UncertaintyEllipse                           | 6.1.6.2.22     | Ellipse with uncertainty                                                              |
| UeLcsCapability                              | 6.1.6.2.23     | Indicates the LCS capability supported by the UE.                                     |
| PeriodicEventInfo                            | 6.1.6.2.24     | Indicates the information of periodic event reporting                                 |
| AreaEventInfo                                | 6.1.6.2.25     | Indicates the information of area based event reporting                               |
| ReportingArea                                | 6.1.6.2.26     | Indicates an area for event reporting                                                 |
| MotionEventInfo                              | 6.1.6.2.27     | Indicates the information of motion based event reporting                             |
| CancelLocData                                | 6.1.6.2.29     | Information within Cancel Location Request                                            |
| LocContextData                               | 6.1.6.2.30     | Information within Transfer Location Context Request                                  |
| EventReportMessage                           | 6.1.6.2.31     | Indicates an event report message                                                     |
| EventReportingStatus                         | 6.1.6.2.32     | Indicates the status of event reporting                                               |
| UeLocationInfo                               | 6.1.6.2.33     | Indicates location information of a UE                                                |
| EventNotifyData                              | 6.1.6.2.34     | Information within Event Notify Request                                               |
| UeConnectivityState                          | 6.1.6.2.35     | Indicates the connectivity state of a UE                                              |
| LocalOrigin                                  | 6.1.6.2.36     | Indicates a Local origin in a reference system.                                       |
| RelativeCartesianLocation                    | 6.1.6.2.37     | Relative Cartesian Location                                                           |
| Local2dPointUncertaintyEllipse               | 6.1.6.2.38     | Local 2D point with uncertainty ellipse                                               |
| Local3dPointUncertaintyEllipsoid             | 6.1.6.2.39     | Local 3D point with uncertainty ellipsoid                                             |
| UncertaintyEllipsoid                         | 6.1.6.2.40     | Ellipsoid with uncertainty                                                            |
| LocalArea                                    | 6.1.6.2.41     | Local area specified by different shape                                               |
| UeAreaIndication                             | 6.1.6.2.42     | UE area Indication                                                                    |
| MinorLocationQoS                             | 6.1.6.2.43     | Minor Location QoS                                                                    |
| MbsrInfo                                     | 6.1.6.2.44     | MBSR Information                                                                      |
| LocMeasurementReq                            | 6.1.6.2.45     | Location Measurement Request                                                          |
| LocMeasurementResp                           | 6.1.6.2.46     | Location Measurement Response                                                         |
| LocMeasurements                              | 6.1.6.2.47     | Location Measurements result                                                          |
| HighAccuracyGnssMetrics                      | 6.1.6.2.48     | High Accuracy GNSS Metrics                                                            |
| UpNotifyData                                 | 6.1.6.2.49     | UP Notify Data                                                                        |
| UpSubscription                               | 6.1.6.2.50     | UP Subscription                                                                       |
| RelatedUe                                    | 6.1.6.2.51     | Specifies information for related UE for ranging and sidelink positioning             |
| UpConfig                                     | 6.1.6.2.52     | UP Config                                                                             |
| RangeDirection                               | 6.1.6.2.53     | Represents the straight line distance (range) and direction between two points.       |

|                                 |            |                                                                                                      |
|---------------------------------|------------|------------------------------------------------------------------------------------------------------|
| 2DRelativeLocation              | 6.1.6.2.54 | Represents 2D local co-ordinates with origin corresponding to another known point.                   |
| 3DRelativeLocation              | 6.1.6.2.55 | Represents 3D local co-ordinates with origin corresponding to another known point.                   |
| AddLocationDatas                | 6.1.6.2.56 | Contains one or more LocationData.                                                                   |
| AddEventNotifyDatas             | 6.1.6.2.57 | Contains one or more EventNotifyData.                                                                |
| LocationDataExt                 | 6.1.6.2.58 | Represents location data with one or more corresponding UEs.                                         |
| EventNotifyDataExt              | 6.1.6.2.59 | Represents notified data with one or more corresponding UEs, containing one or more EventNotifyData. |
| MappedLocationQoSeps            | 6.1.6.2.60 | Mapped Location QoS for EPS                                                                          |
| AdditionalUeInfo                | 6.1.6.2.61 | MBSR UE or MWAB UE Information                                                                       |
| AddLocationContext              | 6.1.6.2.62 | LCS context information                                                                              |
| TimeWindowsNrppa                | 6.1.6.2.63 | Time windows for scheduling of PRU measurements when network assisted positioning is used.           |
| RelativeVelocityWithUncertainty | 6.1.6.2.64 | Represents relative velocity with uncertainty                                                        |
| RadialVelocity                  | 6.1.6.2.65 | Represents rate of change of a range                                                                 |
| AngularVelocity                 | 6.1.6.2.66 | Represents rate of change of an angle                                                                |
| TrpParameter                    | 6.1.6.2.67 | TRP Parameter                                                                                        |
| Altitude                        | 6.1.6.3.2  | Indicates value of altitude                                                                          |
| Angle                           | 6.1.6.3.2  | Indicates value of angle                                                                             |
| Uncertainty                     | 6.1.6.3.2  | Indicates value of uncertainty                                                                       |
| Orientation                     | 6.1.6.3.2  | Indicates value of orientation angle                                                                 |
| HorizAxesOrientation            | 6.1.6.3.2  | Orientation of Horizontal Coordinate System                                                          |
| Confidence                      | 6.1.6.3.2  | Indicates value of confidence                                                                        |
| Accuracy                        | 6.1.6.3.2  | Indicates value of accuracy                                                                          |
| InnerRadius                     | 6.1.6.3.2  | Indicates value of the inner radius                                                                  |
| CorrelationID                   | 6.1.6.3.2  | LCS Correlation ID                                                                                   |
| AgeOfLocationEstimate           | 6.1.6.3.2  | Indicates value of the age of the location estimate                                                  |
| HorizontalSpeed                 | 6.1.6.3.2  | Indicates value of horizontal speed                                                                  |
| VerticalSpeed                   | 6.1.6.3.2  | Indicates value of vertical speed                                                                    |
| SpeedUncertainty                | 6.1.6.3.2  | Indicates value of speed uncertainty                                                                 |
| BarometricPressure              | 6.1.6.3.2  | Specifies the measured uncompensated atmospheric pressure                                            |
| LcsServiceType                  | 6.1.6.3.2  | LCS service type                                                                                     |
| LdrReference                    | 6.1.6.3.2  | LDR Reference                                                                                        |
| LirReference                    | 6.1.6.3.2  | LIR Reference                                                                                        |
| ReportingAmount                 | 6.1.6.3.2  | Number of required periodic event reports                                                            |
| ReportingInterval               | 6.1.6.3.2  | Event reporting periodic interval                                                                    |
| MinimumInterval                 | 6.1.6.3.2  | Minimum interval between event reports                                                               |
| MaximumInterval                 | 6.1.6.3.2  | Maximum interval between event reports                                                               |
| SamplingInterval                | 6.1.6.3.2  | Maximum time interval between consecutive evaluations by a UE of a trigger event                     |
| ReportingDuration               | 6.1.6.3.2  | Maximum duration of event reporting                                                                  |
| LinearDistance                  | 6.1.6.3.2  | Minimum straight line distance moved by a UE to trigger a motion event report                        |
| LMFIdentification               | 6.1.6.3.2  | LMF identification                                                                                   |
| EventReportCounter              | 6.1.6.3.2  | Number of event reports received from the target UE                                                  |
| EventReportDuration             | 6.1.6.3.2  | Duration of event reporting                                                                          |
| UePositioningCapabilities       | 6.1.6.3.2  | Indicates the positioning capabilities supported by the UE.                                          |
| TimeWindow                      | 6.1.6.3.2  | Indicates the Time Window for scheduling of PRU measurements when UE assisted positioning is used.   |
| TimeWindowInfoMeasurementList   | 6.1.6.3.2  | Contains the Time Window Information Measurement List when network assisted positioning is used.     |
| TimeWindowInfoSrsList           | 6.1.6.3.2  | Contains the Time Window Information SRS List when network assisted positioning is used.             |
| RadialVelocityValue             | 6.1.6.3.2  | Indicates the value of radial velocity                                                               |
| AngularVelocityValue            | 6.1.6.3.2  | Indicates the value of angular velocity                                                              |

|                             |            |                                                                                  |
|-----------------------------|------------|----------------------------------------------------------------------------------|
| AngularVelocityUncertainty  | 6.1.6.3.2  | Indicates the value of angular velocity uncertainty                              |
| RadialVelocityUncertainty   | 6.1.6.3.2  | Indicates the value of linear velocity uncertainty                               |
| TrpId                       | 6.1.6.3.2  | Indicates the identifier of TRP                                                  |
| ExternalClientType          | 6.1.6.3.3  | Indicates types of External Clients                                              |
| SupportedGADShapes          | 6.1.6.3.4  | Indicates supported GAD shapes                                                   |
| ResponseTime                | 6.1.6.3.5  | Indicates acceptable delay of location request                                   |
| PositioningMethod           | 6.1.6.3.6  | Indicates supported positioning methods                                          |
| PositioningMode             | 6.1.6.3.7  | Indicates supported modes used for positioning method                            |
| GnssId                      | 6.1.6.3.8  | Global Navigation Satellite System (GNSS) ID                                     |
| Usage                       | 6.1.6.3.9  | Indicates usage made of the location measurement                                 |
| LcsPriority                 | 6.1.6.3.10 | Indicates priority of the LCS client                                             |
| VelocityRequested           | 6.1.6.3.11 | Indicates velocity requirement                                                   |
| AccuracyFulfilmentIndicator | 6.1.6.3.12 | Indicates fulfilment of requested accuracy                                       |
| VerticalDirection           | 6.1.6.3.13 | Indicates direction of vertical speed                                            |
| LdrType                     | 6.1.6.3.14 | Indicates LDR types                                                              |
| ReportingAreaType           | 6.1.6.3.15 | Indicates type of event reporting area                                           |
| OccurrenceInfo              | 6.1.6.3.16 | Specifies occurrence of event reporting                                          |
| ReportingAccessType         | 6.1.6.3.17 | Specifies access types of event reporting                                        |
| EventClass                  | 6.1.6.3.18 | Specifies event classes                                                          |
| ReportedEventType           | 6.1.6.3.19 | Specifies type of event reporting                                                |
| TerminationCause            | 6.1.6.3.20 | Specifies causes of event reporting termination                                  |
| LcsQosClass                 | 6.1.6.3.21 | Specifies LCS QoS class                                                          |
| UeLocationServiceInd        | 6.1.6.3.22 | Specifies location service types requested by UE                                 |
| IndoorOutdoorInd            | 6.1.6.3.23 | Indoor Outdoor Indication                                                        |
| FixType                     | 6.1.6.3.24 | Fix Type                                                                         |
| LosNlosMeasureInd           | 6.1.6.3.25 | LOS/NLOS measurement indication                                                  |
| UpConnectionStatus          | 6.1.6.3.26 | UP Connection Status                                                             |
| RangingSIResult             | 6.1.6.3.27 | Specifies result type for ranging and sidelink positioning                       |
| RelatedUeType               | 6.1.6.3.28 | Specifies type of related UE for ranging and sidelink positioning                |
| LcsUpConnectionInd          | 6.1.6.3.29 | LCS UP Connection Indication                                                     |
| UeUpPositioningCapabilities | 6.1.6.3.30 | Indicates the user plane positioning capabilities supported by the UE.           |
| SIPositioningCapability     | 6.1.6.3.31 | Indicates the Ranging and sidelink positioning capabilities supported by the UE. |
| UnitsLinearVelocity         | 6.1.6.3.32 | Indicates the units of linear velocity.                                          |
| UnitsAngularVelocity        | 6.1.6.3.33 | Indicates the units of angular velocity.                                         |

Table 6.1.6.1-2 specifies data types re-used by the Nlmf\_Location service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nlmf\_Location service based interface.

**Table 6.1.6.1-2: Nlmf\_Location re-used Data Types**

| Data type             | Reference           | Comments                                                          |
|-----------------------|---------------------|-------------------------------------------------------------------|
| Supi                  | 3GPP TS 29.571 [8]  | Subscription Permanent Identifier                                 |
| Pei                   | 3GPP TS 29.571 [8]  | Permanent Equipment Identifier                                    |
| Gpsi                  | 3GPP TS 29.571 [8]  | Generic Public Subscription Identifier                            |
| Ecgi                  | 3GPP TS 29.571 [8]  | E-UTRA Cell Identity                                              |
| Ncgi                  | 3GPP TS 29.571 [8]  | NR Cell Identity                                                  |
| NfnInstanceId         | 3GPP TS 29.571 [8]  | Network Function Instance ID                                      |
| Uri                   | 3GPP TS 29.571 [8]  | Uniform Resource Identifier                                       |
| RefToBinaryData       | 3GPP TS 29.571 [8]  | Reference to binary data                                          |
| AccessType            | 3GPP TS 29.571 [8]  | Access type                                                       |
| CmState               | 3GPP TS 29.518 [23] | Connection Management State                                       |
| Guami                 | 3GPP TS 29.571 [8]  | GUAMI                                                             |
| SupportedFeatures     | 3GPP TS 29.571 [8]  | Supported Features                                                |
| RedirectResponse      | 3GPP TS 29.571 [8]  | Redirect Response                                                 |
| TwapId                | 3GPP TS 29.571 [8]  | TWAP identifier                                                   |
| TnapId                | 3GPP TS 29.571 [8]  | TNAP identifier                                                   |
| DateTime              | 3GPP TS 29.571 [8]  | Date and Time                                                     |
| DurationSec           | 3GPP TS 29.571 [8]  | Duration Second                                                   |
| Tai                   | 3GPP TS 29.571 [8]  | Tracking Area Identity                                            |
| LpHapType             | 3GPP TS 29.518 [23] | Type of Low Power and/or High Accuracy Positioning                |
| ReportingInd          | 3GPP TS 29.515 [27] | Reporting indication                                              |
| IntegrityRequirements | 3GPP TS 29.515 [27] | Integrity Requirements                                            |
| UpCumEvtRptCriteria   | 3GPP TS 29.515 [27] | Criteria for sending cumulative events reports over control plane |
| IntegrityResult       | 3GPP TS 29.515 [27] | Integrity Result                                                  |
| UpLocRepAddrAf        | 3GPP TS 29.515 [27] | Endpoint address for location reporting over user plane           |
| GlobalRanNodeId       | 3GPP TS 29.571 [8]  | Global identity of the gNodeB                                     |

## 6.1.6.2 Structured data types

### 6.1.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

6.1.6.2.2      Type: InputData

**Table 6.1.6.2.2-1: Definition of type InputData**



| Attribute name     | Data type                 | P | Cardinality | Description                                                                                                                                                                                                                                  | Applicability |
|--------------------|---------------------------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| externalClientType | ExternalClientType        | O | 0..1        | When present, this IE shall carry the external client type of the requester.                                                                                                                                                                 |               |
| correlationID      | CorrelationID             | O | 0..1        | When present, this IE shall carry the correlation ID of the request.                                                                                                                                                                         |               |
| amfId              | NfInstanceId              | O | 0..1        | Indicates the AMF Instance serving the UE. LMF shall use the AMF Instance to forward LCS related N1/N2 messages to the UE/RAN.                                                                                                               |               |
| locationQoS        | LocationQoS               | O | 0..1        | When present, this IE shall carry the QoS of the location request.                                                                                                                                                                           |               |
| supportedGADShapes | array(SupportedGADShapes) | O | 1..N        | When present, this IE shall carry the GAD shapes supported by the requester.                                                                                                                                                                 |               |
| supi               | Supi                      | C | 0..1        | This IE shall present if available. When present, it indicates the SUPI of the target UE. (NOTE 6)                                                                                                                                           |               |
| pei                | Pei                       | O | 0..1        | Indicates the PEI of the target UE.                                                                                                                                                                                                          |               |
| gpsi               | Gpsi                      | O | 0..1        | Indicates the GPSI of the target UE. (NOTE 6)                                                                                                                                                                                                |               |
| ecgi               | Ecgi                      | O | 0..1        | When present, this IE shall indicate the identifier of the E-UTRAN cell serving the UE or the serving cell identifier of the Primary Cell in the Master RAN Node that is an E-UTRAN node on Dual Connectivity scenarios. (NOTE 2)            |               |
| ecgiOnSecondNode   | Ecgi                      | O | 0..1        | When present, the serving cell identifier of the Primary Cell in the Secondary RAN Node that is an E-UTRAN node when available on Dual Connectivity scenarios. (NOTE 3) (NOTE 4)                                                             |               |
| ncgi               | Ncgi                      | O | 0..1        | When present, this IE shall indicate the identifier of the NR cell serving the UE or the serving cell identifier of the Primary Cell in the Master RAN Node that is a NR node on Dual Connectivity scenarios. (NOTE 2)                       |               |
| ncgiOnSecondNode   | Ncgi                      | O | 0..1        | When present, the serving cell identifier of the Primary Cell in the Secondary RAN Node that is a NR node when available on Dual Connectivity scenarios. (NOTE 3) (NOTE 4)                                                                   |               |
| priority           | LcsPriority               | O | 0..1        | When present, this IE shall indicate the priority of the location request.                                                                                                                                                                   |               |
| velocityRequested  | VelocityRequested         | O | 0..1        | When present, this IE shall indicate whether velocity is requested or not.                                                                                                                                                                   |               |
| ueLcsCap           | UeLcsCapability           | O | 0..1        | When present, this IE shall indicate the LCS capability supported by the UE.                                                                                                                                                                 |               |
| lcsServiceType     | LcsServiceType            | O | 0..1        | The LCS service type                                                                                                                                                                                                                         |               |
| ldrType            | LdrType                   | O | 0..1        | The type of LDR                                                                                                                                                                                                                              |               |
| hgmlcCallbackURI   | Uri                       | C | 0..1        | Callback URI of the H-GMLC<br><br>It shall be present, if attribute LdrType is present.<br><br>This IE shall also be present for location service in PNI-NPN with signalling optimisation, as specified in 3GPP TS 23.273 [19] clause 6.1.2. |               |

|                       |                                 |   |      |                                                                                                                                                                                                                                          |  |
|-----------------------|---------------------------------|---|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| lirGmlcCallbackUri    | Uri                             | C | 0..1 | <p>This IE shall be present when the intermediateLocationInd IE is present with the value "true".</p> <p>When present, this IE shall contain callback URI of the GMLC to receive the intermediate location reports.</p>                  |  |
| vgmlcAddress          | Uri                             | C | 0..1 | <p>V-GMLC address that corresponds to the V-GMLC that receives Location Request</p> <p>It shall be present, if attribute LdrType is present and the target UE is in roaming case.</p>                                                    |  |
| ldrReference          | LdrReference                    | C | 0..1 | <p>LDR Reference Number</p> <p>It shall be present, if attribute LdrType is present.</p> <p>This IE shall be present for location service in PNI-NPN with signalling optimisation, as specified in 3GPP TS 23.273 [19] clause 6.1.2.</p> |  |
| lirReference          | LirReference                    | C | 0..1 | <p>This IE shall be present when the intermediateLocationInd IE is present with the value "true".</p> <p>When present, this IE shall contain the LIR Reference Number for a multiple location request</p>                                |  |
| periodicEventInfo     | PeriodicEventInfo               | C | 0..1 | <p>Information for periodic event reporting.</p> <p>This IE shall be present when ldrType is set to "PERIODIC".</p>                                                                                                                      |  |
| areaEventInfo         | AreaEventInfo                   | C | 0..1 | <p>Information for area event reporting.</p> <p>This IE shall be present when ldrType is set to "ENTERING_INTO_AREA", "LEAVING_FROM_AREA" or "BEING_INSIDE_AREA".</p>                                                                    |  |
| motionEventInfo       | MotionEventInfo                 | C | 0..1 | <p>Information for motion event reporting.</p> <p>This IE shall be present when ldrType is set to "MOTION".</p>                                                                                                                          |  |
| reportingAccessTypes  | array(ReportingAccessType)      | O | 1..N | Allowed access types for event reporting                                                                                                                                                                                                 |  |
| ueConnectivityStates  | array(UeConnectivityState)      | O | 1..N | When present, this IE shall indicate the UE connectivity state per access type                                                                                                                                                           |  |
| ueLocationServiceInd  | UeLocationServiceInd            | C | 0..1 | If UE sends an MO-LR Request message, this IE shall be present and indicate the request type for a 5GC-MO-LR.                                                                                                                            |  |
| moAssistanceDataTypes | LcsBroadcastAssistanceTypesData | O | 0..1 | When present, this IE shall indicate a list of one or more types of location assistance data that UE subscribed.                                                                                                                         |  |
| lppMessage            | RefToBinaryData                 | C | 0..1 | <p>If UE includes the first LPP message in MO-LR Request, this IE shall be present and indicate the binary data of LPP message.</p> <p>(NOTE 5)</p>                                                                                      |  |
| lppMessageExt         | array(RefToBinaryData)          | C | 1..N | <p>If UE includes the additional LPP messages (maximum 3) in MO-LR Request, this IE shall be present and indicate the binary data of LPP message.</p> <p>(NOTE 5)</p>                                                                    |  |

|                         |                           |   |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|-------------------------|---------------------------|---|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| supportedFeatures       | SupportedFeatures         | C | 0..1   | This IE shall be present if at least one feature defined in clause 6.1.9 is supported.                                                                                                                                                                                                                                                                                                                                                             |     |
| uePositioningCap        | UePositioningCapabilities | O | 0..1   | When present, this IE shall indicate the positioning capabilities supported by the UE.                                                                                                                                                                                                                                                                                                                                                             |     |
| tnapId                  | TnapId                    | O | 0..1   | When present, this IE shall contain the TNAP Identifier.<br><br>This IE may be present for non-3GPP access.                                                                                                                                                                                                                                                                                                                                        |     |
| twapId                  | TwapId                    | O | 0..1   | When present, This IE shall contain the TWAP Identifier.<br>This IE may be present for non-3GPP access.                                                                                                                                                                                                                                                                                                                                            |     |
| ueCountryDetInd         | boolean                   | O | 0..1   | When present, This IE shall contain an indication of determining the UE geographical area identified by the country, area within a country or international area indication where UE is located for PLMN selection verification.                                                                                                                                                                                                                   | SAT |
| scheduledLocTime        | DateTime                  | O | 0..1   | When present, this IE shall contain the scheduled time (in UTC) that the UE needs to be located.                                                                                                                                                                                                                                                                                                                                                   |     |
| reliableLocReq          | boolean                   | C | 0..1   | This IE shall be included with the value true to indicate that reliable UE location information is required, as specified in 3GPP TS 33.256 [26] clause 5.3.2.<br><br>When present, this IE shall be set as following:<br>- true: the reliable UE location information is required<br>- false (default): the reliable UE location information is not required                                                                                      |     |
| evtRptAllowedAreas      | array(ReportingArea)      | O | 1..250 | When present, this IE shall contain a list of event report allowed areas, where UE is allowed to generate and send the event report to network during the deferred 5GC-MT-LR procedure for UE power saving purpose.                                                                                                                                                                                                                                |     |
| ueUnawareInd            | boolean                   | C | 0..1   | UE Unaware Positioning indication.<br><br>If the UE Unaware Positioning is required, as specified in 3GPP TS 23.273 [19] clause 5.12, this IE shall be included and set to true; otherwise, the IE shall be absent.                                                                                                                                                                                                                                |     |
| intermediateLocationInd | boolean                   | C | 0..1   | This IE shall be included by the AMF if received from the GMLC, during a 5GC-MT-LR multiple location procedure for the regulatory location service (see clause 6.1.3 and clause 6.10.4 of 3GPP TS 23.273 [19]).<br><br>When present, this IE shall indicate the acceptance of intermediate location response at the GMLC:<br>- true: intermediate location response acceptable<br>- false (default): intermediate location response not acceptable |     |

|                          |                                    |   |      |                                                                                                                                                                                                                                                                                                                                                                                                    |            |
|--------------------------|------------------------------------|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| maxRespTime              | DurationSec                        | C | 0..1 | <p>This IE shall be included by the AMF if received from the GMLC.</p> <p>When present, this IE shall contain the maximum response time for the GMLC to receive the final location response.</p> <p>The AMF may overwrite the received maximum response time when passing it to the LMF, e.g., to avoid timeout of the HTTP service request.</p>                                                   |            |
| lpHapType                | LpHapType                          | C | 0..1 | <p>This IE shall be included and set to "LOW_POW_HIGH_ACCU_POS" to request low power and high accuracy positioning, as specified in clause 6.1.2 of 3GPP TS 23.273 [19].</p>                                                                                                                                                                                                                       |            |
| ueUpPosCaps              | array(UeUpPositioningCapabilities) | O | 1..N | <p>When present, this IE shall indicate the user plane positioning capabilities supported by the UE.</p>                                                                                                                                                                                                                                                                                           |            |
| reportingInd             | ReportingInd                       | O | 0..1 | <p>This IE may be present if the evtRptAllowedAreas IE is present.</p> <p>When present, this IE shall indicate whether the UE is allowed to generate and send the reports inside or outside the event report allowed areas:</p> <ul style="list-style-type: none"> <li>- Inside reporting (default)</li> <li>- Outside reporting</li> </ul> <p>(see 3GPP TS 23.273 [19] clause 5.14 and 6.3.1)</p> |            |
| mbsrInfo                 | MbsrInfo                           | O | 0..1 | <p>Indicates that serving cell of the UE belongs to a MBSR</p>                                                                                                                                                                                                                                                                                                                                     | MBSR       |
| additionalUeInfo         | AdditionalUeInfo                   | O | 0..1 | <p>This IE may be present, if the mbsrInfo IE is present. It shall indicate the ULI of the MBSR UE (i.e., ULI of IAB UE) (see clause 5.35A.6 of 3GPP TS 23.501 [2])</p> <p>Or</p> <p>This IE may be present, if the mwabSupportInd IE is present and set to true. It shall indicate the ULI of MWAB UE. See clause 5.49.4 of 3GPP TS 23.501 [2]</p>                                                | MBSR, MWAB |
| mwabSupportInd           | boolean                            | O | 0..1 | <p>This attribute may be present if ncgi is present.</p> <p>When this attribute is present and set to true, it shall indicate that the target UE is served by a MWAB. See clauses 5.19 and 6.1.5 of 3GPP TS 23.273 [19].</p> <p>The presence of this attribute with value false shall be prohibited.</p>                                                                                           | MWAB       |
| integrityRequirements    | IntegrityRequirements              | O | 0..1 | <p>When present, this IE shall indicate the integrity requirements.</p>                                                                                                                                                                                                                                                                                                                            |            |
| requestedRangingSIResult | array(RangingSIResult)             | O | 1..N | <p>This IE shall contain the type of result requested for ranging and sidelink positioning, such as absolute locations, relative locations or straight line distances (ranges) and directions related to the UEs, etc.</p>                                                                                                                                                                         |            |
| relatedUes               | array(RelatedUe)                   | O | 1..N | <p>This IE contains a list of the information for the related UEs for the ranging and sidelink positioning.</p>                                                                                                                                                                                                                                                                                    |            |
| upLocRepAddrAf           | UpLocRepAddrAf                     | C | 0..1 | <p>This IE shall be present if the request is for the location reporting over user plane.</p>                                                                                                                                                                                                                                                                                                      |            |

|                           |                                |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |
|---------------------------|--------------------------------|---|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| upCumEvtRptCriteria       | UpCumEvtRptCriteria            | O | 0..1 | This IE may be present if the upLocRepAddrAf is present.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |
| mappedQoSeps              | MappedLocationQoSeps           | C | 0..1 | This IE shall be present if the Multiple QoS Class is indicated in the locationQoS IE.<br><br>When present, this IE shall indicate the mapped Location QoS applicable to EPS ("BEST_EFFORT" or "ASSURED") based on the Multiple Location QoS (see clause 6.19 of 3GPP TS 23.273 [19]).                                                                                                                                                                                                                                                                                                                                            |            |
| coordinateID              | integer                        | O | 0..1 | This IE may be present when requestedRangingSIResult indicates "ABSOLUTE_LOCATION".<br><br>When present, this IE represents a local coordinate (see clause 6.20.3 of 3GPP TS 23.273 [19]).                                                                                                                                                                                                                                                                                                                                                                                                                                        | Ranging_SL |
| rangingSICapability       | RangingSICapability            | O | 0..1 | When present, this IE shall indicate that the UE supports Ranging/Sidelink Positioning Capability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Ranging_SL |
| slPositioningCapabilities | array(SlPositioningCapability) | C | 1..N | This IE shall be present if the supported Ranging and sidelink positioning capabilities were received in NAS (in 5GMM Capability NAS IE) from the target UE.<br><br>When present, this IE shall indicate the Ranging and sidelink positioning capabilities supported by the target UE.<br><br>The LMF may use the information to determine the candidate Located UE(s) that can communicate with the target UE for ranging and sidelink positioning via PC5.                                                                                                                                                                      | Ranging_SL |
| extendedFacilityInd       | boolean                        | C | 0..1 | This IE shall be present with the value true if the UE has indicated support for "ExtendedFacilityIE for 5G LCS" in NAS (in 5GMM Capability NAS IE) and the network also supports usage of the ExtendedFacility IE for 5GS LCS related Supplementary service messages.<br><br>When this IE is received, the LMF shall use ExtendedFacility IE in 5G LCS related Supplementary service messages, when the component of the messages cannot be carried in Facility IE, i.e. the size of the component exceeds 255 octets, as specified in 3GPP TS 24.080 [20].<br><br>Presence of this IE with the value false shall be prohibited. |            |

NOTE 1: At least one of the attributes defined in this table shall be present in the InputData structure.

NOTE 2: Attribute "ecgi" and "ncgi" shall not be present at the same time.

NOTE 3: Attribute "ecgiOnSecondNode" and "ncgiOnSecondNode" shall not be present at the same time.

NOTE 4: Attribute "ecgiOnSecondNode" or "ncgiOnSecondNode" shall not be present if neither attribute "ecgi" nor "ncgi" is present.

NOTE 5: If 3 LPP messages are received, then first LPP message shall be encoded in lppMessage IE and additional 2 LPP messages shall be encoded in lppMessageExt IE.

NOTE 6: Based on UE user plane positioning capabilities, if the target UE supports the UE user plane positioning capability for LCS-UPP, the AMF shall provide the SUPI and optionally the GPSI of the UE, as defined in 3GPP TS 23.273 [19].

6.1.6.2.3      Type: LocationData

**Table 6.1.6.2.3-1: Definition of type LocationData**

| Attribute name              | Data type                            | P | Cardinality | Description                                                                                                                                                                                       | Applicability |
|-----------------------------|--------------------------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| locationEstimate            | GeographicArea                       | M | 1           | For a request for triggered location where location estimates are not required, the location estimate can be based on current serving cell.                                                       |               |
| accuracyFulfilmentIndicator | AccuracyFulfilmentIndicator          | O | 0..1        | When present, this IE shall indicate fulfilment of required accuracy.                                                                                                                             |               |
| ageOfLocationEstimate       | AgeOfLocationEstimate                | O | 0..1        | When present, this IE shall indicate age of the location estimate.                                                                                                                                |               |
| timestampOfLocationEstimate | DateTime                             | O | 0..1        | When present, this IE shall indicate the estimated UTC time when the location estimate corresponded to the UE location (i.e. when the location estimate and the actual UE location was the same). |               |
| velocityEstimate            | VelocityEstimate                     | O | 0..1        | When present, this IE shall indicate velocity estimate.                                                                                                                                           |               |
| civicAddress                | CivicAddress                         | O | 0..1        | When present, this IE shall indicate a civic address.                                                                                                                                             |               |
| localLocationEstimate       | LocalArea                            | O | 0..1        | When present, this IE shall indicate a local area in reference system.                                                                                                                            |               |
| positioningDataList         | array(PositioningMethodAndUsage)     | O | 1..N        | When present, this IE shall include a list of data related to positioning methods.                                                                                                                |               |
| gnssPositioningDataList     | array(GnssPositioningMethodAndUsage) | O | 1..N        | When present, this IE shall include a list of data related to GNSS positioning methods.                                                                                                           |               |
| ecgi                        | Ecgi                                 | O | 0..1        | When present, this IE shall indicate the ID of the E-UTRAN cell serving the UE.                                                                                                                   |               |
| ncgi                        | Ncgi                                 | O | 0..1        | When present, this IE shall indicate the ID of the NR cell serving the target UE, or the NR cell serving the relay UE if remoteUeInd IE is present with the value true.                           |               |
| remoteUeInd                 | <u>boolean</u>                       | C | <u>0..1</u> | This IE shall be present with the value true when the UE is a Remote UE.<br><br>Presence of this IE with the value false shall be prohibited.                                                     |               |
| altitude                    | Altitude                             | O | 0..1        | Altitude of the positioning estimate. When the shape used in "locationEstimate" supports conveying the altitude parameter, this IE shall be absent.                                               |               |
| barometricPressure          | BarometricPressure                   | O | 0..1        | If present, this IE contains the barometric pressure measurement as reported by the target UE.                                                                                                    |               |
| servingLMFidentification    | LMFIdentification                    | O | 0..1        | When present, this IE shall indicate the identity of the serving LMF                                                                                                                              |               |
| uePositioningCap            | UePositioningCapabilities            | O | 0..1        | When present, this IE shall indicate the positioning capabilities supported by the UE.                                                                                                            |               |

|                           |                   |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
|---------------------------|-------------------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| supportedFeatures         | SupportedFeatures | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.9 is supported.                                                                                                                                                                                                                                                                                                                                                    |         |
| ueAreaInd                 | UeAreaIndication  | O | 0..1 | When present, this IE shall contain a country, area in a country or international area indication where UE is located.<br><br>If UE is outside of the area of any known country, i.e. international area, then the LMF shall provide the International Area Indication without a country.                                                                                                                                                 | SAT     |
| achievedQos               | MinorLocationQoS  | O | 0..1 | When present, this IE shall contain the achieved Location QoS Accuracy of the estimated location.<br><br>This IE may be present if the LCS QoS Class required in the request message is "MULTIPLE_QOS".<br><br>If it's absent, LCS QoS Class required in the request message is "MULTIPLE_QOS" and AccuracyFulfilmentIndicator is "REQUESTED_ACCURACY_FULFILLED", it indicates that the location QoS in the highest priority is achieved. | MUTIQOS |
| directReportInd           | boolean           | C | 0..1 | This IE shall be present, if the H-GMLC contact address and the Notification Correlation ID are received in the request, i.e. the location estimates will be sent directly to the H-GMLC.<br><br>When present, this IE shall be set for the following value:<br>- true: location determination will be sent by LMF to GMLC directly<br>- false (default): location determination will not be sent by LMF to GMLC directly                 |         |
| indoorOutdoorInd          | IndoorOutdoorInd  | O | 0..1 | When present, this IE shall indicate whether the UE is indoor or outdoor.                                                                                                                                                                                                                                                                                                                                                                 |         |
| acceptedPeriodicEventInfo | PeriodicEventInfo | C | 0..1 | This IE shall be present if PeriodicEventInfo was received in the request which includes reportingAmountInf IE and/or reportingIntervalMs IE.<br><br>When present, this IE shall provide the accepted periodic event reporting information.                                                                                                                                                                                               |         |



|                           |                         |   |      |                                                                                                                                                                                                                                                                                                                                                                                     |            |
|---------------------------|-------------------------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| haGnssMetrics             | HighAccuracyGnssMetrics | O | 0..1 | When present, this IE shall indicate the high accuracy GNSS metrics received from the device in the LPP <i>HA-GNSS-Metrics-r17</i> IE as specified in 3GPP TS 37.355 [21].                                                                                                                                                                                                          |            |
| losNlosMeasureInd         | LosNlosMeasureInd       | O | 0..1 | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used.                                                                                                                                                                                                                                                                                           |            |
| rangingSICapability       | RangingSICapability     | O | 0..1 | When present, this IE shall indicate that the UE supports Ranging/Sidelink Positioning Capability.                                                                                                                                                                                                                                                                                  | Ranging_SL |
| relatedApplicationlayerId | ApplicationlayerId      | O | 0..1 | Identifies the application layer ID of the related UE for ranging and sidelink positioning, such as located UE, reference UE, etc.                                                                                                                                                                                                                                                  | Ranging_SL |
| distanceDirection         | RangeDirection          | O | 0..1 | When present, this IE identifies a straight line distance (range) and direction from a point A to a point B, comprising a straight line distance (range) from point A to point B, an azimuth direction from point A to point B and an elevation direction from point A to point B.                                                                                                  | Ranging_SL |
| 2dRelativeLocation        | 2DRelativeLocation      | O | 0..1 | When present, this IE identifies a relative 2D location with uncertainty ellipse, characterised by a point described in 2D local co-ordinates with origin corresponding to another known point, distances r1 and r2 and an angle of orientation A.                                                                                                                                  | Ranging_SL |
| 3dRelativeLocation        | 3DRelativeLocation      | O | 0..1 | When present, this IE identifies a relative 3D location with uncertainty ellipsoid, characterised by a point described in 3D local co-ordinates with origin corresponding to another known point, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis"). | Ranging_SL |
| relativeVelocity          | VelocityEstimate        | O | 0..1 | When present, this IE identifies UE velocity relative to the UE identified with relatedApplicationlayerId.                                                                                                                                                                                                                                                                          | Ranging_SL |
| integrityResult           | IntegrityResult         | C | 0..1 | This IE shall be present when the integrity requirements are present in the request.<br><br>When present, this IE shall indicate the integrity result.                                                                                                                                                                                                                              | INTRES     |

|                  |         |   |      |                                                                                                                                                                                                                                  |  |
|------------------|---------|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| nrppaPeriodicInd | boolean | C | 0..1 | <p>This IE shall be present with the value true, if the LMF has determined to generate Periodic Location Estimates based on NRPPa Periodic Measurement.</p> <p>Presence of this IE with the value false shall be prohibited.</p> |  |
|------------------|---------|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

#### 6.1.6.2.4 Type: GeographicalCoordinates

**Table 6.1.6.2.4-1: Definition of type GeographicalCoordinates**

| Attribute name | Data type | P | Cardinality | Description                                                                                                     |
|----------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------|
| lon            | number    | M | 1           | <p>Longitude (Double-precision float value):</p> <p>Format: double</p> <p>Minimum: -180</p> <p>Maximum: 180</p> |
| lat            | number    | M | 1           | <p>Latitude (Double-precision float value):</p> <p>Format: double</p> <p>Minimum: -90</p> <p>Maximum: 90</p>    |

#### 6.1.6.2.5 Type: GeographicArea

**Table 6.1.6.2.5-1: Definition of type GeographicArea as a list of mutually exclusive alternatives**

| Data type                                                                                                                                                                                                                                                                                                                                                                                                    | Cardinality | Discriminator property name | Discriminator mapping      | Description                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------|----------------------------|----------------------------------------------------------------------------------------------|
| Point                                                                                                                                                                                                                                                                                                                                                                                                        | 1           | shape                       | POINT                      | Geographical area consisting of a single point, represented by its longitude and latitude.   |
| PointUncertaintyCircle                                                                                                                                                                                                                                                                                                                                                                                       | 1           | shape                       | POINT_UNCERTAINTY_CIRCLE   | Geographical area consisting of a point and an uncertainty value.                            |
| PointUncertaintyEllipse                                                                                                                                                                                                                                                                                                                                                                                      | 1           | shape                       | POINT_UNCERTAINTY_ELLIPSE  | Geographical area consisting of a point, plus an uncertainty ellipse and a confidence value. |
| Polygon                                                                                                                                                                                                                                                                                                                                                                                                      | 1           | shape                       | POLYGON                    | Geographical area consisting of a list of points (between 3 to 15 points).                   |
| PointAltitude                                                                                                                                                                                                                                                                                                                                                                                                | 1           | shape                       | POINT_ALTITUDE             | Geographical area consisting of a point and an altitude value.                               |
| PointAltitudeUncertainty                                                                                                                                                                                                                                                                                                                                                                                     | 1           | shape                       | POINT_ALTITUDE_UNCERTAINTY | Geographical area consisting of a point, an altitude value and an uncertainty value.         |
| EllipsoidArc                                                                                                                                                                                                                                                                                                                                                                                                 | 1           | shape                       | ELLIPSOID_ARC              | Geographical are consisting of an ellipsoid arc.                                             |
| <p>NOTE: The "anyOf" keyword (instead of "oneOf" keyword which is normally used for mutually exclusive alternatives) is used for GeographicArea type in yaml file to avoid validation failure of OpenAPI. According to current definition, a PointUncertaintyCircle object will always pass the validation with both PointUncertaintyCircle and Point, which fails the qualification of "oneOf" keyword.</p> |             |                             |                            |                                                                                              |

## 6.1.6.2.6 Type: Point

**Table 6.1.6.2.6-1: Definition of type Point**

| Attribute name | Data type               | P | Cardinality | Description                                                             |
|----------------|-------------------------|---|-------------|-------------------------------------------------------------------------|
| shape          | SupportedGADShapes      | M | 1           | It shall take the value "POINT".                                        |
| point          | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude. |

## 6.1.6.2.7 Type: PointUncertaintyCircle

**Table 6.1.6.2.7-1: Definition of type PointUncertaintyCircle**

| Attribute name | Data type               | P | Cardinality | Description                                                             |
|----------------|-------------------------|---|-------------|-------------------------------------------------------------------------|
| shape          | SupportedGADShapes      | M | 1           | It shall take the value "POINT UNCERTAINTY CIRCLE".                     |
| point          | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude. |
| uncertainty    | Uncertainty             | M | 1           | Indicates the uncertainty value.                                        |

## 6.1.6.2.8 Type: PointUncertaintyEllipse

**Table 6.1.6.2.8-1: Definition of type PointUncertaintyEllipse**

| Attribute name     | Data type               | P | Cardinality | Description                                                             |
|--------------------|-------------------------|---|-------------|-------------------------------------------------------------------------|
| shape              | SupportedGADShapes      | M | 1           | It shall take the value "POINT UNCERTAINTY ELLIPSE".                    |
| point              | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude. |
| uncertaintyEllipse | UncertaintyEllipse      | M | 1           | Indicates an uncertainty ellipse.                                       |
| confidence         | Confidence              | M | 1           | Indicates the value of confidence.                                      |

## 6.1.6.2.9 Type: Polygon

**Table 6.1.6.2.9-1: Definition of type Polygon**

| Attribute name | Data type                      | P | Cardinality | Description                                              |
|----------------|--------------------------------|---|-------------|----------------------------------------------------------|
| shape          | SupportedGADShapes             | M | 1           | It shall take the value "POLYGON".                       |
| pointList      | array(GeographicalCoordinates) | M | 3..15       | Array with up to 15 items, where each item is a "point". |

## 6.1.6.2.10 Type: PointAltitude

**Table 6.1.6.2.10-1: Definition of type PointAltitude**

| Attribute name | Data type               | P | Cardinality | Description                                                             |
|----------------|-------------------------|---|-------------|-------------------------------------------------------------------------|
| shape          | SupportedGADShapes      | M | 1           | It shall take the value "POINT_ALTITUDE".                               |
| point          | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude. |
| altitude       | Altitude                | M | 1           | Indicates the value of altitude.                                        |

## 6.1.6.2.11 Type: PointAltitudeUncertainty

**Table 6.1.6.2.11-1: Definition of type PointAltitudeUncertainty**

| Attribute name      | Data type               | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                             |
|---------------------|-------------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shape               | SupportedGADShapes      | M | 1           | It shall take the value "POINT_ALTITUDE_UNCERTAINTY".                                                                                                                                                                                                                                                                                                   |
| point               | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude.                                                                                                                                                                                                                                                                                 |
| altitude            | Altitude                | M | 1           | Indicates the value of altitude.                                                                                                                                                                                                                                                                                                                        |
| uncertaintyEllipse  | UncertaintyEllipse      | M | 1           | Indicates the uncertainty ellipse                                                                                                                                                                                                                                                                                                                       |
| uncertaintyAltitude | Uncertainty             | M | 1           | Indicates the uncertainty of the altitude.                                                                                                                                                                                                                                                                                                              |
| confidence          | Confidence              | M | 1           | Indicates the value of confidence.<br><br>If the vConfidence IE is present, this IE shall indicate the value of horizontal confidence.                                                                                                                                                                                                                  |
| vConfidence         | Confidence              | C | 0..1        | This IE shall be present when vertical confidence is available, and the value of vertical and horizontal confidence are different.<br><br>This IE may be present when vertical confidence is available, and the value of vertical and horizontal confidence are the same.<br><br>When present, this IE shall Indicate the value of vertical confidence. |

## 6.1.6.2.12 Type: EllipsoidArc

Table 6.1.6.2.12-1: Definition of type EllipsoidArc

| Attribute name    | Data type               | P | Cardinality | Description                                                             |
|-------------------|-------------------------|---|-------------|-------------------------------------------------------------------------|
| shape             | SupportedGADShapes      | M | 1           | It shall take the value "ELLIPSOID_ARC".                                |
| point             | GeographicalCoordinates | M | 1           | Indicates a geographic point represented by its longitude and latitude. |
| innerRadius       | InnerRadius             | M | 1           | Indicates the value of inner radius of the Ellipsoid Arc.               |
| uncertaintyRadius | Uncertainty             | M | 1           | Indicates the uncertainty radius of the Ellipsoid Arc.                  |
| offsetAngle       | Angle                   | M | 1           | Indicates the offset angle of the Ellipsoid Arc.                        |
| includedAngle     | Angle                   | M | 1           | Indicates the included angle of the Ellipsoid Arc.                      |
| confidence        | Confidence              | M | 1           | Indicates the value of confidence.                                      |

## 6.1.6.2.13 Type: LocationQoS

Table 6.1.6.2.13-1: Definition of type LocationQoS

| Attribute name | Data type               | P | Cardinality | Description                                                                                                                                                                                                                                           | Applicability |
|----------------|-------------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| hAccuracy      | Accuracy                | O | 0..1        | Horizontal accuracy                                                                                                                                                                                                                                   |               |
| vAccuracy      | Accuracy                | O | 0..1        | Vertical accuracy                                                                                                                                                                                                                                     |               |
| vertRequested  | boolean                 | O | 0..1        | Vertical accuracy requested (yes/no)                                                                                                                                                                                                                  |               |
| responseTime   | ResponseTime            | O | 0..1        | No delay, Low delay or Delay tolerant                                                                                                                                                                                                                 |               |
| minorLocQoses  | array(MinorLocationQoS) | C | 1..2        | If present, this IE shall contain a list of MinorLocationQoS in priority order.<br><br>When the lcsQoSClass sets to "MULTIPLE_QOS", this IE shall be present, and when lcsQoSClass sets to either "BEST_EFFORT" or "ASSURED" this IE shall be absent. | MUTIQOS       |
| lcsQoSClass    | LcsQoSClass             | C | 0..1        | LCS QoS Class, see clause 4.1b of 3GPP TS 23.273 [19].<br>This IE shall be absent if neither hAccuracy nor vAccuracy is included.                                                                                                                     |               |

NOTE: If the lcsQoSClass set to "MULTIPLE\_QOS", at least one of attributes hAccuracy, vAccuracy shall be present.

6.1.6.2.14      Type: CivicAddress

**Table 6.1.6.2.14-1: Definition of type CivicAddress**

| Attribute name | Data type | P | Cardinality | Description                                                                                        |
|----------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------|
| country        | string    | M | 1           | The two-letter ISO 3166 country code in capital ASCII letters, e.g., DE or US<br>IETF RFC 4776 [6] |
| A1             | string    | O | 0..1        | National subdivisions (state, canton, region, province, prefecture)<br>IETF RFC 4776 [6]           |
| A2             | string    | O | 0..1        | County, parish, gun (JP), district (IN)<br>IETF RFC 4776 [6]                                       |
| A3             | string    | O | 0..1        | City, township, shi (JP)<br>IETF RFC 4776 [6]                                                      |
| A4             | string    | O | 0..1        | City division, borough, city district, ward, chou (JP)<br>IETF RFC 4776 [6]                        |
| A5             | string    | O | 0..1        | Neighbourhood, block<br>IETF RFC 4776 [6]                                                          |
| A6             | string    | O | 0..1        | Group of streets below the neighbourhood level<br>IETF RFC 4776 [6]                                |
| PRD            | string    | O | 0..1        | Leading street direction<br>IETF RFC 4776 [6]                                                      |
| POD            | string    | O | 0..1        | Trailing street suffix<br>IETF RFC 4776 [6]                                                        |
| STS            | string    | O | 0..1        | Street suffix or type<br>IETF RFC 4776 [6]                                                         |
| HNO            | string    | O | 0..1        | House number<br>IETF RFC 4776 [6]                                                                  |
| HNS            | string    | O | 0..1        | House number suffix<br>IETF RFC 4776 [6]                                                           |
| LMK            | string    | O | 0..1        | Landmark or vanity address<br>IETF RFC 4776 [6]                                                    |
| LOC            | string    | O | 0..1        | Additional location information<br>IETF RFC 4776 [6]                                               |
| NAM            | string    | O | 0..1        | Name (residence and office occupant)<br>IETF RFC 4776 [6]                                          |
| PC             | string    | O | 0..1        | Postal/zip code<br>IETF RFC 4776 [6]                                                               |
| BLD            | string    | O | 0..1        | Building (structure)<br>IETF RFC 5139 [7]                                                          |
| UNIT           | string    | O | 0..1        | Unit (apartment, suite)<br>IETF RFC 5139 [7]                                                       |
| FLR            | string    | O | 0..1        | Floor<br>IETF RFC 4776 [6]                                                                         |
| ROOM           | string    | O | 0..1        | Room<br>IETF RFC 5139 [7]                                                                          |
| PLC            | string    | O | 0..1        | Place-type<br>IETF RFC 5139 [7]                                                                    |
| PCN            | string    | O | 0..1        | Postal community name<br>IETF RFC 5139 [7]                                                         |
| POBOX          | string    | O | 0..1        | Post office box (P.O. box)<br>IETF RFC 5139 [7]                                                    |
| ADDCODE        | string    | O | 0..1        | Additional code<br>IETF RFC 5139 [7]                                                               |
| SEAT           | string    | O | 0..1        | Seat (desk, cubicle, workstation)<br>IETF RFC 5139 [7]                                             |
| RD             | string    | O | 0..1        | Primary road or street<br>IETF RFC 5139 [7]                                                        |
| RDSEC          | string    | O | 0..1        | Road clause<br>IETF RFC 5139 [7]                                                                   |
| RDBR           | string    | O | 0..1        | Road branch<br>IETF RFC 5139 [7]                                                                   |
| RDSUBBR        | string    | O | 0..1        | Road sub-branch<br>IETF RFC 5139 [7]                                                               |
| PRM            | string    | O | 0..1        | Road pre-modifier<br>IETF RFC 5139 [7]                                                             |
| POM            | string    | O | 0..1        | Road post-modifier<br>IETF RFC 5139 [7]                                                            |

|            |        |   |      |                                                                                                                                              |
|------------|--------|---|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| usageRules | string | O | 0..1 | When present, this IE shall carry the value of "usage-rules" Element of the PIDL-LO XML document, with UTF-8 encoding.<br>IETF RFC 4119 [25] |
| method     | string | O | 0..1 | When present, this IE shall contain the method token, carried by the "method" Element of the PIDL-LO XML document.<br>IETF RFC 4119 [25]     |
| providedBy | string | O | 0..1 | When present, this IE shall carry the value of "provided-by" Element of the PIDL-LO XML document, with UTF-8 encoding.<br>IETF RFC 4119 [25] |

**EXAMPLE:** The above structure follows the same label naming as in the XML schema shown in IETF RFC 5139 [7]. The same example shown in XML in that RFC, in clause 5, would be equivalent to the following JSON document:

```
{
  "country": "AU",
  "A1": "NSW",
  "A3": "Wollongong",
  "A4": "North Wollongong",
  "RD": "Flinders",
  "STS": "Street",
  "RDBR": "Campbell Street",
  "LMK": "Gilligan's Island",
  "LOC": "Corner",
  "NAM": "Video Rental Store",
  "PC": "2500",
  "ROOM": "Westerns and Classics",
  "PLC": "store",
  "POBOX": "Private Box 15"
}
```

#### 6.1.6.2.15 Type: PositioningMethodAndUsage

**Table 6.1.6.2.15-1: Definition of type PositioningMethodAndUsage**

| Attribute name | Data type         | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|-------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| method         | PositioningMethod | M | 1           | Indicates the related positioning method                                                                                                                                                                                                                                                                                                                                                                                        |
| mode           | PositioningMode   | M | 1           | Indicates the mode of the location measurement from the related positioning method.                                                                                                                                                                                                                                                                                                                                             |
| usage          | Usage             | M | 1           | Indicates the usage of the location measurement from the related positioning method.                                                                                                                                                                                                                                                                                                                                            |
| methodCode     | integer           | C | 0..1        | This IE shall be present when the <i>method</i> IE is with value "NETWORK_SPECIFIC".<br><br>When present, this IE shall carry the code value of the network specific positioning method in decimal which encodes the binary value "10000 to 11111" (bits 8-4 of "Positioning Method and Usage" IE within "Positioning Data" parameter, as specified in clause 7.4.13 of 3GPP TS 29.171 [24].)<br><br>Minimum: 16<br>Maximum: 31 |



## 6.1.6.2.16 Type: GnssPositioningMethodAndUsage

**Table 6.1.6.2.16-1: Definition of type GnssPositioningMethodAndUsage**

| Attribute name | Data type       | P | Cardinality | Description                                                                           |
|----------------|-----------------|---|-------------|---------------------------------------------------------------------------------------|
| mode           | PositioningMode | M | 1           | Indicates the mode of location measurement from the related GNSS positioning method.  |
| gnss           | GnssId          | M | 1           | Indicates the related GNSS positioning method                                         |
| usage          | Usage           | M | 1           | Indicates the usage of the location measurement from related GNSS positioning method. |

## 6.1.6.2.17 Type: VelocityEstimate

**Table 6.1.6.2.17-1: Definition of type VelocityEstimate as a list of mutually exclusive alternatives**

| Data type                                                                                                                                                                                                                                                                                                                                                                                                                            | Cardinality | Description                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HorizontalVelocity                                                                                                                                                                                                                                                                                                                                                                                                                   | 1           | Velocity estimate including horizontal speed and bearing.                                                                                                                                                                                                                       |
| HorizontalWithVerticalVelocity                                                                                                                                                                                                                                                                                                                                                                                                       | 1           | Velocity estimate including horizontal speed and bearing, and also vertical speed and vertical direction.                                                                                                                                                                       |
| HorizontalVelocityWithUncertainty                                                                                                                                                                                                                                                                                                                                                                                                    | 1           | Velocity estimate including horizontal speed and bearing; it also includes an uncertainty value.                                                                                                                                                                                |
| HorizontalWithVerticalVelocityAndUncertainty                                                                                                                                                                                                                                                                                                                                                                                         | 1           | Velocity estimate including horizontal speed and bearing, and also vertical speed and vertical direction; it also includes uncertainty value for horizontal and vertical speeds.                                                                                                |
| RelativeVelocityWithUncertainty                                                                                                                                                                                                                                                                                                                                                                                                      | 1           | Velocity estimate including radial velocity characterized by a rate of change of a range, and transverse velocity characterized by the change rate of an angle of azimuth and the change rate of an angle of elevation; each rate of change also includes an uncertainty value. |
| NOTE: The "anyOf" keyword (instead of "oneOf" keyword which is normally used for mutually exclusive alternatives) is used for VelocityEstimate type in yaml file to avoid validation failure of OpenAPI. According to current definition, a HorizontalWithVerticalVelocity object will always pass the validation with both HorizontalWithVerticalVelocity and HorizontalVelocity, which fails the qualification of "oneOf" keyword. |             |                                                                                                                                                                                                                                                                                 |

## 6.1.6.2.18 Type: HorizontalVelocity

**Table 6.1.6.2.18-1: Definition of type HorizontalVelocity**

| Attribute name | Data type       | P | Cardinality | Description                                              |
|----------------|-----------------|---|-------------|----------------------------------------------------------|
| hSpeed         | HorizontalSpeed | M | 1           | Horizontal speed in kilometres per hour.                 |
| bearing        | Angle           | M | 1           | Bearing angle in degrees, measured clockwise from North. |

## 6.1.6.2.19 Type: HorizontalWithVerticalVelocity

**Table 6.1.6.2.19-1: Definition of type HorizontalWithVerticalVelocity**

| Attribute name | Data type         | P | Cardinality | Description                                              |
|----------------|-------------------|---|-------------|----------------------------------------------------------|
| hSpeed         | HorizontalSpeed   | M | 1           | Horizontal speed in kilometres per hour.                 |
| bearing        | Angle             | M | 1           | Bearing angel in degrees, measured clockwise from North. |
| vSpeed         | VerticalSpeed     | M | 1           | Vertical Seed in kilometres per hour.                    |
| vDirection     | VerticalDirection | M | 1           | Vertical Direction: upward or downward.                  |

## 6.1.6.2.20 Type: HorizontalVelocityWithUncertainty

**Table 6.1.6.2.20-1: Definition of type HorizontalVelocityWithUncertainty**

| Attribute name | Data type        | P | Cardinality | Description                                              |
|----------------|------------------|---|-------------|----------------------------------------------------------|
| hSpeed         | HorizontalSpeed  | M | 1           | Speed in kilometres per hour.                            |
| bearing        | Angle            | M | 1           | Bearing angel in degrees, measured clockwise from North. |
| uncertainty    | SpeedUncertainty | M | 1           | Uncertainty of horizontal speed in kilometres per hour.  |

## 6.1.6.2.21 Type: HorizontalWithVerticalVelocityAndUncertainty

**Table 6.1.6.2.21-1: Definition of type HorizontalWithVerticalVelocityAndUncertainty**

| Attribute name | Data type         | P | Cardinality | Description                                              |
|----------------|-------------------|---|-------------|----------------------------------------------------------|
| hspeed         | HorizontalSpeed   | M | 1           | Speed in kilometres per hour.                            |
| bearing        | Angle             | M | 1           | Bearing angel in degrees, measured clockwise from North. |
| vSpeed         | VerticalSpeed     | M | 1           | Vertical Seed in kilometres per hour.                    |
| vDirection     | VerticalDirection | M | 1           | Vertical Direction: upwards or downwards.                |
| hUncertainty   | SpeedUncertainty  | M | 1           | Uncertainty of horizontal speed in kilometres per hour.  |
| vUncertainty   | SpeedUncertainty  | M | 1           | Uncertainty of vertical speed in kilometres per hour.    |

## 6.1.6.2.22 Type: UncertaintyEllipse

**Table 6.1.6.2.22-1: Definition of type UncertaintyEllipse**

| Attribute name   | Data type   | P | Cardinality | Description                                               |
|------------------|-------------|---|-------------|-----------------------------------------------------------|
| semiMajor        | Uncertainty | M | 1           | Indicates the semi-major axis of the uncertainty ellipse. |
| semiMinor        | Uncertainty | M | 1           | Indicates the semi-minor axis of the uncertainty ellipse. |
| orientationMajor | Orientation | M | 1           | Indicates the orientation angle of the major axis.        |

## 6.1.6.2.23 Type: UeLcsCapability

**Table 6.1.6.2.23-1: Definition of type UeLcsCapability**

| Attribute name   | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lppSupport       | boolean   | O | 0..1        | Indicates whether the UE supports LPP or not.<br><br>- true (default): LPP supported by the UE<br>- false: LPP not supported by the UE                                                                                                                                                                                                                                                                         |
| ciotOptimisation | boolean   | O | 0..1        | Indicates whether the UE supports and is allowed to use Control Plane CIoT 5GS Optimisation to send an event report for periodic or triggered location or not. Refer to 3GPP TS 23.273 [19] clause 6.7 for more detail.<br><br>- true: Control Plane CIoT 5GS Optimisation is supported by the UE and allowed<br>- false (default): Control Plane CIoT 5GS Optimisation not supported by the UE or not allowed |

## 6.1.6.2.24 Type: PeriodicEventInfo

Table 6.1.6.2.24-1: Definition of type PeriodicEventInfo

| Attribute name                                                                                                                                            | Data type           | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reportingAmount                                                                                                                                           | ReportingAmount     | M | 1           | Number of event reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| reportingInterval                                                                                                                                         | ReportingInterval   | M | 1           | Interval of event reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| reportingInfiniteInd                                                                                                                                      | boolean             | O | 0..1        | <p>When present, this IE shall be set to the value true, indicating that as many reports as possible to be generated during allowed duration (8639999 seconds).</p> <p>When this IE is present in a request, the reportingAmount should be set to the largest possible value (see NOTE) for backward compatibility consideration. An LMF supporting this IE shall ignore the reportingAmount IE and an LMF not supporting this IE will use reportingAmount IE to generate indicated number of reports as legacy.</p> <p>When this IE is present in a response, it indicates that this IE is supported and accepted by the LMF.</p> |
| reportingIntervalMs                                                                                                                                       | ReportingIntervalMs | O | 0..1        | <p>When present, this IE shall indicate the Interval of event reports in milliseconds.</p> <p>When this IE is present in a request, the reportingInterval is set to 1 for backward compatible consideration. An LMF supporting this IE shall ignore the reportingInterval IE; an LMF not supporting this IE will use reportingInterval IE to generate report with minimal interval as legacy, i.e. 1 second.</p> <p>When this IE is present in a response, it indicates that this IE is supported by the LMF and it shall indicate the value accepted by the LMF.</p>                                                              |
| NOTE: reportingAmount x reportingInterval shall not exceed 8639999 (99 days, 23 hours, 59 minutes and 59 seconds) for compatibility with OMA MLP and RLP. |                     |   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## 6.1.6.2.25 Type: AreaEventInfo

Table 6.1.6.2.25-1: Definition of type AreaEventInfo

| Attribute name       | Data type            | P | Cardinality | Description                                                                                                                                                                                                                                  |
|----------------------|----------------------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| areaDefinition       | array(ReportingArea) | M | 1..250      | One or more reporting areas                                                                                                                                                                                                                  |
| occurrenceInfo       | OccurrenceInfo       | O | 0..1        | One time only report indication                                                                                                                                                                                                              |
| minimumInterval      | MinimumInterval      | O | 0..1        | Minimum interval between event reports.<br>This IE shall not be included if occurrenceInfo is present and set to one time event.                                                                                                             |
| maximumInterval      | MaximumInterval      | O | 0..1        | Maximum interval between event reports.<br>This IE shall not be included if occurrenceInfo is present and set to one time event.                                                                                                             |
| samplingInterval     | SamplingInterval     | O | 0..1        | Maximum time interval between consecutive evaluations by a UE of a trigger event.                                                                                                                                                            |
| reportingDuration    | ReportingDuration    | O | 0..1        | Maximum duration of event reporting.                                                                                                                                                                                                         |
| reportingLocationReq | boolean              | C | 0..1        | This IE shall be present and set to true if a location estimate is required for each event report.<br><br>When present, it shall be set as follows:<br>- true: location report is required.<br><br>- false: location report is not required. |

## 6.1.6.2.26 Type: ReportingArea

Table 6.1.6.2.26-1: Definition of type ReportingArea

| Attribute name | Data type         | P | Cardinality | Description                                                                                       |
|----------------|-------------------|---|-------------|---------------------------------------------------------------------------------------------------|
| areaType       | ReportingAreaType | M | 1           | Type of reporting area.                                                                           |
| tai            | Tai               | C | 1           | TAI for EPS or 5GS.<br>This IE shall be present if the reporting area type is EPS TAI or 5GS TAI. |
| ecgi           | Ecgi              | C | 1           | ECGI.<br>This IE shall be present if the reporting area type is ECGI.                             |
| ncgi           | Ncgi              | C | 1           | NCGI.<br>This IE shall be present if the reporting area type is NCGI.                             |

NOTE: One of tai, ecgi or ncgi shall be included.

## 6.1.6.2.27 Type: MotionEventInfo

Table 6.1.6.2.27-1: Definition of type MotionEventInfo

| Attribute name       | Data type         | P | Cardinality | Description                                                                                                                                                                                                                                  |
|----------------------|-------------------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| linearDistance       | LinearDistance    | M | 1           | Minimum linear (straight line) distance for motion event reports.                                                                                                                                                                            |
| occurrenceInfo       | OccurrenceInfo    | O | 0..1        | One time only report indication                                                                                                                                                                                                              |
| minimumInterval      | MinimumInterval   | O | 0..1        | Minimum interval between event reports.<br>This IE shall not be included if occurrenceInfo is present and set to one time event.                                                                                                             |
| maximumInterval      | MaximumInterval   | O | 0..1        | Maximum interval between event reports.<br>This IE shall not be included if occurrenceInfo is present and set to one time event.                                                                                                             |
| samplingInterval     | SamplingInterval  | O | 0..1        | Maximum time interval between consecutive evaluations by a UE of a trigger event.                                                                                                                                                            |
| reportingDuration    | ReportingDuration | O | 0..1        | Maximum duration of event reporting.                                                                                                                                                                                                         |
| reportingLocationReq | boolean           | C | 0..1        | This IE shall be present and set to true if a location estimate is required for each event report.<br><br>When present, it shall be set as follows:<br>- true: location report is required.<br><br>- false: location report is not required. |

## 6.1.6.2.28 Void

## 6.1.6.2.29 Type: CancelLocData

Table 6.1.6.2.29-1: Definition of type CancelLocData

| Attribute name                                                                                          | Data type         | P | Cardinality | Description                                                                                                                                                                                                                      | Applicability |
|---------------------------------------------------------------------------------------------------------|-------------------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| hgmlcCallbackURI                                                                                        | Uri               | M | 1           | Callback URI of the H-GMLC                                                                                                                                                                                                       |               |
| ldrReference                                                                                            | LdrReference      | M | 1           | LDR Reference, used to identify the deferred 5G-MT-LR procedure that needs to be cancelled.<br>(NOTE)                                                                                                                            |               |
| lcsCorrelationID                                                                                        | CorrelationID     | C | 0..1        | It shall be present if the request is to cancel the 5G-MO-LR / 5G-MT-LR procedure.<br>When present, it indicates the LCS Correlation Id used to identify the 5G-MO-LR / 5G-MT-LR procedures that need to be cancelled.<br>(NOTE) | CIMLR         |
| supportedFeatures                                                                                       | SupportedFeatures | C | 0..1        | This IE shall be present if at least one feature defined in clause 6.1.9 is supported.                                                                                                                                           |               |
| NOTE: When the lcsCorrelationId is present, the hgmlcCallbackURI and the ldrReference shall be ignored. |                   |   |             |                                                                                                                                                                                                                                  |               |

6.1.6.2.30      Type: LocContextData

**Table 6.1.6.2.30-1: Definition of type LocContextData**

| Attribute name       | Data type                 | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                                            | Applicability |
|----------------------|---------------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| amfId                | NfInstanceId              | M | 1           | Indicates the AMF Instance serving the UE. LMF shall use the AMF Instance to forward LCS related N1/N2 messages to the UE/RAN.                                                                                                                                                                                                                                         |               |
| locationQoS          | LocationQoS               | C | 0..1        | This IE shall contain the location QoS if available.                                                                                                                                                                                                                                                                                                                   |               |
| supportedGADShapes   | array(SupportedGADShapes) | C | 1..N        | This IE shall contain the supported GAD shapes if available.                                                                                                                                                                                                                                                                                                           |               |
| supi                 | Supi                      | C | 0..1        | This IE shall contain the SUPI if available.<br>(NOTE 3)                                                                                                                                                                                                                                                                                                               |               |
| gpsi                 | Gpsi                      | C | 0..1        | This IE shall contain the GPSI if available.                                                                                                                                                                                                                                                                                                                           |               |
| addLocContextList    | array(AddLocationContext) | O | 1..N        | When present, this IE shall contain a list of locationContext(s) to be transferred from source LMF to target LMF. (NOTE 2)                                                                                                                                                                                                                                             | MCTX          |
| ldrType              | LdrType                   | M | 1           | The type of LDR                                                                                                                                                                                                                                                                                                                                                        |               |
| hgmlcCallbackURI     | Uri                       | M | 1           | Callback URI of the H-GMLC                                                                                                                                                                                                                                                                                                                                             |               |
| ldrReference         | LdrReference              | M | 1           | LDR Reference                                                                                                                                                                                                                                                                                                                                                          |               |
| periodicEventInfo    | PeriodicEventInfo         | C | 0..1        | Information for periodic event reporting. (NOTE 1)                                                                                                                                                                                                                                                                                                                     |               |
| areaEventInfo        | AreaEventInfo             | C | 0..1        | Information for area event reporting. (NOTE 1)                                                                                                                                                                                                                                                                                                                         |               |
| motionEventInfo      | MotionEventInfo           | C | 0..1        | Information for motion event reporting. (NOTE 1)                                                                                                                                                                                                                                                                                                                       |               |
| eventReportMessage   | EventReportMessage        | M | 1           | Contains an embedded event report<br><br>When the event report message from the UE is not available for the location context, e.g. during LCS-UPP connection modification procedure, the EventClass of this IE shall be set to "DUMMY" and the target LMF shall ignore the received IE with EventClass set to the value "DUMMY".                                       |               |
| eventReportingStatus | EventReportingStatus      | O | 0..1        | Status of event reporting                                                                                                                                                                                                                                                                                                                                              |               |
| ueLocationInfo       | UELocationInfo            | O | 0..1        | Location information for the target UE                                                                                                                                                                                                                                                                                                                                 |               |
| clot5GSOptimisation  | boolean                   | C | 0..1        | This IE shall be present if it was received from AMF. When present, it shall be set as follows:<br>- true: Control Plane Clot 5GS Optimisation was used and no signalling or data is currently pending for the UE at the AMF.<br>- false (default): Control Plane Clot 5GS Optimisation was not used or signalling or data is currently pending for the UE at the AMF. |               |
| ecgi                 | Ecgi                      | C | 0..1        | When present, this IE shall indicate the identifier of the E-UTRAN cell serving the UE. This IE shall be present if it was received from AMF.                                                                                                                                                                                                                          |               |

|                     |                                    |   |      |                                                                                                                                                                                                                                                                                              |  |
|---------------------|------------------------------------|---|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| ncgi                | Ncgi                               | C | 0..1 | When present, this IE shall indicate the identifier of the NR cell serving the UE.<br>This IE shall be present if it was received from AMF.                                                                                                                                                  |  |
| guami               | Guami                              | C | 0..1 | This IE shall be present if it was received from AMF.<br><br>When present, it shall contain the GUAMI serving the UE.                                                                                                                                                                        |  |
| supportedFeatures   | SupportedFeatures                  | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.9 is supported.                                                                                                                                                                                                       |  |
| uePositioningCap    | UePositioningCapabilities          | O | 0..1 | When present, this IE shall indicate the positioning capabilities supported by the UE.                                                                                                                                                                                                       |  |
| ueUpPosCaps         | array(UeUpPositioningCapabilities) | C | 1..N | This IE shall be present if the source LMF has received the UE user plane positioning capabilities. The IE shall contain the supported UE user plane positioning capabilities.                                                                                                               |  |
| scheduledLocTime    | DateTime                           | O | 0..1 | When present, this IE shall contain the scheduled time (in UTC) that the UE needs to be located.                                                                                                                                                                                             |  |
| indoorOutdoorInd    | IndoorOutdoorInd                   | O | 0..1 | When present, this IE shall indicate whether the UE is indoor or outdoor.                                                                                                                                                                                                                    |  |
| losNlosMeasureInd   | LosNlosMeasureInd                  | O | 0..1 | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used.                                                                                                                                                                                                    |  |
| upCumEvtRptCriteria | UpCumEvtRptCriteria                | O | 0..1 | When present, this IE shall include the cumulative event report timer or the maximum number of location reporting over user plane or both.                                                                                                                                                   |  |
| lcsUppExistInd      | boolean                            | C | 0..1 | This IE shall be present with the value true, if the location context being transferred is associated with the LCS-UPP connection between the UE and the source LMF.<br>See clause 6.18 of 3GPP TS 23.273 [19] (NOTE 3)<br><br>Presence of this IE with the value false shall be prohibited. |  |
| srcLmfRoutingInfo   | LMFIdentification                  | C | 0..1 | This IE shall be present, if the lcsUppExistInd is present and the UE supports multiple LCS-UPP connections.<br><br>When present, this IE shall indicate the source LMF routing information (i.e. the source LMF Identification).                                                            |  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| extendedFacilityInd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | boolean | C | 0..1 | <p>This IE shall be included when received from the AMF during determine location service operation for this location context or from an old LMF.</p> <p>When this IE is received, the LMF shall use ExtendedFacility IE in 5G LCS related Supplementary service messages, when the component of the messages cannot be carried in Facility IE, i.e. the size of the component exceeds 255 octets, as specified in 3GPP TS 24.080 [20].</p> <p>Presence of this IE with the value false shall be prohibited.</p> |  |
| <p>NOTE 1: At least one of periodicEventInfo, areaEventInfo or motionEventInfo shall be present in the LocContextData structure.</p> <p>NOTE 2: If the source and target LMF support MCTX feature and the source LMF contains more than one LCS context for the UE with LCS UP connection, this attribute shall include the additional location context(s) of the UE. The first LCS context shall be in the LocContextData data type and shall not be repeated in the addLocContextList attribute.</p> <p>NOTE 3: If lcsUppExistInd is present and set to true, the supi IE shall be present.</p> |         |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |

#### 6.1.6.2.31 Type: EventReportMessage

**Table 6.1.6.2.31-1: Definition of type EventReportMessage**

| Attribute name | Data type       | P | Cardinality | Description                                                                              |
|----------------|-----------------|---|-------------|------------------------------------------------------------------------------------------|
| eventClass     | EventClass      | M | 1           | This IE shall contain the event class for the message content specified in eventContent. |
| eventContent   | RefToBinaryData | M | 1           | This IE shall reference the event report binary data corresponding to the eventClass.    |

#### 6.1.6.2.32 Type: EventReportingStatus

**Table 6.1.6.2.32-1: Definition of type EventReportingStatus**

| Attribute name      | Data type           | P | Cardinality | Description                                            |
|---------------------|---------------------|---|-------------|--------------------------------------------------------|
| eventReportCounter  | EventReportCounter  | O | 0..1        | This IE shall contain a count of event reports.        |
| eventReportDuration | EventReportDuration | O | 0..1        | This IE shall contain the duration of event reporting. |

## 6.1.6.2.33      Type: UELocationInfo

**Table 6.1.6.2.33-1: Definition of type UELocationInfo**

| Attribute name              | Data type             | P | Cardinality | Description                                                                                                                                                                                       |
|-----------------------------|-----------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| locationEstimate            | GeographicArea        | O | 0..1        | Previous location estimate for the target UE.                                                                                                                                                     |
| ageOfLocationEstimate       | AgeOfLocationEstimate | O | 0..1        | Age of previous location estimate.                                                                                                                                                                |
| timestampOfLocationEstimate | DateTime              | O | 0..1        | When present, this IE shall indicate the estimated UTC time when the location estimate corresponded to the UE location (i.e. when the location estimate and the actual UE location was the same). |
| velocityEstimate            | VelocityEstimate      | O | 0..1        | Previous velocity estimate for the target UE.                                                                                                                                                     |
| ageOfVelocityEstimate       | AgeOfLocationEstimate | O | 0..1        | Age of previous velocity estimate.                                                                                                                                                                |
| timestampOfVelocityEstimate | DateTime              | O | 0..1        | When present, this IE shall indicate the estimated UTC time when the velocity estimate corresponded to the UE velocity (i.e. when the velocity estimate and the actual UE velocity was the same). |

6.1.6.2.34      Type: EventNotifyData

**Table 6.1.6.2.34-1: Definition of type EventNotifyData**

| Attribute name              | Data type                            | P | Cardinality | Description                                                                                                                                                                                                            | Applicability |
|-----------------------------|--------------------------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| reportedEventType           | ReportedEventType                    | M | 1           | This IE shall contain the type of event being reported.                                                                                                                                                                |               |
| supi                        | Supi                                 | C | 0..1        | This IE shall contain the SUPI if available.                                                                                                                                                                           |               |
| gpsi                        | Gpsi                                 | C | 0..1        | This IE shall contain the GPSI if available.                                                                                                                                                                           |               |
| hgmlcCallbackURI            | Uri                                  | C | 0..1        | Callback URI of the H-GMLC (NOTE 1)                                                                                                                                                                                    |               |
| ldrReference                | LdrReference                         | M | 1           | LDR Reference<br><br>When the ReportedEventType is "INTERMEDIATE_EVENT", the LMF shall set this IE to the value "NULL" and shall be ignored by the receiver.                                                           |               |
| lirReference                | LirRefernece                         | C | 0..1        | This IE shall be present when the ReportedEventType is set to the value "INTERMEDIATE_EVENT".<br><br>When present, this IE shall include the LIR Reference number received in the 5GC-MR-LR multiple location request. |               |
| locationEstimate            | GeographicArea                       | O | 0..1        | If present, this IE shall contain an estimate of the location of the UE in universal coordinates and the accuracy of the estimate.                                                                                     |               |
| ageOfLocationEstimate       | AgeOfLocationEstimate                | O | 0..1        | If present, this IE shall contain an indication of how long ago the location estimate was obtained.                                                                                                                    |               |
| timestampOfLocationEstimate | DateTime                             | O | 0..1        | When present, this IE shall indicate the estimated UTC time when the location estimate corresponded to the UE location (i.e. when the location estimate and the actual UE location was the same).                      |               |
| civicAddress                | CivicAddress                         | O | 0..1        | If present, this IE shall contain a civic address.                                                                                                                                                                     |               |
| localLocationEstimate       | LocalArea                            | O | 0..1        | When present, this IE shall indicate a local area in reference system.                                                                                                                                                 |               |
| positioningDataList         | array(PositioningMethodAndUsage)     | O | 1..N        | If present, this IE shall indicate the usage of each non-GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully.                                        |               |
| gnssPositioningDataList     | array(GnssPositioningMethodAndUsage) | O | 1..N        | If present, this IE shall indicate the usage of each GANSS positioning method that was attempted to determine the location estimate, either successfully or unsuccessfully.                                            |               |
| servingLMFIdentification    | LMFIdentification                    | C | 0..1        | This IE shall be included to identify an LMF which acts as a serving LMF if a serving LMF is used.                                                                                                                     |               |
| terminationCause            | TerminationCause                     | C | 0..1        | This IE shall be included if event reporting has been terminated                                                                                                                                                       |               |
| velocityEstimate            | VelocityEstimate                     | O | 0..1        | If present, this IE shall contain an estimate of the velocity of the target UE, composed by horizontal speed, vertical speed, and their respective uncertainty.                                                        |               |

|                           |                         |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                |
|---------------------------|-------------------------|---|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| altitude                  | Altitude                | O | 0..1 | If present, this IE indicates the altitude of the positioning estimate. When the shape used in "locationEstimate" supports conveying the altitude parameter, this IE shall be absent.                                                                                                                                                                                                                                                                               |                |
| achievedQos               | MinorLocationQoS        | O | 0..1 | When present, this IE shall contain the achieved Location QoS Accuracy of the estimated location.<br><br>This IE may be present if the required LCS QoS Class in the location request procedure is "MULTIPLE_QOS".<br><br>If it's absent, the required LCS QoS Class in the location request procedure is "MULTIPLE_QOS" and AccuracyFulfilmentIndicator is "REQUESTED_ACCURACY_FULFILLED", it indicates that the location QoS in the highest priority is achieved. | MUTIQ<br>OS    |
| supportedFeatures         | SupportedFeatures       | C | 0..1 | This IE shall be present if at least one feature defined in clause 6.1.9 is supported.                                                                                                                                                                                                                                                                                                                                                                              |                |
| indoorOutdoorInd          | IndoorOutdoorInd        | O | 0..1 | When present, this IE shall indicate whether the UE is indoor or outdoor.                                                                                                                                                                                                                                                                                                                                                                                           |                |
| haGnssMetrics             | HighAccuracyGnssMetrics | O | 0..1 | When present, this IE shall indicate the high accuracy GNSS metrics received from the device in the LPP <i>HA-GNSS-Metrics-r17</i> IE as specified in 3GPP TS 37.355 [21].                                                                                                                                                                                                                                                                                          |                |
| losNlosMeasureInd         | LosNlosMeasureInd       | O | 0..1 | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used.                                                                                                                                                                                                                                                                                                                                                                           |                |
| upLocRepStatAf            | integer                 | O | 0..1 | When present, this IE contains the number of event reports have transferred over user plane. If the cumulative event report has been sent previously, this IE contains the number of event reports have transferred over user plane since the last cumulative event report was sent.                                                                                                                                                                                |                |
| relatedApplicationlayerId | ApplicationlayerId      | O | 0..1 | Identifies the application layer ID of the related UE for ranging and sidelink positioning, such as located UE, reference UE, etc.                                                                                                                                                                                                                                                                                                                                  | Rangin<br>g_SL |
| distanceDirection         | RangeDirection          | O | 0..1 | When present, this IE identifies a straight line distance (range) and direction from a point A to a point B, comprising a straight line distance (range) from point A to point B, an azimuth direction from point A to point B and an elevation direction from point A to point B.                                                                                                                                                                                  | Rangin<br>g_SL |
| 2dRelativeLocation        | 2DRelativeLocation      | O | 0..1 | When present, this IE identifies a relative 2D location with uncertainty ellipse, characterised by a point described in 2D local co-ordinates with origin corresponding to another known point, distances r1 and r2 and an angle of orientation A.                                                                                                                                                                                                                  | Rangin<br>g_SL |

|                                                                                                                                              |                    |   |      |                                                                                                                                                                                                                                                                                                                                                                                     |            |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 3dRelativeLocation                                                                                                                           | 3DRelativeLocation | O | 0..1 | When present, this IE identifies a relative 3D location with uncertainty ellipsoid, characterised by a point described in 3D local co-ordinates with origin corresponding to another known point, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis"). | Ranging_SL |
| relativeVelocity                                                                                                                             | VelocityEstimate   | O | 0..1 | When present, this IE identifies UE velocity relative to the UE identified with relatedApplicationlayerId.                                                                                                                                                                                                                                                                          | Ranging_SL |
| integrityResult                                                                                                                              | IntegrityResult    | C | 0..1 | This IE shall be present when the integrity requirements are present in the request.<br><br>When present, this IE shall indicate the integrity result.                                                                                                                                                                                                                              | INTRES     |
| NOTE 1: If the reportedEventType is not "INTERMEDIATE_EVENT", the hgmlcCallbackURI shall be included when the consumer NF is not the H-GMLC. |                    |   |      |                                                                                                                                                                                                                                                                                                                                                                                     |            |

#### 6.1.6.2.35 Type: UeConnectivityState

**Table 6.1.6.2.35-1: Definition of type UeConnectivityState**

| Attribute name    | Data type  | P | Cardinality | Description                                                                             |
|-------------------|------------|---|-------------|-----------------------------------------------------------------------------------------|
| accessType        | AccessType | M | 1           | Shall indicate the access type of the UE.                                               |
| connectivitystate | CmState    | O | 0..1        | When present, it shall indicate the UE connectivity state in the indicated access type. |

#### 6.1.6.2.36 Type: LocalOrigin

**Table 6.1.6.2.36-1: Definition of type LocalOrigin**

| Attribute name                                                                        | Data type               | P | Cardinality | Description                                                                                                                   |
|---------------------------------------------------------------------------------------|-------------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| coordinateId                                                                          | string                  | M | 1           | This IE defines a known reference point which configured by the PLMN operator                                                 |
| point                                                                                 | GeographicalCoordinates | O | 0..1        | When present, this IE shall indicate the local origin as a geographic point represented by its longitude and latitude. (NOTE) |
| area                                                                                  | GeographicArea          | O | 0..1        | When present, this IE shall indicate the geographic area where the local origin is located. (NOTE)                            |
| horizAxesOrientation                                                                  | HorizAxesOrientation    | O | 0..1        | Indicates the rotation in degrees of the horizontal coordinate system from northing. If not present, the rotation is zero.    |
| NOTE: Either the point IE or the area IE may be present to indicate the local origin. |                         |   |             |                                                                                                                               |

## 6.1.6.2.37 Type: RelativeCartesianLocation

**Table 6.1.6.2.37-1: Definition of type RelativeCartesianLocation**

| Attribute name | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                        |
|----------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x              | Float     | M | 1           | Indicates the value (in the unit of meters) on x-axis of the relative location in the cartesian system.<br><br>Positive value represents easting from reference point (origin), unless horizAxesOrientation is provided in the LocalOrigin, in which case the local coordinate system is rotated clockwise horizontally by horizAxesOrientation).  |
| y              | Float     | M | 1           | Indicates the value (in the unit of meters) on y-axis of the relative location in the cartesian system.<br><br>Positive value represents northing from reference point (origin), unless horizAxesOrientation is provided in the LocalOrigin, in which case the local coordinate system is rotated clockwise horizontally by horizAxesOrientation). |
| z              | Float     | O | 0..1        | Indicates the value (in the unit of meters) on z-axis of the relative location in the cartesian system for a 3D-Point.<br><br>Positive value represents height above reference point (origin).                                                                                                                                                     |

## 6.1.6.2.38 Type: Local2dPointUncertaintyEllipse

**Table 6.1.6.2.38-1: Definition of type Local2dPointUncertaintyEllipse**

| Attribute name     | Data type                 | P | Cardinality | Description                                                                                                                                                                        |
|--------------------|---------------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shape              | SupportedGADShapes        | M | 1           | It shall take the value "LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE".                                                                                                                      |
| localOrigin        | LocalOrigin               | M | 1           | Indicates the local origin in the local Cartesian co-ordinates system configured by the PLMN operator.                                                                             |
| point              | RelativeCartesianLocation | M | 1           | Indicates a 2D-point (specified by "x" and "y" coordinates) relative to origin in reference system.                                                                                |
| uncertaintyEllipse | UncertaintyEllipse        | M | 1           | Indicates an uncertainty ellipse in relation to the local Cartesian coordinate system defined by the localOrigin, where the orientationMajor angle is clockwise from the "y" axis. |
| confidence         | Confidence                | M | 1           | Indicates the value of confidence.                                                                                                                                                 |

## 6.1.6.2.39 Type: Local3dPointUncertaintyEllipsoid

**Table 6.1.6.2.39-1: Definition of type Local3dPointUncertaintyEllipsoid**

| Attribute name       | Data type                 | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|---------------------------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shape                | SupportedGADShapes        | M | 1           | It shall take the value "LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID".                                                                                                                                                                                                                                                                                                |
| localOrigin          | LocalOrigin               | M | 1           | Indicates the local origin in the local Cartesian co-ordinates system configured by the PLMN operator.                                                                                                                                                                                                                                                         |
| point                | RelativeCartesianLocation | M | 1           | Indicates a 3D-point (specified by "x", "y" and "z" coordinates) relative to origin in reference system.                                                                                                                                                                                                                                                       |
| uncertaintyEllipsoid | UncertaintyEllipsoid      | M | 1           | Indicates the uncertainty ellipsoid in relation to the local Cartesian coordinate system defined by the localOrigin, where the orientationMajor angle is clockwise from the "y" axis.                                                                                                                                                                          |
| confidence           | Confidence                | M | 1           | Indicates the value of confidence.                                                                                                                                                                                                                                                                                                                             |
| vConfidence          | Confidence                | C | 0..1        | <p>This IE shall be present when vertical confidence is available, and the value of vertical and horizontal confidence are different.</p> <p>This IE may be present when vertical confidence is available, and the value of vertical and horizontal confidence are the same.</p> <p>When present, this IE shall Indicate the value of vertical confidence.</p> |

## 6.1.6.2.40 Type: UncertaintyEllipsoid

**Table 6.1.6.2.40-1: Definition of type UncertaintyEllipsoid**

| Attribute name   | Data type   | P | Cardinality | Description                                                 |
|------------------|-------------|---|-------------|-------------------------------------------------------------|
| semiMajor        | Uncertainty | M | 1           | Indicates the semi-major axis of the uncertainty ellipsoid. |
| semiMinor        | Uncertainty | M | 1           | Indicates the semi-minor axis of the uncertainty ellipsoid. |
| vertical         | Uncertainty | M | 1           | Indicates the vertical axis of the uncertainty ellipsoid.   |
| orientationMajor | Orientation | M | 1           | Indicates the orientation angle of the major axis.          |



## 6.1.6.2.41 Type: LocalArea

**Table 6.1.6.2.41-1: Definition of type LocalArea as a list of mutually exclusive alternatives**

| Data type                        | Cardinality | Discriminator property name | Discriminator mapping                | Description                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------|-------------|-----------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local2dPointUncertaintyEllipse   | 1           | shape                       | LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE   | Local area consisting of a point described in 2D local coordinates relative to an origin in a reference system, plus an uncertainty ellipse and a confidence value.                                                                                                                                                                           |
| Local3dPointUncertaintyEllipsoid | 1           | shape                       | LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID | Local area consisting of a point described in 3D local coordinates relative to an origin in a reference system, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis") and horizontal/vertical confidence value(s). |

## 6.1.6.2.42 Type: UeAreaIndication

**Table 6.1.6.2.42-1: Definition of type UeAreaIndication**

| Attribute name                                                 | Data type | P | Cardinality | Description                                                                                                                                                                                                                      |
|----------------------------------------------------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| country                                                        | string    | C | 0..1        | This IE shall be present if the country or the area of country where the UE is located is determined. When present, this IE contains two-letter ISO 3166 country code in capital ASCII letters, e.g., DE or US IETF RFC 4776 [6] |
| internationalAreaInd                                           | boolean   | C | 0..1        | Indicates international area<br><br>When present, it shall be set as follows:<br>- true: UE is located in international area.<br><br>- false (default): UE is not located in international area.                                 |
| NOTE: Either country or internationalAreaInd shall be present. |           |   |             |                                                                                                                                                                                                                                  |

## 6.1.6.2.43 Type: MinorLocationQoS

**Table 6.1.6.2.43-1: Definition of type MinorLocationQoS**

| Attribute name | Data type | P | Cardinality | Description         |
|----------------|-----------|---|-------------|---------------------|
| hAccuracy      | Accuracy  | O | 0..1        | Horizontal accuracy |
| vAccuracy      | Accuracy  | O | 0..1        | Vertical accuracy   |

## 6.1.6.2.44 Type: MbsrInfo

**Table 6.1.6.2.44-1: Definition of type MbsrInfo**

| Attribute name | Data type | P | Cardinality | Description          | Applicability |
|----------------|-----------|---|-------------|----------------------|---------------|
| ecgi           | Ecgi      | O | 0..1        | E-UTRA Cell Identity |               |
| ncgi           | Ncgi      | O | 0..1        | NR Cell Identity     |               |

## 6.1.6.2.45 Type: LocMeasurementReq

Table 6.1.6.2.45-1: Definition of type LocMeasurementReq

| Attribute name                                                                                    | Data type           | P | Cardinality | Description                                                                                         |
|---------------------------------------------------------------------------------------------------|---------------------|---|-------------|-----------------------------------------------------------------------------------------------------|
| ecgi                                                                                              | Ecgi                | O | 0..1        | E-UTRA Cell Identity (NOTE 1)                                                                       |
| ncgi                                                                                              | Ncgi                | O | 0..1        | NR Cell Identity (NOTE 1)                                                                           |
| preCalcuLocEstimate                                                                               | GeographicArea      | O | 0..1        | Pre-calculated location estimate of target UE                                                       |
| timestampOfPreCalcuLocEstimate                                                                    | DateTime            | O | 0..1        | Timestamp (in UTC) of pre-calculated location estimate of target UE                                 |
| timeWindows                                                                                       | array(TimeWindow)   | C | 1..N        | Time windows for scheduling of PRU measurements when UE assisted positioning is used. (NOTE 2)      |
| timeWindowsNrppa                                                                                  | TimeWindowsNrppa    | C | 0..1        | Time windows for scheduling of PRU measurements when network assisted positioning is used. (NOTE 2) |
| method                                                                                            | PositioningMethod   | O | 0..1        | Indicates the related positioning method                                                            |
| trpParameters                                                                                     | array(TrpParameter) | O | 1..N        | The identifiers of TRPs if the network assisted positioning is used                                 |
| NOTE 1: Either the "ecgi" attribute or the "ncgi" attribute shall be included.                    |                     |   |             |                                                                                                     |
| NOTE 2: Either the "timeWindows" attribute or the "timeWindowsNrppa" attribute shall be included. |                     |   |             |                                                                                                     |

## 6.1.6.2.46 Type: LocMeasurementResp

Table 6.1.6.2.46-1: Definition of type LocMeasurementResp

| Attribute name  | Data type              | P | Cardinality | Description                                    |
|-----------------|------------------------|---|-------------|------------------------------------------------|
| locMeasurements | array(LocMeasurements) | M | 1..N        | Location measurements and estimates of PRU(s). |

## 6.1.6.2.47 Type: LocMeasurements

Table 6.1.6.2.47-1: Definition of type LocMeasurements

| Attribute name | Data type | P | Cardinality | Description                                                                                                                                                                                      |
|----------------|-----------|---|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| locInfo        | Bytes     | M | 1           | This attribute shall indicate the PRU location measurements and associated PRU known location, and encode the "ProvideLocationInformation" IE as specified in clause 6.3 of 3GPP TS 37.355 [21]. |

## 6.1.6.2.48 Type: HighAccuracyGnssMetrics

**Table 6.1.6.2.48-1: Definition of type HighAccuracyGnssMetrics**

| Attribute name     | Data type | P | Cardinality | Description                                                                                                                                                           |
|--------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nrOfUsedSatellites | integer   | M | 1           | Indicates the number of satellites used for the high accuracy GNSS positioning method.<br><br>Minimum: 0. Maximum: 64.                                                |
| hdopi              | Integer   | O | 0..1        | When present, this IE shall indicate the horizontal dilution of precision for the location estimate, scale factor 0.1.<br><br>Minimum: 1. Maximum: 256.               |
| pdopi              | integer   | O | 0..1        | When present, this IE shall indicate the 3D position dilution of precision, scale factor 0.1.<br><br>Minimum: 1. Maximum: 256.                                        |
| age                | integer   | O | 0..1        | When present, this IE shall indicate the age of the most recent used assistance data for high accuracy GNSS, scale factor 0.1 second.<br><br>Minimum: 0. Maximum: 99. |
| fixType            | FixType   | O | 0..1        | When present, this IE shall indicate the positioning fix type.                                                                                                        |

## 6.1.6.2.49 Type: UpNotifyData

**Table 6.1.6.2.49-1: Definition of type UpNotifyData**

| Attribute name     | Data type          | P | Cardinality | Description                                                                                                                               |
|--------------------|--------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| notifCorrelationId | CorrelationID      | M | 1           | Notification correlation ID.                                                                                                              |
| upConnectionStatus | UpConnectionStatus | M | 1           | UP Connection Status                                                                                                                      |
| targetLMFId        | LMFIdentification  | O | 0..1        | This IE may be present if the UpConnectionStatus is set to "MOVE".<br><br>When present, this IE shall indicate the Target LMF identifier. |

## 6.1.6.2.50 Type: UpSubscription

**Table 6.1.6.2.50-1: Definition of type UpSubscription**

| Attribute name      | Data type     | P | Cardinality | Description                             |
|---------------------|---------------|---|-------------|-----------------------------------------|
| upNotifyCallBackUri | Uri           | M | 1           | Callback URI of the NF Service Consumer |
| notifCorrelationId  | CorrelationID | M | 1           | Notification correlation ID.            |
| supi                | Supi          | M | 1           | SUPI                                    |
| gpsi                | Gpsi          | C | 0..1        | GPSI shall be included if available.    |

## 6.1.6.2.51 Type: RelatedUe

**Table 6.1.6.2.51-1: Definition of type RelatedUe**

| Attribute name     | Data type          | P | Cardinality | Description                                                                                             |
|--------------------|--------------------|---|-------------|---------------------------------------------------------------------------------------------------------|
| applicationlayerId | ApplicationlayerId | M | 1           | The application layer identification of the UE                                                          |
| relatedUeType      | RelatedUeType      | M | 1           | The type of the related UE for ranging and sidelink positioning, such as located UE, reference UE, etc. |

## 6.1.6.2.52 Type: UpConfig

Table 6.1.6.2.52-1: Definition of type UpConfig

| Attribute name                                                                                                       | Data type                           | P | Cardinality | Description                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| upNotifyCallbackUri                                                                                                  | Uri                                 | M | 1           | Callback URI of the NF Service Consumer                                                                                                                             |
| notifCorrelationId                                                                                                   | CorrelationID                       | M | 1           | Notification correlation ID.                                                                                                                                        |
| supi                                                                                                                 | Supi                                | C | 0..1        | It shall be present if available.<br>When present, it indicates the SUPI of the target UE. (NOTE)                                                                   |
| gpsi                                                                                                                 | Gpsi                                | O | 0..1        | Indicates the GPSI of the target UE. (NOTE)                                                                                                                         |
| amfReallocationInd                                                                                                   | boolean                             | O | 0..1        | Indicates AMF reallocation indication.<br><br>When present, it shall be set as follows:<br>- true: AMF reallocated.<br>- false (default): AMF not reallocated.      |
| lcsUpConnectionInd                                                                                                   | LcsUpConnectionInd                  | O | 0..1        | LCS-UP connection indication                                                                                                                                        |
| targetLMFId                                                                                                          | LMFIdentification                   | O | 0..1        | Target LMF identifier.                                                                                                                                              |
| ueUpPosCaps                                                                                                          | array(UeUpPositioning Capabilities) | O | 1..N        | This IE shall be present if available. It shall indicate the user plane positioning capabilities supported by the UE.<br>See clause 5.5.1.3 of 3GPP TS 24.501 [22]. |
| NOTE: An AMF complying with this version of the specification shall provide the supi IE and may provide the gpsi IE. |                                     |   |             |                                                                                                                                                                     |

## 6.1.6.2.53 Type: RangeDirection

Table 6.1.6.2.53-1: Definition of type RangeDirection

| Attribute name     | Data type | P | Cardinality | Description                                                                                | Applicability |
|--------------------|-----------|---|-------------|--------------------------------------------------------------------------------------------|---------------|
| distance           | number    | O | 0..1        | Identifies the straight line distance (range) expressed in metres from point A to point B. |               |
| azimuthDirection   | Angle     | O | 0..1        | Identifies the azimuth direction from point A to point B.                                  |               |
| elevationDirection | Angle     | O | 0..1        | Identifies the elevation direction from point A to point B.                                |               |

## 6.1.6.2.54 Type: 2DRelativeLocation

Table 6.1.6.2.54-1: Definition of type 2DRelativeLocation

| Attribute name   | Data type   | P | Cardinality | Description                                               | Applicability |
|------------------|-------------|---|-------------|-----------------------------------------------------------|---------------|
| semiMinor        | Uncertainty | M | 1           | Indicates the semi-minor axis of the uncertainty ellipse. |               |
| semiMajor        | Uncertainty | M | 1           | Indicates the semi-major axis of the uncertainty ellipse. |               |
| orientationAngle | Angle       | M | 1           | Identifies the angle of orientation A.                    |               |

## 6.1.6.2.55 Type: 3DRelativeLocation

**Table 6.1.6.2.55-1: Definition of type 3DRelativeLocation**

| Attribute name      | Data type   | P | Cardinality | Description                                               | Applicability |
|---------------------|-------------|---|-------------|-----------------------------------------------------------|---------------|
| semiMinor           | Uncertainty | M | 1           | Indicates the semi-minor axis of the uncertainty ellipse. |               |
| semiMajor           | Uncertainty | M | 1           | Indicates the semi-major axis of the uncertainty ellipse. |               |
| verticalUncertainty | Uncertainty | M | 1           | Indicates the vertical uncertainty.                       |               |
| orientationAngle    | Angle       | M | 1           | Identifies the angle of orientation A.                    |               |

## 6.1.6.2.56 Type: AddLocationDatas

**Table 6.1.6.2.56-1: Definition of type AddLocationDatas**

| Attribute name   | Data type           | P | Cardinality | Description                        | Applicability |
|------------------|---------------------|---|-------------|------------------------------------|---------------|
| addLocationDatas | array(LocationData) | O | 1..N        | Contains one or more LocationData. |               |

## 6.1.6.2.57 Type: AddEventNotifyDatas

**Table 6.1.6.2.57-1: Definition of type AddEventNotifyDatas**

| Attribute name      | Data type              | P | Cardinality | Description                           | Applicability |
|---------------------|------------------------|---|-------------|---------------------------------------|---------------|
| addEventNotifyDatas | array(EventNotifyData) | O | 1..N        | Contains one or more EventNotifyData. |               |

## 6.1.6.2.58 Type: LocationDataExt

**Table 6.1.6.2.58-1: Definition of type LocationDataExt as a list of data types to be combined**

| Data type        | Cardinality | Description              | Applicability |
|------------------|-------------|--------------------------|---------------|
| LocationData     | 1           | Location Data            |               |
| AddLocationDatas | 1           | Additional Location Data |               |

## 6.1.6.2.59 Type: EventNotifyDataExt

**Table 6.1.6.2.59-1: Definition of type EventNotifyDataExt as a list of data types to be combined**

| Data type           | Cardinality | Description                            | Applicability |
|---------------------|-------------|----------------------------------------|---------------|
| EventNotifyData     | 1           | Positioning event notify               |               |
| AddEventNotifyDatas | 1           | Additional positioning event(s) notify |               |

## 6.1.6.2.60 Type: MappedLocationQoSEps

**Table 6.1.6.2.60-1: Definition of type MappedLocationQoSEps**

| Attribute name | Data type | P | Cardinality | Description                        | Applicability |
|----------------|-----------|---|-------------|------------------------------------|---------------|
| hAccuracy      | Accuracy  | M | 1           | Mapped Horizontal accuracy for EPS |               |
| vAccuracy      | Accuracy  | O | 0..1        | Mapped Vertical accuracy for EPS   |               |

## 6.1.6.2.61 Type: AdditionalUeInfo

Table 6.1.6.2.61-1: Definition of type AdditionalUeInfo

| Attribute name | Data type | P | Cardinality | Description                                                                                                                                                                                                                | Applicability |
|----------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| ncgi           | Ncgi      | O | 0..1        | NR Cell Identity of the serving cell of the MBSR UE (i.e., IAB UE) or NR Cell Identity of the serving cell of the MWAB UE reflecting the location of the MWAB that serves the UE. See clause 5.49.4 of 3GPP TS 23.501 [2]) |               |
| tai            | Tai       | O | 0..1        | Tracking Area Identity reflecting the location of the MWAB that serves the UE. See clause 5.49.4 of 3GPP TS 23.501 [2])                                                                                                    |               |

## 6.1.6.2.62 Type: AddLocationContext

Table 6.1.6.2.62-1: Definition of type AddLocationContext

| Attribute name                                                                                                                | Data type            | P | Cardinality | Description                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------|----------------------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| locationQoS                                                                                                                   | LocationQoS          | C | 0..1        | This IE shall contain the location QoS if available.                                                                                                                                                                                                             |
| ldrType                                                                                                                       | LdrType              | M | 1           | The type of LDR                                                                                                                                                                                                                                                  |
| hgmIcCallbackURI                                                                                                              | Uri                  | M | 1           | Callback URI of the H-GMLC                                                                                                                                                                                                                                       |
| ldrReference                                                                                                                  | LdrReference         | M | 1           | LDR Reference                                                                                                                                                                                                                                                    |
| periodicEventInfo                                                                                                             | PeriodicEventInfo    | C | 0..1        | Information for periodic event reporting. (NOTE 1)                                                                                                                                                                                                               |
| areaEventInfo                                                                                                                 | AreaEventInfo        | C | 0..1        | Information for area event reporting. (NOTE 1)                                                                                                                                                                                                                   |
| motionEventInfo                                                                                                               | MotionEventInfo      | C | 0..1        | Information for motion event reporting. (NOTE 1)                                                                                                                                                                                                                 |
| eventReportMessage                                                                                                            | EventReportMessage   | C | 1           | This IE shall be present when the event report message from the UE is available for the location context being transferred to the target LMF.<br><br>When present, this IE shall contain the event report message received from the UE for the location context. |
| eventReportingStatus                                                                                                          | EventReportingStatus | O | 0..1        | Status of event reporting                                                                                                                                                                                                                                        |
| ueLocationInfo                                                                                                                | UELocationInfo       | O | 0..1        | Location information for the target UE                                                                                                                                                                                                                           |
| scheduledLocTime                                                                                                              | DateTime             | O | 0..1        | When present, this IE shall contain the scheduled time (in UTC) that the UE needs to be located.                                                                                                                                                                 |
| indoorOutdoorInd                                                                                                              | IndoorOutdoorInd     | O | 0..1        | When present, this IE shall indicate whether the UE is indoor or outdoor.                                                                                                                                                                                        |
| losNlosMeasureInd                                                                                                             | LosNlosMeasureInd    | O | 0..1        | When present, this IE shall indicate whether LOS measurement or NLOS measurement is used.                                                                                                                                                                        |
| upCumEvtRptCriteria                                                                                                           | UpCumEvtRptCriteria  | O | 0..1        | When present, this IE shall include the cumulative event report timer or the maximum number of location reporting over user plane or both.                                                                                                                       |
| lcsUppExistInd                                                                                                                | boolean              | C | 0..1        | This IE shall be present with the value true, if the location context being transferred is associated with the LCS-UPP connection between the UE and the source LMF.<br><br>Presence of this IE with the value false shall be prohibited.                        |
| NOTE 1: At least one of periodicEventInfo, areaEventInfo or motionEventInfo shall be present in the LocContextList structure. |                      |   |             |                                                                                                                                                                                                                                                                  |

## 6.1.6.2.63 Type: TimeWindowsNrppa

**Table 6.1.6.2.63-1: Definition of type TimeWindowsNrppa**

| Attribute name                                                              | Data type                     | P | Cardinality | Description                                                   | Applicability |
|-----------------------------------------------------------------------------|-------------------------------|---|-------------|---------------------------------------------------------------|---------------|
| measurementList                                                             | TimeWindowInfoMeasurementList | O | 0..1        | Contains the Time Window Information Measurement List. (NOTE) |               |
| srsList                                                                     | TimeWindowInfoSrsList         | O | 0..1        | Contains the Time Window Information SRS List. (NOTE)         |               |
| NOTE: At least one of attributes measurementList, srsList shall be present. |                               |   |             |                                                               |               |

## 6.1.6.2.64 Type: RelativeVelocityWithUncertainty

**Table 6.1.6.2.64-1: Definition of type RelativeVelocityWithUncertainty**

| Attribute name                                                                                                                                                                                                                                                                                        | Data type       | P | Cardinality | Description                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| rVelocity                                                                                                                                                                                                                                                                                             | RadialVelocity  | O | 0..1        | Rate of change of a range between the device A and device B, measured in metres per second or centimeters per second.                |
| aTransverseVelocity                                                                                                                                                                                                                                                                                   | AngularVelocity | O | 0..1        | Rate of change of an angle of azimuth to the device B from the device A, measured in degrees per second or 0.1 degrees per second.   |
| eTransverseVelocity                                                                                                                                                                                                                                                                                   | AngularVelocity | O | 0..1        | Rate of change of an angle of elevation to the device B from the device A, measured in degrees per second or 0.1 degrees per second. |
| NOTE: The rate of change of the range (rVelocity), the rate of change of angles of azimuth (aTransverseVelocity) and the rate of change of elevation (eTransverseVelocity) for relative velocity can be each independently included or excluded in the relative velocity and each has an uncertainty. |                 |   |             |                                                                                                                                      |

## 6.1.6.2.65 Type: RadialVelocity

**Table 6.1.6.2.65-1: Definition of type RadialVelocity**

| Attribute name       | Data type                 | P | Cardinality | Description                                                                                                       |
|----------------------|---------------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------|
| unitsRadialVelocity  | UnitsLinearVelocity       | M | 1           | Units of linear velocity, which can be metres per second or centimeters per second.                               |
| radialVelocity       | RadialVelocityValue       | M | 1           | Value for rate of change of a range between the device A and device B in unit determined via unitsRadialVelocity. |
| rVelocityUncertainty | RadialVelocityUncertainty | M | 1           | Value for uncertainty of radial velocity in unit determined via unitsRadialVelocity.                              |

6.1.6.2.66      Type: AngularVelocity

**Table 6.1.6.2.66-1: Definition of type AngularVelocity**

| Attribute name       | Data type                  | P | Cardinality | Description                                                                                                         |
|----------------------|----------------------------|---|-------------|---------------------------------------------------------------------------------------------------------------------|
| unitsAngularVelocity | UnitsAngularVelocity       | M | 1           | Units of angular velocity, which can be degrees per second or 0.1 degrees per second.                               |
| angularVelocity      | AngularVelocityValue       | M | 1           | Value for rate of change of an angle to the device B from the device A in unit determined via unitsAngularVelocity. |
| aVelocityUncertainty | AngularVelocityUncertainty | M | 1           | Uncertainty of the change rate of an angle in unit determined via unitsAngularVelocity.                             |

6.1.6.2.67      Type: TrpParameter

**Table 6.1.6.2.67-1: Definition of type TrpParameter**

| Attribute name | Data type           | P | Cardinality | Description                   | Applicability |
|----------------|---------------------|---|-------------|-------------------------------|---------------|
| trpIds         | array(TrpId)        | O | 1..N        | TRP IDs                       |               |
| globalGnbId    | GlobalRanNode<br>Id | O | 0..1        | Global identity of the gNodeB |               |

6.1.6.3      Simple data types and enumerations

6.1.6.3.1      Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.1.6.3.2      Simple data types

The simple data types defined in table 6.1.6.3.2-1 shall be supported.



Table 6.1.6.3.2-1: Simple data types

| Type Name             | Type Definition | Description                                                                                                                                                                                                                                                                                   |
|-----------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Altitude              | number          | Double-precision float value of the altitude, expressed in meters.<br>Minimum: -32767. Maximum: 32767.<br>Format: double.                                                                                                                                                                     |
| Angle                 | integer         | Integer value of the angle, expressed in degrees.<br>Minimum: 0. Maximum: 360.                                                                                                                                                                                                                |
| Uncertainty           | number          | Float value of uncertainty, expressed in meters.<br>Minimum: 0<br>Format: float.                                                                                                                                                                                                              |
| Orientation           | integer         | Integer value of the orientation angle, expressed in degrees.<br>Minimum: 0. Maximum: 180.                                                                                                                                                                                                    |
| HorizAxesOrientation  | integer         | Integer value of the "y" axis orientation angle clockwise from northing, expressed in 0.1 degrees.<br>Minimum: 0. Maximum: 3600.                                                                                                                                                              |
| Confidence            | integer         | Integer value of the confidence, expressed in percentage value.<br>Minimum: 0. Maximum: 100.                                                                                                                                                                                                  |
| Accuracy              | number          | Float value of accuracy, expressed in meters.<br>Minimum: 0<br>Format: float.                                                                                                                                                                                                                 |
| InnerRadius           | integer         | Integer value of the inner radius, expressed in meters.<br>Minimum: 0. Maximum: 327675.<br>Format: int32.                                                                                                                                                                                     |
| CorrelationID         | string          | LCS Correlation ID. The correlation ID shall be of a minimum length of 1 character and maximum length of 255 characters.                                                                                                                                                                      |
| AgeOfLocationEstimate | integer         | Integer value of the age of the location estimate, expressed in minutes.<br>Minimum: 0. Maximum: 32767.                                                                                                                                                                                       |
| HorizontalSpeed       | number          | Float value of horizontal speed, expressed in kilometres per hour.<br>Minimum: 0. Maximum: 2047.<br>Format: float.                                                                                                                                                                            |
| VerticalSpeed         | number          | Float value of horizontal speed, expressed in kilometres per hour.<br>Minimum: 0. Maximum: 255.<br>Format: float.                                                                                                                                                                             |
| SpeedUncertainty      | number          | Float value of speed uncertainty, expressed in kilometres per hour.<br>Minimum: 0. Maximum: 255.<br>Format: float.                                                                                                                                                                            |
| BarometricPressure    | integer         | This IE specifies the measured uncompensated atmospheric pressure in units of Pascal (Pa).<br>Minimum: 30000. Maximum: 115000.                                                                                                                                                                |
| LcsServiceType        | integer         | The LCS service type as defined in 3GPP TS 22.071 [17] and clause 17.7.8 of 3GPP TS 29.002 [18].<br>Minimum: 0. Maximum: 127.                                                                                                                                                                 |
| LdrReference          | string          | LDR Reference encoded as a string of hexadecimal characters, representing the LDR Reference Number in octets. The LdrReference shall be of a minimum length of 2 characters and maximum length of 510 characters, i.e. representing 1 to 255 octets.<br>pattern: "^([0-9a-fA-F]{2}){1,255}\$" |

|                           |         |                                                                                                                                                                                                                                                                            |
|---------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LirReference              | string  | LIR Reference encoded as a string of hexadecimal characters. The LirReference shall be of a minimum length of 2 characters and maximum length of 510 characters.<br>pattern: "[0-9a-fA-F]{2,510}\$"                                                                        |
| ReportingAmount           | integer | Number of required periodic event reports.<br>Minimum: 1. Maximum: 8639999.                                                                                                                                                                                                |
| ReportingInterval         | integer | Event reporting periodic interval in seconds.<br>Minimum: 1. Maximum: 8639999.<br>ReportingInterval x ReportingAmount shall not exceed 8639999.                                                                                                                            |
| ReportingIntervalMs       | integer | Event reporting periodic interval in milliseconds.<br>Minimum: 1. Maximum: 999.                                                                                                                                                                                            |
| MinimumInterval           | integer | Minimum interval between event reports in seconds.<br>Minimum: 1. Maximum: 32767.                                                                                                                                                                                          |
| MaximumInterval           | integer | Maximum interval between event reports in seconds.<br>Minimum: 1. Maximum: 86400.                                                                                                                                                                                          |
| SamplingInterval          | integer | Maximum time interval between consecutive evaluations by a UE of a trigger event, in seconds.<br>Minimum: 1. Maximum: 3600                                                                                                                                                 |
| ReportingDuration         | integer | Maximum duration of event reporting, in seconds.<br>Minimum: 1. Maximum: 8640000.                                                                                                                                                                                          |
| LinearDistance            | integer | The minimum straight line distance moved by a UE to trigger a motion event report, in meters.<br>Minimum: 1. Maximum: 10000.                                                                                                                                               |
| LMFIdentification         | string  | The serving LMF identification as defined in 3GPP TS 23.273 [19], encoded as a string of hexadecimal characters.                                                                                                                                                           |
| EventReportCounter        | integer | Number of event reports received from the target UE.<br>Minimum: 1. Maximum: 8640000.<br>Note: the current event report is included in the count.                                                                                                                          |
| EventReportDuration       | integer | Duration of event reporting, in seconds.<br>Minimum: 0. Maximum: 8640000.<br>Note: the duration starts when event reporting is activated in the UE and extends to the current time.                                                                                        |
| UePositioningCapabilities | Bytes   | Positioning capabilities supported by the UE.<br><br>String with format "byte" as defined in OpenAPI Initiative [14], i.e. base64-encoded characters, encoding the "ProvideCapabilities-r9-IEs" IE as specified in clause 6.3 of 3GPP TS 37.355 [21] (start from octet 1). |

|                               |         |                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TimeWindow                    | Bytes   | Time window of the target UE when UE assisted positioning is used.<br><br>String with format "byte" as defined in OpenAPI Initiative [14], i.e. base64-encoded characters, encoding the "NR-DL-PRS-MeasurementTimeWindowsConfig-r18" IE as specified in clause 6.4.3 of 3GPP TS 37.355 [21] (start from octet 1).                              |
| TimeWindowInfoMeasurementList | Bytes   | Contains the Time Window Information Measurement List when network assisted positioning is used.<br><br>String with format "byte" as defined in OpenAPI Initiative [14], i.e. base64-encoded characters, encoding the "Time Window Information Measurement List" IE as specified in clause 9.2.91 of 3GPP TS 38.455 [31] (start from octet 1). |
| TimeWindowInfoSrsList         | Bytes   | Contains the Time Window Information SRS List when network assisted positioning is used.<br><br>String with format "byte" as defined in OpenAPI Initiative [14], i.e. base64-encoded characters, encoding the "Time Window Information SRS List" IE as specified in clause 9.2.90 of 3GPP TS 38.455 [31] (start from octet 1).                 |
| RangingSICapability           | Bytes   | Ranging/Sidelink Positioning Capability supported by the UE.<br><br>String with format "byte" as defined in OpenAPI Initiative [14], i.e. base64-encoded characters, encoding the "ProvideCapabilities" IE as specified in clause 6.2.2 of 3GPP TS 38.355 [30] (start from octet 1).                                                           |
| RadialVelocityValue           | integer | Value of rate of change of a range between the device A and device B. Minimum: -2048. Maximum: 2047.                                                                                                                                                                                                                                           |
| AngularVelocityValue          | integer | Value of rate of change of an angle. Minimum: -1024. Maximum: 1023.                                                                                                                                                                                                                                                                            |
| AngularVelocityUncertainty    | integer | Value of uncertainty for rate of change of an angle. Minimum: 0. Maximum: 255.                                                                                                                                                                                                                                                                 |
| RadialVelocityUncertainty     | integer | Value of uncertainty for rate of change of an linear velocity. Minimum: 0. Maximum: 255.                                                                                                                                                                                                                                                       |
| TrpId                         | integer | Identifiers of TRP. Minimum: 1. Maximum: 65535.                                                                                                                                                                                                                                                                                                |

#### 6.1.6.3.3 Enumeration: ExternalClientType

The enumeration ExternalClientType represents the different types of clients of the location service.

**Table 6.1.6.3.3-1: Enumeration ExternalClientType**

| Enumeration value                         | Description                                                 |
|-------------------------------------------|-------------------------------------------------------------|
| "EMERGENCY_SERVICES"                      | External client for emergency services                      |
| "VALUE_ADDED_SERVICES"                    | External client for value added services                    |
| "PLMN_OPERATOR_SERVICES"                  | External client for PLMN operator services                  |
| "LAWFUL_INTERCEPT_SERVICES"               | External client for Lawful Intercept services               |
| "PLMN_OPERATOR_BROADCAST_SERVICES"        | External client for PLMN Operator Broadcast services        |
| "PLMN_OPERATOR_OM"                        | External client for PLMN Operator O&M                       |
| "PLMN_OPERATOR_ANONYMOUS_STATISTICS"      | External client for PLMN Operator anonymous statistics      |
| "PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT" | External client for PLMN Operator target MS service support |
| "SL_POS"                                  | External client for ranging and sidelink positioning        |

#### 6.1.6.3.4 Enumeration: SupportedGADShapes

The enumeration SupportedGADShapes represents the different types, or shapes, of geographic areas supported by the system.

**Table 6.1.6.3.4-1: Enumeration SupportedGADShapes**

| Enumeration value                            | Description                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "POINT"                                      | Ellipsoid Point                                                                                                                                                                                                                                                                                                                                  |
| "POINT_UNCERTAINTY_CIRCLE"                   | Ellipsoid point with uncertainty circle                                                                                                                                                                                                                                                                                                          |
| "POINT_UNCERTAINTY_ELLIPSE"                  | Ellipsoid point with uncertainty ellipse                                                                                                                                                                                                                                                                                                         |
| "POLYGON"                                    | Polygon                                                                                                                                                                                                                                                                                                                                          |
| "POINT_ALTITUDE"                             | Ellipsoid point with altitude                                                                                                                                                                                                                                                                                                                    |
| "POINT_ALTITUDE_UNCERTAINTY"                 | Ellipsoid point with altitude and uncertainty ellipsoid                                                                                                                                                                                                                                                                                          |
| "ELLIPSOID_ARC"                              | Ellipsoid Arc                                                                                                                                                                                                                                                                                                                                    |
| "LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE"         | Local 2D point with uncertainty ellipse                                                                                                                                                                                                                                                                                                          |
| "LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID"       | Local 3D point with uncertainty ellipsoid                                                                                                                                                                                                                                                                                                        |
| "DISTANCE_DIRECTION"                         | Distance and direction from a point A to a point B, comprising a distance from point A to point B, an azimuth direction from point A to point B and an elevation direction from point A to point B                                                                                                                                               |
| "RELATIVE_2D_LOCATION_UNCERTAINTY_ELLIPSE"   | Relative 2D location with uncertainty ellipse, characterised by a point described in 2D local co-ordinates with origin corresponding to another known point, distances r1 and r2 and an angle of orientation A.                                                                                                                                  |
| "RELATIVE_3D_LOCATION_UNCERTAINTY_ELLIPSOID" | Relative 3D location with uncertainty ellipsoid, characterised by a point described in 3D local co-ordinates with origin corresponding to another known point, distances r1 (the "semi-major uncertainty"), r2 (the "semi-minor uncertainty") and r3 (the "vertical uncertainty") and an angle of orientation A (the "angle of the major axis"). |

#### 6.1.6.3.5 Enumeration: ResponseTime

The enumeration ResponseTime represents the acceptable delay in the determination of the location of the UE.

**Table 6.1.6.3.5-1: Enumeration ResponseTime**

| Enumeration value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| "LOW_DELAY"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Location request is expected with low delay level.   |
| "DELAY_TOLERANT"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Location request is delay tolerant.                  |
| "NO_DELAY "                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Location request is expected with no delay<br>(NOTE) |
| NOTE: The value is only used in the interface between GMLC and AF/LCS client via NEF, not further delivered to other NFs in the network. After receiving the enumeration value, the GMLC shall immediately return any location estimate, local location or civic location that it currently has. The GMLC shall return either the Initial or Last Known Location of the Target UE. If no location estimate or Dispatchable Location is available, the GMLC shall return the failure indication and may optionally initiate procedures to obtain a location estimate or Dispatchable Location (e.g. to be available for a later request). |                                                      |

### 6.1.6.3.6 Enumeration: PositioningMethod

The enumeration PositioningMethod represents the method used to determine the location of the UE.

**Table 6.1.6.3.6-1: Enumeration PositioningMethod**

| Enumeration value     | Description                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| "CELLID"              | Cell ID positioning method                                                                                                        |
| "ECID"                | Enhanced cell ID methods based on LTE signals                                                                                     |
| "OTDOA"               | Observed time difference of arrival positioning based on LTE signals                                                              |
| "BAROMETRIC_PRESSURE" | Positioning method based on barometric Pressure Sensor                                                                            |
| "WLAN"                | WLAN positioning                                                                                                                  |
| "BLUETOOTH"           | Bluetooth positioning                                                                                                             |
| "MBS"                 | Terrestrial Beacon System (TBS) positioning based on MBS signals                                                                  |
| "MOTION_SENSOR"       | Positioning method based on motion Sensor                                                                                         |
| "DL_TDOA"             | Downlink Time Difference of Arrival (DL-TDOA) based on NR signals                                                                 |
| "DL_AOD"              | Downlink Angle-of-Departure (DL-AoD) based on NR signals                                                                          |
| "MULTI-RTT"           | Multi-Round Trip Time Positioning (Multi-RTT based on NR signals).                                                                |
| "NR_ECID"             | NR enhanced cell ID methods (NR E-CID) based on NR signals.                                                                       |
| "UL_TDOA"             | Uplink Time Difference of Arrival (UL-TDOA) based on NR signals                                                                   |
| "UL_AOA"              | Uplink Angle of Arrival (UL-AoA), including the Azimuth of Arrival (A-AoA) and the Zenith of Arrival (Z-AoA) based on NR signals. |
| "NETWORK_SPECIFIC"    | Network specific position methods.                                                                                                |
| "SL_TDOA"             | Sidelink Time Difference of Arrival (TDOA) based on Sidelink NR PC5 radio signals                                                 |
| "SL_TOA"              | Sidelink Time Of Arrival based on NR PC5 radio signals                                                                            |
| "SL_AoA"              | Sidelink Angle-of-Arrival based on NR PC5 radio signals                                                                           |
| "SL_RT"               | Sidelink Round Trip based on NR PC5 radio signals                                                                                 |
| "AIML"                | The positioning method using a trained ML Model, as specified in clause 4.2b of 3GPP TS 23.273 [19].                              |

NOTE: Some of the values in the enumeration deviate from the naming conventions indicated in clause 5.1.4 of 3GPP TS 29.501 [5] (i.e. to use UPPER\_WITH\_UNDERSCORE); however, it is kept as currently defined in this specification to maintain backwards compatibility

#### 6.1.6.3.7 Enumeration: PositioningMode

The enumeration PositioningMode represents the mode used to determine the location of the UE when a certain positioning method is used.

**Table 6.1.6.3.7-1: Enumeration PositioningMode**

| Enumeration value | Description       |
|-------------------|-------------------|
| "UE_BASED"        | UE-based mode     |
| "UE_ASSISTED"     | UE-assisted mode  |
| "CONVENTIONAL"    | Conventional mode |

#### 6.1.6.3.8 Enumeration: GnssId

The enumeration GnssId represents the different GNSS systems.

**Table 6.1.6.3.8-1: Enumeration GnssId**

| Enumeration value | Description                          |
|-------------------|--------------------------------------|
| "GPS"             | GPS                                  |
| "GALILEO"         | Galileo                              |
| "SBAS"            | Space Based Augmentation Systems     |
| "MODERNIZED_GPS"  | Modernized GPS                       |
| "QZSS"            | Quasi Zenith Satellite System        |
| "GLONASS"         | Global Navigation Satellite System   |
| "BDS"             | BeiDou Navigation Satellite System   |
| "NAVIC"           | Navigation with Indian Constellation |

#### 6.1.6.3.9 Enumeration: Usage

The enumeration Usage represents the type of usage made of the location measurement from the UE.

**Table 6.1.6.3.9-1: Enumeration Usage**

| Enumeration value                           | Description                                              |
|---------------------------------------------|----------------------------------------------------------|
| "UNSUCCESS"                                 | Not successful                                           |
| "SUCCESS_RESULTS_NOT_USED"                  | Successful result not used                               |
| "SUCCESS_RESULTS_USED_TO_VERIFY_LOCATION"   | Successful result used to verify the location estimate   |
| "SUCCESS_RESULTS_USED_TO_GENERATE_LOCATION" | Successful result used to generate the location estimate |
| "SUCCESS_METHOD_NOT_DETERMINED"             | Successful method not determined                         |

#### 6.1.6.3.10 Enumeration: LcsPriority

The enumeration LcsPriority represents the priority of the LCS client.

**Table 6.1.6.3.10-1: Enumeration LcsPriority**

| Enumeration value  | Description                      |
|--------------------|----------------------------------|
| "HIGHEST_PRIORITY" | LCS client with highest priority |
| "NORMAL_PRIORITY"  | LCS client with normal priority  |

#### 6.1.6.3.11 Enumeration: VelocityRequested

The enumeration VelocityRequested represents the indication of velocity requirement.

**Table 6.1.6.3.11-1: Enumeration VelocityRequested**

| Enumeration value           | Description                       |
|-----------------------------|-----------------------------------|
| "VELOCITY_IS_NOT_REQUESTED" | velocity estimate is required     |
| "VELOCITY_IS_REQUESTED"     | velocity estimate is not required |

#### 6.1.6.3.12 Enumeration: AccuracyFulfilmentIndicator

The enumeration AccuracyFulfilmentIndicator represents whether the requested accuracy was fulfilled or not.

**Table 6.1.6.3.12-1: Enumeration AccuracyFulfilmentIndicator**

| Enumeration value                  | Description                         |
|------------------------------------|-------------------------------------|
| "REQUESTED_ACCURACY_FULFILLED"     | requested accuracy is fulfilled     |
| "REQUESTED_ACCURACY_NOT_FULFILLED" | requested accuracy is not fulfilled |

#### 6.1.6.3.13 Enumeration: VerticalDirection

The enumeration VerticalDirection represents the direction (upward/downward) of the vertical speed.

**Table 6.1.6.3.13-1: Enumeration VerticalDirection**

| Enumeration value | Description                |
|-------------------|----------------------------|
| "UPWARD"          | Vertical speed is upward   |
| "DOWNWARD"        | Vertical speed is downward |

#### 6.1.6.3.14 Enumeration: LdrType

**Table 6.1.6.3.14-1: Enumeration LdrType**

| Enumeration value    | Description             |
|----------------------|-------------------------|
| "UE_AVAILABLE"       | UE available event      |
| "PERIODIC"           | Periodic event          |
| "ENTERING INTO AREA" | Entering area event     |
| "LEAVING FROM AREA"  | Leaving area event      |
| "BEING INSIDE AREA"  | Being inside area event |
| "MOTION"             | Motion event            |

#### 6.1.6.3.15 Enumeration: ReportingAreaType

The enumeration ReportingAreaType indicates the type of a reporting area.

**Table 6.1.6.3.15-1: Enumeration ReportingAreaType**

| Enumeration value                    | Description |
|--------------------------------------|-------------|
| "EPS_TRACKING_AREA_IDENTITY"         | EPS TAI     |
| "E-UTRAN_CELL_GLOBAL_IDENTIFICATION" | ECGI        |
| "5GS_TRACKING_AREA_IDENTITY"         | 5GS TAI     |
| "NR_CELL_GLOBAL_IDENTITY"            | NCGI        |



NOTE: Some of the values in the enumeration deviate from the naming conventions indicated in clause 5.1.4 of 3GPP TS 29.501 [5] (i.e. to use UPPER\_WITH\_UNDERSCORE); however, it is kept as currently defined in this specification to maintain backwards compatibility

#### 6.1.6.3.16 Enumeration: OccurrenceInfo

The enumeration OccurrenceInfo indicates whether event reporting is one time.

**Table 6.1.6.3.16-1: Enumeration OccurrenceInfo**

| Enumeration value     | Description                         |
|-----------------------|-------------------------------------|
| "ONE_TIME_EVENT"      | Event to be reported one-time only  |
| "MULTIPLE_TIME_EVENT" | Event to be reported multiple times |

#### 6.1.6.3.17 Enumeration: ReportingAccessType

The enumeration ReportingAccessType indicates an allowed access type for event reporting.

**Table 6.1.6.3.17-1: Enumeration ReportingAccessType**

| Enumeration value           | Description                      |
|-----------------------------|----------------------------------|
| "NR"                        | NG Radio access                  |
| "EUTRA_CONNECTED_TO_5GC"    | E-URTAN access connected to 5GC  |
| "NON_3GPP_CONNECTED_TO_5GC" | Non-3GPP access connected to 5GC |
| "NR_LEO"                    | NR (LEO) satellite access        |
| "NR_MEO"                    | NR (MEO) satellite access        |
| "NR_GEO"                    | NR (GEO) satellite access        |
| "NR_OTHER_SAT"              | NR (OTHERSAT) satellite access   |
| "EUTRA_CONNECTED_TO_EPC"    | E-URTAN access connected to EPC  |

#### 6.1.6.3.18 Enumeration: EventClass

**Table 6.1.6.3.18-1: Enumeration EventClass**

| Enumeration value        | Description                                                                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "SUPPLEMENTARY_SERVICES" | A supplementary services message containing an argument for an lcs-EventReport operation as defined in 3GPP TS 24.080 [20].                                                                |
| "DUMMY"                  | A dummy message.<br><br>A dummy message is used when no event report message is available at the sender but the service operation requires an event report message object as mandatory IE. |

## 6.1.6.3.19 Enumeration: ReportedEventType

**Table 6.1.6.3.19-1: Enumeration ReportedEventType**

| Enumeration value                   | Description                                    |
|-------------------------------------|------------------------------------------------|
| "PERIODIC_EVENT"                    | Periodic reporting event                       |
| "ENTERING_AREA_EVENT"               | Entering area reporting event                  |
| "LEAVING_AREA_EVENT"                | Leaving area reporting event                   |
| "BEING_INSIDE_AREA_EVENT"           | Being inside area reporting event              |
| "MOTION_EVENT"                      | Motion reporting event                         |
| "MAXIMUM_INTERVAL_EXPIRATION_EVENT" | Expiration of maximum reporting interval event |
| "LOCATION_CANCELLATION_EVENT"       | Cancellation of location reporting event       |
| "INTERMEDIATE_EVENT"                | Intermediate location reporting event          |
| "DIRECT_REPORT_EVENT"               | Direct location reporting event                |
| "CUMULATIVE_EVENT_REPORT"           | Cumulative event report for events reported    |

## 6.1.6.3.20 Enumeration: TerminationCause

**Table 6.1.6.3.20-1: Enumeration TerminationCause**

| Enumeration value        | Description                           |
|--------------------------|---------------------------------------|
| "TERMINATION_BY_UE"      | Event reporting terminated by UE      |
| "TERMINATION_BY_NETWORK" | Event reporting terminated by Network |
| "NORMAL_TERMINATION"     | Normal Termination                    |

## 6.1.6.3.21 Enumeration: LcsQosClass

**Table 6.1.6.3.21-1: Enumeration LcsQosClass**

| Enumeration value | Description        |
|-------------------|--------------------|
| "BEST_EFFORT"     | Best Effort Class  |
| "ASSURED"         | Assured Class      |
| "MULTIPLE_QOS"    | Multiple QoS Class |

## 6.1.6.3.22 Enumeration: UeLocationServiceInd

**Table 6.1.6.3.22-1: Enumeration UeLocationServiceInd**

| Enumeration value          | Description                      |
|----------------------------|----------------------------------|
| "LOCATION_ESTIMATE"        | Request location estimate        |
| "LOCATION_ASSISTANCE_DATA" | Request location assistance data |

## 6.1.6.3.23 Enumeration: IndoorOutdoorInd

**Table 6.1.6.3.23-1: Enumeration IndoorOutdoorInd**

| Enumeration value | Description |
|-------------------|-------------|
| "INDOOR"          | UE indoor.  |
| "OUTDOOR"         | UE outdoor. |

## 6.1.6.3.24 Enumeration: FixType

**Table 6.1.6.3.24-1: Enumeration FixType**

| Enumeration value     | Description                                                  |
|-----------------------|--------------------------------------------------------------|
| "CARRIER_PHASE_FLOAT" | Converging carrier phase floating point ambiguity resolution |
| "CARRIER_PHASE_FIX"   | Converged carrier phase integer ambiguity resolution         |

## 6.1.6.3.25 Enumeration: LosNlosMeasureInd

**Table 6.1.6.3.25-1: Enumeration LosNlosMeasureInd**

| Enumeration value | Description       |
|-------------------|-------------------|
| "LOS"             | LOS measurement.  |
| "NLOS"            | NLOS measurement. |

## 6.1.6.3.26 Enumeration: UpConnectionStatus

The enumeration UpConnectionStatus indicates the UP Connection Status.

**Table 6.1.6.3.26-1: Enumeration UpConnectionStatus**

| Enumeration value | Description                     |
|-------------------|---------------------------------|
| "ESTABLISHED"     | UP Connection Established       |
| "RELEASED"        | UP Connection Released          |
| "MOVE"            | UP Connection Change Indication |

## 6.1.6.3.27 Enumeration: RangingSIResult

The enumeration RangingSIResult represents the type of result requested for ranging and sidelink positioning.

**Table 6.1.6.3.27-1: Enumeration RangingSIResult**

| Enumeration value   | Description                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "ABSOLUTE_LOCATION" | This value indicates that the absolute location of the target UE is required.                                                                                                          |
| "RELATIVE_LOCATION" | This value indicates that the position of the target UE relative to other UEs is required.                                                                                             |
| "RANGING_DIRECTION" | This value indicates that the straight line distance (range) between two UEs or more UEs and the direction of one UE (i.e. Target UE) from another UE (i.e. Reference UE) is required. |
| "RANGING"           | This value indicates that the distance between two UEs or more UEs is required.                                                                                                        |
| "DIRECTION"         | This value indicates that the direction of one UE (i.e. Target UE) from another UE (i.e. Reference UE) is required.                                                                    |
| "VELOCITY"          | This value indicates that the velocities of the target UE is required.                                                                                                                 |
| "RELATIVE_VELOCITY" | This value indicates that the velocities of the target UE relative to other UEs is required.                                                                                           |

#### 6.1.6.3.28 Enumeration: RelatedUeType

The enumeration RelatedUeType represents the different roles of UE for ranging and sidelink positioning service.

**Table 6.1.6.3.28-1: Enumeration RelatedUeType**

| Enumeration value | Description                                                         |
|-------------------|---------------------------------------------------------------------|
| "LOCATED_UE"      | UE as located UE for the ranging and sidelink positioning service   |
| "REFERENCE_UE"    | UE as reference UE for the ranging and sidelink positioning service |

#### 6.1.6.3.29 Enumeration: LcsUpConnectionInd

The enumeration LcsUpConnectionInd indicates the LCS UP Connection Indication.

**Table 6.1.6.3.29-1: Enumeration LcsUpConnectionInd**

| Enumeration value | Description                                 |
|-------------------|---------------------------------------------|
| "TERMINATION"     | LCS-UP connection termination indication    |
| "SETUP"           | LCS-UP connection set up request indication |

#### 6.1.6.3.30 Enumeration: UeUpPositioningCapabilities

The enumeration UeUpPositioningCapabilities represents the user plane positioning capabilities supported by the UE.

**Table 6.1.6.3.30-1: UeUpPositioningCapabilities**

| Enumeration value  | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| "LCS_UPP"          | Indicates the capability to support LCS-UPP user plane positioning              |
| "SUPL"             | Indicates the capability to support SUPL user plane positioning                 |
| "MULTIPLE_LCS_UPP" | Indicates the capability to support multiple LCS secured user plane connections |

#### 6.1.6.3.31 Enumeration: SIPositioningCapability

The enumeration SIPositioningCapability represents the ranging and sidelink positioning capabilities supported by the UE.

**Table 6.1.6.3.31-1: SIPositioningCapability**

| Enumeration value | Description                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------|
| "V2X"             | Indicates that the UE is supporting Ranging and sidelink positioning with V2X capable UE      |
| "5G_PROSE"        | Indicates that the UE is supporting Ranging and sidelink positioning with 5G ProSe capable UE |

#### 6.1.6.3.32 Enumeration: UnitsLinearVelocity

The enumeration UnitsLinearVelocity represents the units of linear velocity.

**Table 6.1.6.3.32-1: UnitsLinearVelocity**

| Enumeration value | Description                                  |
|-------------------|----------------------------------------------|
| "MPERS"           | Indicates the unit of metres per second      |
| "CMPERS"          | Indicates the unit of centimeters per second |

#### 6.1.6.3.33 Enumeration: UnitsAngularVelocity

The enumeration UnitsAngularVelocity represents the units of angular velocity.

**Table 6.1.6.3.33-1: UnitsAngularVelocity**

| Enumeration value | Description                                  |
|-------------------|----------------------------------------------|
| "DEGPERS1"        | Indicates the unit of degrees per second     |
| "DEGPERS01"       | Indicates the unit of 0.1 degrees per second |

### 6.1.6.4 Binary data

#### 6.1.6.4.1 Introduction

This clause defines the binary data that shall be supported in a binary body part in an HTTP multipart message (see clauses 6.1.2.2.2 and 6.1.2.4).

#### 6.1.6.4.2 LPP Message

LPP Message shall encode a LPP message as specified in 3GPP TS 37.355 [21], using the vnd.3gpp.lpp content-type.

## 6.1.7 Error Handling

### 6.1.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of 3GPP TS 29.500 [4].

### 6.1.7.2 Protocol Errors

Protocol errors handling shall be supported as specified in clause 5.2.7 of 3GPP TS 29.500 [4].

### 6.1.7.3 Application Errors

The common application errors defined in Table 5.2.7.2-1 in 3GPP TS 29.500 [4] may be used for the Nlmf\_Location. The application errors defined for the Nlmf\_Location service are listed in Table 6.1.7.3-1.

**Table 6.1.7.3-1: Application errors**

| Application Error               | HTTP status code          | Description                                                                                                |
|---------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------|
| POSITIONING_DENIED              | 403 Forbidden             | The positioning procedure was denied.                                                                      |
| UNSPECIFIED                     | 403 Forbidden             | The request is rejected due to unspecified reasons.                                                        |
| UNSUPPORTED_BY_UE               | 403 Forbidden             | A request for periodic or triggered location is not supported by the UE.                                   |
| PAGING_NOT_ALLOWED              | 403 Forbidden             | The UE cannot be paged during the UE Unaware Positioning procedure.                                        |
| LOCATION_SESSION_UNKNOWN        | 403 Forbidden             | The location context was not found.                                                                        |
| LOCATION_TRANSFER_NOT_SUPPORTED | 403 Forbidden             | Transfer of a location context is not supported                                                            |
| INSUFFICIENT_RESOURCES          | 403 Forbidden             | Insufficient resources for location context transfer                                                       |
| EVENT_REPORT_UNRECOGNIZED       | 403 Forbidden             | The event report is unrecognized or cannot be parsed.                                                      |
| LOCATION_MEASUREMENT_UNKNOWN    | 403 Forbidden             | The location measurements were not found.                                                                  |
| POSITIONING_FAILED              | 500 Internal Server Error | The positioning procedure failed.                                                                          |
| UNSUPPORTED_EVENT_TYPE          | 501 Not Implemented       | The request for creation of a subscription is rejected because none of the events is supported by the LMF. |
| UNREACHABLE_USER                | 504 Gateway Timeout       | The user could not be reached in order to perform positioning procedure.                                   |

## 6.1.8 Security

As indicated in 3GPP TS 33.501 [9], the access to the Nlmf\_Location API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [10]), using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [11]) plays the role of the authorization server.

If OAuth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Nlmf\_Location API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [11], clause 5.4.2.2.

**NOTE:** When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nlmf\_Location service.

The Nlmf\_Location API defines the following scopes for OAuth2 authorization as specified in 3GPP TS 33.501 [9]:

**Table 6.1.8-1: OAuth2 scopes defined in Nlmf\_Location API**

| Scope                                       | Description                           |
|---------------------------------------------|---------------------------------------|
| "nlmf-loc"                                  | Access to the Nlmf_Location API.      |
| "nlmf-loc:determine-location:invoke"        | Access to invoke Determine Location   |
| "nlmf-loc:cancel-location:invoke"           | Access to invoke Cancel Location      |
| "nlmf-loc:location-context-transfer:invoke" | Access to invoke Location Context     |
| "nlmf-loc:measure-location:invoke"          | Access to invoke Location Measurement |
| "nlmf-loc:up-subscriptions:invoke"          | Access to invoke UP Subscription      |
| "nlmf-loc:configure-up:invoke"              | Access to invoke UP Config            |

### 6.1.9 Feature Negotiation

The features in table 6.1.9-1 are defined for the Nlmf\_Location API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

Table 6.1.9-1: Supported Features

| Feature number | Feature Name | M/O | Description                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|--------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | ES3XX        | M   | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. AMF) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Location service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15.                                   |
| 2              | SAT          | O   | Satellite Access<br><br>Support of this feature implies the LMF shall be able to determine the geographical area identified by a country, area of a country or International area where UE is located for PLMN selection verification.                                                                                                                                                  |
| 3              | MUTIQOS      | O   | Support of Multiple Location QoSes.<br><br>This feature bit indicates the support of more than one Location QoSes during consuming location service.                                                                                                                                                                                                                                    |
| 4              | MBSR         | O   | Support of Mobile Base Station Relay.<br><br>This feature indicates the support of location service capability for MBSR.                                                                                                                                                                                                                                                                |
| 5              | Ranging_SL   | O   | Support of Ranging and Sidelink Positioning<br><br>This feature indicates the support of Ranging and Sidelink Positioning.                                                                                                                                                                                                                                                              |
| 6              | INTRES       | O   | Integrity Result<br><br>Support of this feature implies the LMF shall be able to provide the integrity result for location estimates when integrity requirement was received in the location request.                                                                                                                                                                                   |
| 7              | CIMLR        | O   | Cancel Immediate Location<br><br>This feature indicates the support of cancellation of immediate location i.e. 5G-MT-LR / 5G-MO-LR.                                                                                                                                                                                                                                                     |
| 8              | MCTX         | O   | Support of Multiple LCS contexts transfer.<br><br>This feature is supported if both the source LMF and target LMF are capable of transferring multiple LCS contexts. Otherwise, the Transfer location context procedures are executed multiple times to transfer each LCS context, the target LMF maintains an LCS contexts container where it adds each newly transferred LCS context. |
| 9              | MWAB         | O   | Support of Mobile gNB with Wireless Access Backhauling.<br><br>This feature indicates the support of location service capability for MWAB.                                                                                                                                                                                                                                              |

### 6.1.10 HTTP redirection

An HTTP request may be redirected to a different LMF service instance, within the same LMF or a different LMF of an LMF set, e.g. when an LMF service instance is part of an LMF (service) set or when using indirect communications (see 3GPP TS 29.500 [4]). See also the ES3XX feature in clause 6.1.9.

An SCP that reselects a different LMF producer instance will return the NF Instance ID of the new LMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of 3GPP TS 29.500 [4].

If an LMF within an LMF set redirects a service request to a different LMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new LMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of 3GPP TS 29.500 [4].



## 6.2 Nlmf\_Broadcast Service API

### 6.2.1 API URI

The Nlmf\_Broadcast service shall use the Nlmf\_Broadcast API.

The API URI of the Nlmf\_Broadcast API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "nlmf-broadcast".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.2.3.

### 6.2.2 Usage of HTTP

#### 6.2.2.1 General

HTTP/2, as defined in IETF RFC 9113 [12], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

HTTP messages and bodies for the Nlmf\_Broadcast service shall comply with the OpenAPI [14] specification contained in Annex A.

#### 6.2.2.2 HTTP Standard Headers

##### 6.2.2.2.1 General

##### 6.2.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [13], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of 3GPP TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [15]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

#### 6.2.2.3 HTTP custom headers

##### 6.2.2.3.1 General

The following HTTP custom headers shall be supported:

- 3gpp-Sbi-Message-Priority: See 3GPP TS 29.500 [4], clause 5.2.3.2.2.

This API does not define any new HTTP custom headers.

### 6.2.3 Resources

#### 6.2.3.1 Overview

The structure of the Resource URIs of the Nlmf\_Broadcast service is shown in figure 6.2.3.1-1.

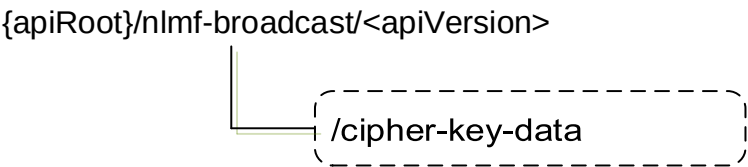


Figure 6.2.3.1-1: Resource URI structure of the Nlmf\_Broadcast API

### 6.2.4 Custom Operations without associated resources

#### 6.2.4.1 Overview

The URI structure for Custom Operations without associated resources is included as part of the Figure 6.2.3.1-1

Table 6.2.4.1-1: Custom operations without associated resources

| Operation Name  | Custom operation URI | Mapped HTTP method | Description        |
|-----------------|----------------------|--------------------|--------------------|
| cipher-key-data | /cipher-key-data     | POST               | Ciphering Key Data |

#### 6.2.4.4 Operation: cipher-key-data

##### 6.2.4.4.1 Description

This clause describes the custom operation and what it is used for.

##### 6.2.4.4.2 Operation Definition

This operation shall support the request and response data structures and response codes specified in table 6.2.4.4.2-1 and table 6.2.4.4.2-2.

Table 6.2.4.4.2-1: Data structures supported by the POST Request Body on this resource

| Data type         | P | Cardinality | Description                                            |
|-------------------|---|-------------|--------------------------------------------------------|
| CipherRequestData | M | 1           | Input parameters to the "Ciphering Key Data" operation |

**Table 6.2.4.4.2-2: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CipherResponseData                                                                                                                                                                                                                                    | M | 1           | 200 OK                 | This case represents a successful request for ciphering key data.<br><br>Upon success, a response body is returned indicating whether the LMF has ciphering key data. The ciphering key data is returned separately in a CipheringKeyData notification. |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                                                                                      |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                                                                                      |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be set to one of the following application errors:<br>- UNSPECIFIED<br>- BROADCAST_CIPHERING_KEYS_NOT_SUPPORTED<br><br>See table 6.2.7.3-1 for the description of this error.                                                 |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                         |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                                                                                         |

**Table 6.2.4.4.2-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

**Table 6.2.4.4.2-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                   |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                     |

## 6.2.5 Notifications

### 6.2.5.1 CipheringKeyData

#### 6.2.5.1.1 Description

The CipheringKeyData operation is used to notify the occurrence of new ciphering key information to a consumer NF (e.g. AMF).

#### 6.2.5.1.2 Notification Definition

Callback URI: {amfCallBackURI}

See clause 5.3.2.2.2 for the description of how the LMF obtains the Callback URI of the NF Service Consumer (i.e. AMF).

### 6.2.5.1.3 Notification Standard Methods

#### 6.2.5.1.3.1 POST

This method sends a ciphering key data notify to the NF Service Consumer.

This method shall support the request and response data structures and response codes specified in table 6.2.5.1.3.1-1 and table 6.2.5.1.3.1-2.

**Table 6.2.5.1.3.1-1: Data structures supported by the POST Request Body**

| Data type        | P | Cardinality | Description                                            |
|------------------|---|-------------|--------------------------------------------------------|
| CipheringKeyInfo | M | 1           | Input parameters to the "Ciphering Key Data" operation |

**Table 6.2.5.1.3.1-2: Data structures supported by the POST Response Body**

| Data type                                                                                                                                                                                                                                             | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CipheringKeyResponse                                                                                                                                                                                                                                  | M | 1           | 200 OK                 | This case represents successful or partially successful storage of ciphering key information by the service consumer NF.<br><br>A response body is returned containing the following parameters:<br>- List of Ciphering Set IDs successfully stored<br>- List of Ciphering Set IDs not successfully stored |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 307 Temporary Redirect | Temporary redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                         |
| RedirectResponse                                                                                                                                                                                                                                      | O | 0..1        | 308 Permanent Redirect | Permanent redirection.<br>(NOTE 2)                                                                                                                                                                                                                                                                         |
| ProblemDetails                                                                                                                                                                                                                                        | O | 0..1        | 403 Forbidden          | The "cause" attribute may be set to one of the following application errors:<br>- UNSPECIFIED<br>- UNABLE_TO_STORE_CIPHERING_KEY_DATA<br><br>See table 6.2.7.3-1 for the description of this error.                                                                                                        |
| NOTE 1: The mandatory HTTP error status codes for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] other than those specified in the table above also apply, with a ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                                                                            |
| NOTE 2: RedirectResponse may be inserted by an SCP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                                        |   |             |                        |                                                                                                                                                                                                                                                                                                            |

**Table 6.2.5.1.3.1-3: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

**Table 6.2.5.1.3.1-4: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of NF service consumer to which the notification should be sent. For the case when a request is redirected to the same target resource via a different SCP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the notification is redirected                                                                                                                                        |

## 6.2.6 Data Model

### 6.2.6.1 General

This clause specifies the application data model supported by the API.

Table 6.2.6.1-1 specifies the data types defined for the Nlmf\_Broadcast service based interface protocol.

**Table 6.2.6.1-1: Nlmf\_Broadcast specific Data Types**

| Data type            | Clause defined | Description                                                 |
|----------------------|----------------|-------------------------------------------------------------|
| CipheringKeyInfo     | 6.2.6.2.2      | Information within Ciphering Key Data Notification request  |
| CipheringKeyResponse | 6.2.6.2.3      | Information within Ciphering Key Data Notification Response |
| CipheringDataSet     | 6.2.6.2.4      | Represents a Ciphering Data Set                             |
| CipheringSetReport   | 6.2.6.2.5      | Represents a report of Ciphering Data Set storage           |
| CipherRequestData    | 6.2.6.2.6      | Information within Ciphering Key Data request               |
| CipherResponseData   | 6.2.6.2.7      | Information within Ciphering Key Data Response              |
| CipheringSetID       | 6.2.6.3.2      | Ciphering Data Set ID                                       |
| CipheringKey         | 6.2.6.3.2      | Ciphering Key                                               |
| C0                   | 6.2.6.3.2      | First component of the initial ciphering counter            |
| ValidityDuration     | 6.2.6.3.2      | Validity Duration of the Ciphering Data Set                 |
| StorageOutcome       | 6.2.6.3.3      | Indicates the result of Ciphering Data Set storage          |
| DataAvailability     | 6.2.6.3.4      | Indicates availability of ciphering key data at an LMF      |

Table 6.2.6.1-2 specifies data types re-used by the Nlmf\_Broadcast service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Nlmf\_Broadcast service based interface.

**Table 6.2.6.1-2: Nlmf\_Broadcast re-used Data Types**

| Data type         | Reference          | Comments                                         |
|-------------------|--------------------|--------------------------------------------------|
| Bytes             | 3GPP TS 29.571 [8] | Binary data encoded as a base64 character string |
| DateTime          | 3GPP TS 29.571 [8] | Date and Time                                    |
| Uri               | 3GPP TS 29.571 [8] | Uniform Resource Identifier                      |
| SupportedFeatures | 3GPP TS 29.571 [8] | Supported Features                               |
| RedirectResponse  | 3GPP TS 29.571 [8] | Redirect Response                                |

### 6.2.6.2 Structured data types

#### 6.2.6.2.1 Introduction

This clause defines the structures to be used in resource representations.

## 6.2.6.2.2 Type: CipheringKeyInfo

**Table 6.2.6.2.2-1: Definition of type CipheringKeyInfo**

| Attribute name    | Data type               | P | Cardinality | Description                                                                                                                 |
|-------------------|-------------------------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| cipheringData     | array(CipheringDataSet) | M | 1..N        | This IE contains one or more ciphering data sets, where each ciphering data set contains information for one ciphering key. |
| supportedFeatures | SupportedFeatures       | C | 0..1        | This IE shall be present if at least one feature defined in clause 6.2.9 is supported.                                      |

## 6.2.6.2.3 Type: CipheringKeyResponse

**Table 6.2.6.2.3-1: Definition of type CipheringKeyResponse**

| Attribute name      | Data type                 | P | Cardinality | Description                                                                                                                                                                                   |
|---------------------|---------------------------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cipheringDataReport | Array(CipheringSetReport) | O | 1..N        | <p>This IE indicates the ciphering data sets which were successfully stored or not stored.</p> <p>The absence of this IE indicates that all ciphering data sets were successfully stored.</p> |

6.2.6.2.4      Type: CipheringDataSet

**Table 6.2.6.2.4-1: Definition of type CipheringDataSet**

| Attribute name | Data type      | P | Cardinality | Description                                                                                        |
|----------------|----------------|---|-------------|----------------------------------------------------------------------------------------------------|
| cipheringSetID | CipheringSetID | M | 1           | Identification of a ciphering data set                                                             |
| cipheringKey   | CipheringKey   | M | 1           | A ciphering key value                                                                              |
| c0             | C0             | M | 1           | First component of the initial ciphering counter as defined in clause 7.4.2 of 3GPP TS 37.355 [21] |



|                |       |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ltePosSibTypes | Bytes | O | 0..1 | <p>This IE contains a bitmap indicating the LTE positioning SIB types for which the ciphering data set is applicable:</p> <ul style="list-style-type: none"> <li>- a bit set to 0 indicates that the ciphering data set is not applicable to the corresponding LTE positioning SIB type</li> <li>- a bit set to 1 indicates that the ciphering data set is applicable to the corresponding LTE positioning SIB type</li> </ul> <p>The bits are ordered as following in the bitmap:</p> <ul style="list-style-type: none"> <li>- first octet is the right-most octet. E.g. in "0x0123", "first octet" would be "0x23" and "second octet" would be "0x01"</li> <li>- bit 1 is the right-most bit and bit 8 is the left-most bit in the octet. E.g. for octet "0x01", "bit 1" would be "1", and "bit 2" to "bit 8" would be "0"</li> </ul> <p>The mapping of the bits to the LTE positioning SIB types is as follows:</p> <ul style="list-style-type: none"> <li>-- bit 8 in the first octet maps to positioning SIB Type 1-1</li> <li>-- bit 7 in the first octet maps to positioning SIB Type 1-2</li> <li>-- bit 6 in the first octet maps to positioning SIB Type 1-3</li> <li>-- bit 5 in the first octet maps to positioning SIB Type 1-4</li> <li>-- bit 4 in the first octet maps to positioning SIB Type 1-5</li> <li>-- bit 3 in the first octet maps to positioning SIB Type 1-6</li> <li>-- bit 2 in the first octet maps to positioning SIB Type 1-7</li> <li>-- bit 1 in the first octet maps to positioning SIB Type 1-8</li> <li>-- bit 8 in the second octet maps to positioning SIB Type 2-1</li> <li>-- bit 7 in the second octet maps to positioning SIB Type 2-2</li> <li>-- bit 6 in the second octet maps to positioning SIB Type 2-3</li> <li>-- bit 5 in the second octet maps to positioning SIB Type 2-4</li> <li>-- bit 4 in the second octet maps to positioning SIB Type 2-5</li> <li>-- bit 3 in the second octet maps to positioning SIB Type 2-6</li> <li>-- bit 2 in the second octet maps to positioning SIB Type 2-7</li> <li>-- bit 1 in the second octet maps to positioning SIB Type 2-8</li> <li>-- bit 8 in the third octet maps to positioning SIB Type 2-9</li> <li>-- bit 7 in the third octet maps to positioning SIB Type 2-10</li> <li>-- bit 6 in the third octet maps to positioning SIB Type 2-11</li> </ul> |
|----------------|-------|---|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <ul style="list-style-type: none"> <li>-- bit 5 in the third octet maps to positioning SIB Type 2-12</li> <li>-- bit 4 in the third octet maps to positioning SIB Type 2-13</li> <li>-- bit 3 in the third octet maps to positioning SIB Type 2-14</li> <li>-- bit 2 in the third octet maps to positioning SIB Type 2-15</li> <li>-- bit 1 in the third octet maps to positioning SIB Type 2-16</li> <br/> <li>-- bit 8 in the fourth octet maps to positioning SIB Type 2-17</li> <li>-- bit 7 in the fourth octet maps to positioning SIB Type 2-18</li> <li>-- bit 6 in the fourth octet maps to positioning SIB Type 2-19</li> <li>-- bit 5 in the fourth octet maps to positioning SIB Type 2-20</li> <li>-- bit 4 in the fourth octet maps to positioning SIB Type 2-21</li> <li>-- bit 3 in the fourth octet maps to positioning SIB Type 2-22</li> <li>-- bit 2 in the fourth octet maps to positioning SIB Type 2-23</li> <li>-- bit 1 in the fourth octet maps to positioning SIB Type 2-24</li> <br/> <li>-- bit 8 in the fifth octet maps to positioning SIB Type 2-25</li> <li>-- bit 7 in the fifth octet maps to positioning SIB Type 3-1</li> <li>-- bit 6 in the fifth octet maps to positioning SIB Type 4-1</li> <li>-- bit 5 in the fifth octet maps to positioning SIB Type 5-1</li> <li>-- bit 4 in the fifth octet maps to positioning SIB Type 1-9</li> <li>-- bit 3 in the fifth octet maps to positioning SIB Type 1-10</li> <li>-- bit 2 in the fifth octet maps to positioning SIB Type 2-17a</li> <li>-- bit 1 in the fifth octet maps to positioning SIB Type 2-18a</li> <br/> <li>-- bit 8 in the sixth octet maps to positioning SIB Type 2-20a</li> <li>-- bit 7 in the sixth octet maps to positioning SIB Type 2-26</li> <li>-- bit 6 in the sixth octet maps to positioning SIB Type 2-27</li> <li>-- bit 5 in the sixth octet maps to positioning SIB Type 1-11</li> <li>-- bit 4 in the sixth octet maps to positioning SIB Type 1-12</li> </ul> <p>Any unassigned bits are spare and shall be coded as zero. Non-included bits shall be treated as being coded as zero.</p> <p>(NOTE 1)</p> |
|--|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |       |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|-------|---|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nrPosSibTypes | Bytes | O | 0..1 | <p>This IE contains a bitmap indicating the NR positioning SIB types for which the ciphering data set is applicable:</p> <ul style="list-style-type: none"> <li>- a bit set to 0 indicates that the ciphering data set is not applicable to the corresponding NR positioning SIB type</li> <li>- a bit set to 1 indicates that the ciphering data set is applicable to the corresponding NR positioning SIB type</li> </ul> <p>The bits are ordered as following in the bitmap:</p> <ul style="list-style-type: none"> <li>- first octet is the right-most octet. E.g. in "0x0123", "first octet" would be "0x23" and "second octet" would be "0x01"</li> <li>- bit 1 is the right-most bit and bit 8 is the left-most bit in the octet. E.g. for octet "0x01", "bit 1" would be "1", and "bit 2" to "bit 8" would be "0"</li> </ul> <p>The mapping of the bits to the NR positioning SIB types is as follows:</p> <ul style="list-style-type: none"> <li>-- bit 8 in the first octet maps to positioning SIB Type 1-1</li> <li>-- bit 7 in the first octet maps to positioning SIB Type 1-2</li> <li>-- bit 6 in the first octet maps to positioning SIB Type 1-3</li> <li>-- bit 5 in the first octet maps to positioning SIB Type 1-4</li> <li>-- bit 4 in the first octet maps to positioning SIB Type 1-5</li> <li>-- bit 3 in the first octet maps to positioning SIB Type 1-6</li> <li>-- bit 2 in the first octet maps to positioning SIB Type 1-7</li> <li>-- bit 1 in the first octet maps to positioning SIB Type 1-8</li> <li>-- bit 8 in the second octet maps to positioning SIB Type 2-1</li> <li>-- bit 7 in the second octet maps to positioning SIB Type 2-2</li> <li>-- bit 6 in the second octet maps to positioning SIB Type 2-3</li> <li>-- bit 5 in the second octet maps to positioning SIB Type 2-4</li> <li>-- bit 4 in the second octet maps to positioning SIB Type 2-5</li> <li>-- bit 3 in the second octet maps to positioning SIB Type 2-6</li> <li>-- bit 2 in the second octet maps to positioning SIB Type 2-7</li> <li>-- bit 1 in the second octet maps to positioning SIB Type 2-8</li> <li>-- bit 8 in the third octet maps to positioning SIB Type 2-9</li> <li>-- bit 7 in the third octet maps to positioning SIB Type 2-10</li> <li>-- bit 6 in the third octet maps to positioning SIB Type 2-11</li> <li>-- bit 5 in the third octet maps to positioning SIB Type 2-12</li> </ul> |
|---------------|-------|---|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  | <p>positioning SIB Type 2-12</p> <p>-- bit 4 in the third octet maps to positioning SIB Type 2-13</p> <p>-- bit 3 in the third octet maps to positioning SIB Type 2-14</p> <p>-- bit 2 in the third octet maps to positioning SIB Type 2-15</p> <p>-- bit 1 in the third octet maps to positioning SIB Type 2-16</p><br><p>-- bit 8 in the fourth octet maps to positioning SIB Type 2-17</p> <p>-- bit 7 in the fourth octet maps to positioning SIB Type 2-18</p> <p>-- bit 6 in the fourth octet maps to positioning SIB Type 2-19</p> <p>-- bit 5 in the fourth octet maps to positioning SIB Type 2-20</p> <p>-- bit 4 in the fourth octet maps to positioning SIB Type 2-21</p> <p>-- bit 3 in the fourth octet maps to positioning SIB Type 2-22</p> <p>-- bit 2 in the fourth octet maps to positioning SIB Type 2-23</p> <p>-- bit 1 in the fourth octet maps to positioning SIB Type 3-1</p><br><p>-- bit 8 in the fifth octet maps to positioning SIB Type 4-1</p> <p>-- bit 7 in the fifth octet maps to positioning SIB Type 5-1</p> <p>-- bit 6 in the fifth octet maps to positioning SIB Type 6-1</p> <p>-- bit 5 in the fifth octet maps to positioning SIB Type 6-2</p> <p>-- bit 4 in the fifth octet maps to positioning SIB Type 6-3</p> <p>-- bit 3 in the fifth octet maps to positioning SIB Type 6-4</p> <p>-- bit 2 in the fifth octet maps to positioning SIB Type 6-5</p> <p>-- bit 1 in the fifth octet maps to positioning SIB Type 6-6</p><br><p>-- bit 8 in the sixth octet maps to positioning SIB Type 1-9</p> <p>-- bit 7 in the sixth octet maps to positioning SIB Type 1-10</p> <p>-- bit 6 in the sixth octet maps to positioning SIB Type 2-24</p> <p>-- bit 5 in the sixth octet maps to positioning SIB Type 2-25</p> <p>-- bit 4 in the sixth octet maps to positioning SIB Type 2-17a</p> <p>-- bit 3 in the sixth octet maps to positioning SIB Type 2-18a</p> <p>-- bit 2 in the sixth octet maps to positioning SIB Type 2-20a</p> <p>-- bit 1 in the sixth octet maps to positioning SIB Type 2-26</p><br><p>-- bit 8 in the seventh octet maps to positioning SIB Type 2-27</p> <p>-- bit 7 in the seventh octet maps to positioning SIB Type 1-11</p> <p>-- bit 6 in the seventh octet maps to positioning SIB Type 1-12</p> <p>-- bit 5 in the seventh octet maps to</p> |
|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                            |                  |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------|------------------|---|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            |                  |   |      | positioning SIB Type 6-7<br>-- bit 4 in the seventh octet maps to positioning SIB Type 7-1<br>-- bit 3 in the seventh octet maps to positioning SIB Type 7-2<br>-- bit 2 in the seventh octet maps to positioning SIB Type 7-3<br>-- bit 1 in the seventh octet maps to positioning SIB Type 7-4<br><br>Any unassigned bits are spare and shall be coded as zero. Non-included bits shall be treated as being coded as zero.<br>(NOTE 1) |
| validityStartTime                                                          | DateTime         | M | 1    | This IE contains the UTC time when the ciphering data set becomes valid.                                                                                                                                                                                                                                                                                                                                                                 |
| validityDuration                                                           | ValidityDuration | M | 1    | The validity duration of the ciphering data set.                                                                                                                                                                                                                                                                                                                                                                                         |
| tailList                                                                   | Bytes            | O | 0..1 | This IE contains the TAIs of the tracking areas for which the ciphering data set is applicable. It is encoded as octets 2 to n of the 5GS tracking area identity list IE specified in clause 9.11.3.9 of 3GPP TS 24.501 [22].<br><br>If this IE is omitted, the ciphering data set is valid in the entire PLMN.                                                                                                                          |
| NOTE 1: At least one of ltsibTypes IE and nrsibTypes IE shall be included. |                  |   |      |                                                                                                                                                                                                                                                                                                                                                                                                                                          |

#### 6.2.6.2.5 Type: CipheringSetReport

**Table 6.2.6.2.5-1: Definition of CipheringSetReport**

| Attribute name | Data type      | P | Cardinality | Description                                                                             |
|----------------|----------------|---|-------------|-----------------------------------------------------------------------------------------|
| cipheringSetID | CipheringSetID | M | 1           | Identification of a ciphering data set                                                  |
| storageOutcome | StorageOutcome | M | 1           | Indication of whether the ciphering data set was successfully stored or was not stored. |

#### 6.2.6.2.6 Type: CipherRequestData

**Table 6.2.6.2.6-1: Definition of CipherRequestData**

| Attribute name    | Data type         | P | Cardinality | Description                                                                            |
|-------------------|-------------------|---|-------------|----------------------------------------------------------------------------------------|
| amfCallbackURI    | Uri               | M | 1           | Callback URI of the NF Service Consumer                                                |
| supportedFeatures | SupportedFeatures | C | 0..1        | This IE shall be present if at least one feature defined in clause 6.2.9 is supported. |

#### 6.2.6.2.7 Type: CipherResponseData

**Table 6.2.6.2.7-1: Definition of CipherResponseData**

| Attribute name   | Data type        | P | Cardinality | Description                                                                                             |
|------------------|------------------|---|-------------|---------------------------------------------------------------------------------------------------------|
| dataAvailability | DataAvailability | M | 1           | An indication of whether the LMF currently has ciphering key data applicable to the NF Service Consumer |

### 6.2.6.3 Simple data types and enumerations

#### 6.2.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

#### 6.2.6.3.2 Simple data types

The simple data types defined in table 6.2.6.3.2-1 shall be supported.

**Table 6.2.6.3.2-1: Simple data types**

| Type Name        | Type Definition | Description                                                       |
|------------------|-----------------|-------------------------------------------------------------------|
| CipheringSetID   | integer         | The ciphering set ID<br>Minimum = 0. Maximum = 65535              |
| CipheringKey     | Bytes           | A 128 bit ciphering key encoded as a base64 character string      |
| C0               | Bytes           | A 128 bit value for C0 encoded as a base64 character string       |
| ValidityDuration | integer         | The validity duration in minutes.<br>Minimum = 1. Maximum = 65535 |

#### 6.2.6.3.3 Enumeration: StorageOutcome

The enumeration StorageOutcome represents the outcome of cipher set data storage at the service consumer NF.

**Table 6.2.6.3.3-1: Enumeration StorageOutcome**

| Enumeration value    | Description                                               |
|----------------------|-----------------------------------------------------------|
| "STORAGE_SUCCESSFUL" | Indicates storage of Ciphering Data Set is successful     |
| "STORAGE_FAILED"     | Indicates storage of Ciphering Data Set is not successful |

#### 6.2.6.3.4 Enumeration: DataAvailability

The enumeration DataAvailability represents the availability of ciphering key data at an LMF.

**Table 6.2.6.3.4-1: Enumeration DataAvailability**

| Enumeration value                  | Description                                          |
|------------------------------------|------------------------------------------------------|
| "CIPHERING_KEY_DATA_AVAILABLE"     | Indicates Ciphering Data Set is available in LMF     |
| "CIPHERING_KEY_DATA_NOT_AVAILABLE" | Indicates Ciphering Data Set is not available in LMF |

## 6.2.7 Error Handling

### 6.2.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of 3GPP TS 29.500 [4].

### 6.2.7.2 Protocol Errors

Protocol errors handling shall be supported as specified in clause 5.2.7 of 3GPP TS 29.500 [4].

### 6.2.7.3 Application Errors

The common application errors defined in Table 5.2.7.2-1 in 3GPP TS 29.500 [4] may be used for the Nlmf\_Broadcast. The application errors defined for the Nlmf\_Broadcast service are listed in table 6.2.7.3-1.

**Table 6.2.7.3-1: Application errors**

| Application Error                      | HTTP status code | Description                                                     |
|----------------------------------------|------------------|-----------------------------------------------------------------|
| UNSPECIFIED                            | 403 Forbidden    | The request is rejected due to unspecified reasons.             |
| UNABLE_TO_STORE_CIPHERING_KEY_DATA     | 403 Forbidden    | The service consumer NF was unable to store ciphering key data. |
| BROADCAST_CIPHERING_KEYS_NOT_SUPPORTED | 403 Forbidden    | Ciphering keys for broadcast are not supported.                 |

## 6.2.8 Security

The Nlmf\_Broadcast API does not define service operations for which additional security is needed in this version of the specification.

## 6.2.9 Feature Negotiation

The features in table 6.2.9-1 are defined for the Nlmf\_Broadcast API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

**Table 6.2.9-1: Supported Features**

| Feature number | Feature Name | M/O | Description                                                                                                                                                                                                                                                                                                                                            |
|----------------|--------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | ES3XX        | M   | Extended Support of HTTP 307/308 redirection<br><br>An NF Service Consumer (e.g. AMF) that supports this feature shall support handling of HTTP 307/308 redirection for any service operation of the Broadcast service. An NF Service Consumer that does not support this feature does only support HTTP redirection as specified for 3GPP Release 15. |

### 6.2.10 HTTP redirection

An HTTP request may be redirected to a different LMF service instance, within the same LMF or a different LMF of an LMF set, e.g. when an LMF service instance is part of an LMF (service) set or when using indirect communications (see 3GPP TS 29.500 [4]). See also the ES3XX feature in clause 6.2.9.

An SCP that reselects a different LMF producer instance will return the NF Instance ID of the new LMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of 3GPP TS 29.500 [4].

If an LMF within an LMF set redirects a service request to a different LMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new LMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of 3GPP TS 29.500 [4].

## 6.3 Nlmf\_DataExposure Service API

### 6.3.1 API URI

The Nlmf\_DataExposure service shall use the Nlmf\_DataExposure API.

The API URI of the Nlmf\_DataExposure API shall be:

**{apiRoot}/<apiName>/<apiVersion>**

The request URI used in HTTP requests from the NF service consumer towards the NF service producer shall have the Resource URI structure defined in clause 4.4.1 of 3GPP TS 29.501 [5], i.e.:

**{apiRoot}/<apiName>/<apiVersion>/<apiSpecificResourceUriPart>**

with the following components:

- The {apiRoot} shall be set as described in 3GPP TS 29.501 [5].
- The <apiName> shall be "nlmf-dataexposure".
- The <apiVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as described in clause 6.3.3.

## 6.3.2 Usage of HTTP

### 6.3.2.1 General

HTTP/2, as defined in IETF RFC 9113 [12], shall be used as specified in clause 5 of 3GPP TS 29.500 [4].

HTTP/2 shall be transported as specified in clause 5.3 of 3GPP TS 29.500 [4].

HTTP messages and bodies for the Nlmf\_DataExposure service shall comply with the OpenAPI [14] specification contained in Annex A.

### 6.3.2.2 HTTP Standard Headers

#### 6.3.2.2.1 General

#### 6.3.2.2.2 Content type

The following content types shall be supported:

- JSON, as defined in IETF RFC 8259 [13], shall be used as content type of the HTTP bodies specified in the present specification as indicated in clause 5.4 of 3GPP TS 29.500 [4].
- The Problem Details JSON Object (IETF RFC 9457 [15]). The use of the Problem Details JSON object in a HTTP response body shall be signalled by the content type "application/problem+json".

### 6.3.2.3 HTTP custom headers

#### 6.3.2.3.1 General

This API does not define any new HTTP custom headers.

### 6.3.2.4 HTTP multipart messages

HTTP multipart messages shall include one JSON body part and one or more binary body parts.

The JSON body part shall be the "root" body part of the multipart message. It shall be encoded as the first body part of the multipart message. The "Start" parameter does not need to be included.

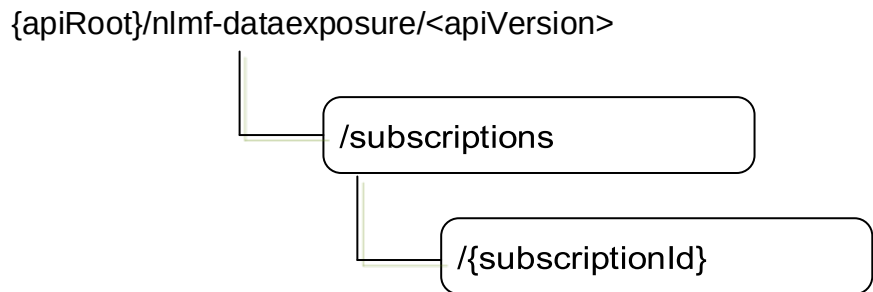
The multipart message shall include a "type" parameter (see IETF RFC 2387 [33]) specifying the media type of the root body part, i.e. "application/json". Please refer to clause 6.3.6.4 for more details.



### 6.3.3 Resources

#### 6.3.3.1 Overview

The structure of the Resource URIs of the Nlmf\_DataExposure service is shown in figure 6.3.3.1-1



**Figure 6.3.3.1-1: Resource URI structure of the Nlmf\_DataExposure API**

Table 6.3.3.1-1 provides an overview of the resources and applicable HTTP methods.

**Table 6.3.3.1-1: Resources and methods overview**

| Resource name              | Resource URI                    | HTTP method or custom operation | Description                                                              |
|----------------------------|---------------------------------|---------------------------------|--------------------------------------------------------------------------|
| subscriptions (Collection) | /subscriptions                  | POST                            | Mapped to the service operation subscribe, when to create a subscription |
| subscriptionId (Document)  | /subscriptions/{subscriptionId} | PATCH                           | Modify the subscription identified by {subscriptionId}                   |
|                            |                                 | DELETE                          | Mapped to the service operation Unsubscribe                              |

#### 6.3.3.2 Resource: subscriptions (Collection)

##### 6.3.3.2.1 Description

This resource represents a collection of subscriptions created by NF service consumers of Nlmf\_DataExposure service.

This resource is modelled as the Collection resource archetype (see clause C.2 of 3GPP TS 29.501 [5]).

##### 6.3.3.2.2 Resource Definition

Resource URI: **{apiRoot}/nlmf-dataexposure/<apiVersion>/subscriptions**

This resource shall support the resource URI variables defined in table 6.3.3.2.2-1.

**Table 6.2.3.2.2-1: Resource URI variables for this resource**

| Name    | Data type | Definition       |
|---------|-----------|------------------|
| apiRoot | string    | See clause 6.3.1 |

##### 6.3.3.2.3 Resource Standard Methods

###### 6.3.3.2.3.1 POST

This method shall support the URI query parameters specified in table 6.3.3.2.3.1-1.

**Table 6.3.3.2.3.1-1: URI query parameters supported by the POST method on this resource**

| Name | Data type | P | Cardinality | Description |
|------|-----------|---|-------------|-------------|
| n/a  |           |   |             |             |

This method shall support the request data structures specified in table 6.3.3.2.3.1-2 and the response data structures and response codes specified in table 6.3.3.2.3.1-3.

**Table 6.3.3.2.3.1-2: Data structures supported by the POST Request Body on this resource**

| Data type                   | P | Cardinality | Description                                  |
|-----------------------------|---|-------------|----------------------------------------------|
| LmfDataExposureSubscription | M | 1           | LMF Data Exposure Subscription to be created |

**Table 6.3.3.2.3.1-3: Data structures supported by the POST Response Body on this resource**

| Data type                                                                                                                                                                                                                                | P | Cardinality | Response codes         | Description                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LmfDataExposureSubscription                                                                                                                                                                                                              | M | 1           | 201 Created            | Represents successful creation of an LMF Subscription                                                                                                                    |
| RedirectResponse                                                                                                                                                                                                                         | O | 0..1        | 307 Temporary Redirect | Temporary redirection.                                                                                                                                                   |
| RedirectResponse                                                                                                                                                                                                                         | O | 0..1        | 308 Permanent Redirect | Permanent redirection.                                                                                                                                                   |
| ProblemDetails                                                                                                                                                                                                                           | O | 0..1        | 403 Forbidden          | The "cause" attribute may be used to indicate the following application error:<br>- UNSUPPORTED_ML_MODEL<br><br>See table 6.3.7.3-1 for the description of these errors. |
| NOTE: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                          |

**Table 6.3.3.2.3.1-4: Headers supported by the 201 Response Code on this resource**

| Name     | Data type | P | Cardinality | Description                                                                                                                                         |
|----------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Location | string    | M | 1           | Contains the URI of the newly created resource, according to the structure: {apiRoot}/nlmf-dataexposure/<apiVersion>/subscriptions/{subscriptionId} |

**Table 6.3.3.2.3.1-5: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

**Table 6.3.3.2.3.1-6: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

### 6.3.3.3 Resource: subscriptionId (Document)

#### 6.3.3.3.1 Description

This resource represents a collection of subscriptions created by NF service consumers of Nlmf\_DataExposure service.

This resource is modelled as the Collection resource archetype (see clause C.2 of 3GPP TS 29.501 [5]).

#### 6.3.3.3.2 Resource Definition

Resource URI: **{apiRoot}/nlmf-dataexposure/<apiVersion>/subscriptions/{subscriptionId}**

This resource shall support the resource URI variables defined in table 6.3.3.3.2-1.

**Table 6.3.3.3.2-1: Resource URI variables for this resource**

| Name           | Data type | Definition                                         |
|----------------|-----------|----------------------------------------------------|
| apiRoot        | string    | See clause 6.3.1                                   |
| apiVersion     | string    | See clause 6.3.1.                                  |
| subscriptionId | string    | Represents the identity of a specific subscription |

#### 6.3.3.3.3 Resource Standard Methods

##### 6.3.3.3.3.1 PATCH

This method shall support the URI query parameters specified in table 6.3.3.3.3.1-1.

**Table 6.3.3.3.3.1-1: URI query parameters supported by the PATCH method on this resource**

| Name | Data type | P | Cardinality | Description |
|------|-----------|---|-------------|-------------|
| n/a  |           |   |             |             |

This method shall support the request data structures specified in table 6.3.3.3.3.1-2 and the response data structure and response codes specified in table 6.3.3.3.3.1-3.

**Table 6.3.3.3.3.1-2: Data structures supported by the PATCH Request Body on this resource**

| Data type        | P | Cardinality | Description                                                                                                                         |
|------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| array(PatchItem) | M | 1..N        | It contains the list of changes to be made to the subscription, according to the JSON PATCH format specified in IETF RFC 6902 [32]. |

**Table 6.3.3.3.1-3: Data structures supported by the PATCH Response Body on this resource**

| Data type                                                                                                                                                                                                                                   | P | Cardinality | Response codes         | Description                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n/a                                                                                                                                                                                                                                         |   |             | 204 No Content         | Represents a successful update on LMF Data Exposure Subscription                                                                                                                                                                                                                            |
| RedirectResponse                                                                                                                                                                                                                            | O | 0..1        | 307 Temporary Redirect | Temporary redirection. (NOTE 2)                                                                                                                                                                                                                                                             |
| RedirectResponse                                                                                                                                                                                                                            | O | 0..1        | 308 Permanent Redirect | Permanent redirection. (NOTE 2)                                                                                                                                                                                                                                                             |
| ProblemDetails                                                                                                                                                                                                                              | O | 0..1        | 403 Forbidden          | Indicates the subscription or the modification of subscription has failed due to application error.<br><br>The "cause" attribute may be used to indicate one of the following application errors:<br>- UNSUPPORTED_ML_MODEL<br><br>See table 6.3.7.3-1 for the description of these errors. |
| NOTE 1: The mandatory HTTP error status code for the PATCH method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                                                                                                                                                                                                                                                                             |
| NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                      |   |             |                        |                                                                                                                                                                                                                                                                                             |

**Table 6.3.3.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

**Table 6.3.3.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

#### 6.3.3.3.2 DELETE

This method shall support the URI query parameters specified in table 6.3.3.3.2-1.

**Table 6.3.3.3.2-1: URI query parameters supported by the DELETE method on this resource**

| Name | Data type | P | Cardinality | Description |
|------|-----------|---|-------------|-------------|
| n/a  |           |   |             |             |

This method shall support the request data structures specified in table 6.3.3.3.2-2 and the response data structures and response codes specified in table 6.3.3.3.2-3.

**Table 6.3.6.3.3.2-2: Data structures supported by the DELETE Request Body on this resource**

| Data type | P | Cardinality | Description |
|-----------|---|-------------|-------------|
| n/a       |   |             |             |

**Table 6.3.3.3.3.2-3: Data structures supported by the DELETE Response Body on this resource**

| Data type                                                                                                                                                                                                                                    | P | Cardinality | Response codes         | Description                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|---------------------------------|
| n/a                                                                                                                                                                                                                                          |   |             | 204 No Content         |                                 |
| RedirectResponse                                                                                                                                                                                                                             | O | 0..1        | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse                                                                                                                                                                                                                             | O | 0..1        | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the DELETE method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                 |
| NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                       |   |             |                        |                                 |

**Table 6.3.3.3.3.2-4: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

**Table 6.3.6.3.3.3.2-5: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                                           |
|-----------------------|-----------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | An alternative URI of the resource located on an alternative service instance within the same LMF or LMF (service) set. For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected                                                                                                                                                                             |

## 6.3.4 Custom Operations without associated resources

There are no custom operations without associated resources supported on Nlmf\_DataExposure Service.

## 6.3.5 Notifications

### 6.3.5.1 Description

This clause specifies the notifications provided by the Nlmf\_DataExposure service.

**Table 6.3.5.1-1: Notifications overview**

| Notification                   | Callback URI      | HTTP method or custom operation | Description (service operation) |
|--------------------------------|-------------------|---------------------------------|---------------------------------|
| LMF Data Exposure Notification | {NotificationURI} | POST                            |                                 |

## 6.3.5.2 LMF Data Exposure Notification

A NF Service Consumer may be subscribed to obtain notification about the input data used for ML model training or ML model performance monitoring for LMF-based AI/ML positioning. In this case, the LMF shall notify the NF Service Consumer by using the HTTP POST method and deliver the notification to the call-back URI, following Subscribe/Notify mechanism defined in 3GPP TS 29.501 [5].

### 6.3.5.2.1 Notification Definition

Call-back URI: {**notificationUri**}

Call-back URI is provided by NF Service Consumer during creation of the subscription. The description of the Callback URI of NF consumer is obtained by LMF is provided in clause 5.4.2.3.

### 6.3.5.2.3 Notification Standard Methods

#### 6.3.5.2.3.1 POST

This method shall support the request data structures specified in table 6.3.5.2.3.1-1 and the response data structures and response codes specified in table 6.3.5.2.3.1-2.

**Table 6.3.5.2.3.1-2: Data structures supported by the POST Request Body**

| Data type                   | P | Cardinality | Description                                 |
|-----------------------------|---|-------------|---------------------------------------------|
| LmfDataExposureNotification | M | 1           | Represents the notification to be delivered |

**Table 6.3.5.2.3.1-3: Data structures supported by the POST Response Body**

| Data type                                                                                                                                                                                                                                  | P | Cardinality | Response codes         | Description                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|------------------------|---------------------------------|
| n/a                                                                                                                                                                                                                                        |   |             | 204 No Content         |                                 |
| RedirectResponse                                                                                                                                                                                                                           | O | 0..1        | 307 Temporary Redirect | Temporary redirection. (NOTE 2) |
| RedirectResponse                                                                                                                                                                                                                           | O | 0..1        | 308 Permanent Redirect | Permanent redirection. (NOTE 2) |
| NOTE 1: The mandatory HTTP error status code for the POST method listed in Table 5.2.7.1-1 of 3GPP TS 29.500 [4] also apply, with response body containing an object of ProblemDetails data type (see clause 5.2.7 of 3GPP TS 29.500 [4]). |   |             |                        |                                 |
| NOTE 2: RedirectResponse may be inserted by an SCP or SEPP, see clause 6.10.9.1 of 3GPP TS 29.500 [4].                                                                                                                                     |   |             |                        |                                 |

**Table 6.3.5.2.3.1-4: Headers supported by the 307 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                                                                                                                                                                          |
|-----------------------|-----------|---|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4]. |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected.<br>For the case when a request is redirected to the same target resource via a different SCP or SEPP, see clause 6.10.9.1 in 3GPP TS 29.500 [4].          |

**Table 6.3.5.2.3.1-5: Headers supported by the 308 Response Code on this resource**

| Name                  | Data type | P | Cardinality | Description                                                                                        |
|-----------------------|-----------|---|-------------|----------------------------------------------------------------------------------------------------|
| Location              | string    | M | 1           | A URI pointing to the endpoint of the NF service consumer to which the notification should be sent |
| 3gpp-Sbi-Target-Nf-Id | string    | O | 0..1        | Identifier of the target NF (service) instance ID towards which the request is redirected          |

## 6.3.6 Data Model

### 6.3.6.1 General

This clause specifies the application data model supported by the API.

Table 6.3.6.1-1 specifies the data types defined for the Nlmf\_DataExposure service-based interface protocol.

**Table 6.3.6.1-1: Nlmf\_DataExposure specific Data Types**

| Data type                   | Clause defined | Description                                                             |
|-----------------------------|----------------|-------------------------------------------------------------------------|
| LmfDataExposureSubscription | 6.3.6.2.2      | Represents an individual data collection subscription resource on LMF   |
| LmfDataExposureNotification | 6.3.6.2.3      | Notify within an LMF exposure for data collection notification request. |
| LmfDataExposureReport       | 6.3.6.2.4      | Represents the data collection exposure report.                         |
| LmfDataExposureSamplingData | 6.3.6.2.5      | LMF Data Exposure Sampling Data                                         |
| LmfDataExposureCause        | 6.3.6.3.3      | Cause code for LMF Data Exposure Service                                |
| DataSourceType              | 6.3.6.3.4      | Data source type for measurement data collection                        |

Table 6.3.6.1-2 specifies data types re-used by the Nlmf\_DataExposure service based interface protocol from other specifications or from other service APIs in current specification, including a reference to their respective specifications and when needed, a short description of their use within the Nlmf\_DataExposure service based interface.

**Table 6.3.6.1-2: Nlmf\_DataExposure re-used Data Types**

| Data type        | Reference           | Description           |
|------------------|---------------------|-----------------------|
| PresenceInfo     | 3GPP TS 29.571 [8]  | Area of Interest      |
| Uri              | 3GPP TS 29.571 [8]  | URI                   |
| RedirectResponse | 3GPP TS 29.571 [8]  | Redirect Response     |
| DateTime         | 3GPP TS 29.571 [8]  | Date time             |
| PatchItem        | 3GPP TS 29.571 [8]  | Patch item            |
| UInteger         | 3GPP TS 29.571 [8]  | Unsigned Integer      |
| RecurTime        | 3GPP TS 29.503 [34] | Recurring Time window |
| LocationData     | 6.1.6.2.3           | Location Data         |
| LocationQoS      | 6.1.6.2.13          | Location QoS          |
| CorrelationId    | 6.1.6.3.2           | Correlation Id        |
| LocMeasurements  | 6.1.6.2.47          | Location Measurements |

## 6.3.6.2 Structured data types

### 6.3.6.2.1 Introduction

Structured data types used in Nlmf\_DataExposure service are specified in this clause.



## 6.3.6.2.2 Type: LmfDataExposureSubscription

**Table 6.3.6.2.2-1: Definition of type LmfDataExposureSubscription**

| Attribute name      | Data type              | P | Cardinality | Description                                                                                                                                                                                                                                                                   | Applicability |
|---------------------|------------------------|---|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| notificationUri     | Uri                    | M | 1           | This IE shall be included to identify the recipients of notifications sent by LMF for data exposure services.                                                                                                                                                                 |               |
| notifyCorrelationId | CorrelationID          | M | 1           | This IE shall be included to identify the notification correlation ID. The LMF shall include this ID in the notifications. The value of this IE shall be unique per subscription for a given NF service consumer.                                                             |               |
| aoi                 | PresenceInfo           | M | 1           | The Area of Interest where the sampling data to be provided.                                                                                                                                                                                                                  |               |
| recurTime           | RecurTime              | O | 0..1        | When present, it shall indicate the list of time interval(s) during which LMF needs to initiate the collection of input data samples.                                                                                                                                         |               |
| numOfSamples        | UInteger               | O | 0..1        | When present, this IE shall indicate the required number of sampling data to be provided.<br><br>If the recurTime IE is present, then this IE shall be applied per each time interval inside recurTime IE.                                                                    |               |
| dataSources         | array(DataSource Type) | O | 1..N        | When present, this IE shall indicate the data source(s) from where the measurement data to be collected.                                                                                                                                                                      |               |
| qualityThreshold    | LocationQoS            | O | 0..1        | When present, this IE shall indicate the quality threshold of the Ground Truth data to be provided.<br><br>The LMF shall not report a sampling data in the notification, if the ground truth of the sampling data cannot meet the quality threshold.                          |               |
| mlModelId           | UInteger               | C | 0..1        | This IE shall be present when the subscription is to collect data for performance monitoring for LMF based AI/ML positioning.<br><br>When present, this IE shall indicate the unique identifier of the ML Model for performing monitoring of LMF based AI/ML positioning.     |               |
| expiryTime          | DateTime               | O | 0..1        | When present, it shall indicate the time at which the subscription expires unless it is renewed.                                                                                                                                                                              |               |
| pruIndication       | boolean                | O | 0..1        | When present, this IE shall indicate to collect the location and measurement data of PRUs only.<br><br>When present, it shall be set as follows:<br>- true: collect the location and measurement data of PRUs.<br><br>The presence of this IE with false value is prohibited. |               |

## 6.3.6.2.3 Type: LmfDataExposureNotification

**Table 6.3.6.2.3-1: Definition of type LmfDataExposureNotification**

| Attribute name      | Data type                    | P | Cardinality | Description                                                                                           | Applicability |
|---------------------|------------------------------|---|-------------|-------------------------------------------------------------------------------------------------------|---------------|
| notifyCorrelationId | CorrelationID                | M | 1           | When present, this IE shall indicate the notification correlation ID during notify service operation. |               |
| reports             | array(LmfDataExposureReport) | M | 1..N        | The report for the data collection.                                                                   |               |

## 6.3.6.2.4 Type: LmfDataExposureReport

**Table 6.3.6.2.4-1: Definition of type LmfDataExposureReport**

| Attribute name   | Data type                          | P | Cardinality | Description                                                                                                                                                                                                                                                                          | Applicability |
|------------------|------------------------------------|---|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| samplingDataList | array(LmfDataExposureSamplingData) | C | 1..N        | <p>This IE shall be present when at least one Sampling data to be reported.</p> <p>When present, this IE shall contain a list of sampling data to be reported by the LMF as requested by the NF consumer.</p> <p>Each array item shall contain the sampling data of a PRU or UE.</p> |               |
| mlModelId        | UInteger                           | C | 0..1        | <p>This IE shall be present when it was received in the subscription request.</p> <p>When present, this IE shall indicate the unique identifier of the ML Model for performing monitoring of LMF based AI/ML positioning.</p>                                                        |               |
| cause            | LmfDataExposureCause               | O | 0..1        | When present, this IE shall indicate the cause code for the report.                                                                                                                                                                                                                  |               |

### 6.3.6.2.5 Type: LmfDataExposureSamplingData

**Table 6.3.6.2.5-1: Definition of type LmfDataExposureSamplingData**

| Attribute name   | Data type       | P | Cardinality | Description                                                                                                                                                                                                                                                                                                                                               | Applicability |
|------------------|-----------------|---|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| timeStamp        | DateTime        | M | 1           | This IE shall contain the time stamp (in UTC time) when the returned sampling data are generated.                                                                                                                                                                                                                                                         |               |
| locMeasureData   | LocMeasurements | M | 1           | This IE shall contain the location measurement data collected from the indicated data source for the PRU or UE.                                                                                                                                                                                                                                           |               |
| groundTruth      | LocationData    | M | 1           | This IE shall contain the ground truth (i.e. the location) of the PRU or UE.                                                                                                                                                                                                                                                                              |               |
| qualityIndicator | float           | C | 0..1        | This IE shall be present when the quality threshold was received in the subscription request.<br><br>When present, this IE shall contain the quality indicator metric for the provided ground truth. The value range is from 0 (lowest quality) to 1 (highest quality)                                                                                    |               |
| mlLocEstimate    | LocationData    | C | 0..1        | This IE shall be present if the subscription is for performance monitoring of LMF based AI/ML positioning, i.e. the modelUniqueId IE was received in the subscription request.<br><br>When present, this IE shall contain the location of the PRU or UE estimated by the LMF using AI/ML positioning with the ML Model indicated by the modelUniqueId IE. |               |

## 6.3.6.3 Simple data types and enumerations

### 6.3.6.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

### 6.3.6.3.2 Simple data types

The simple data types defined in table 6.3.6.3.2-1 shall be supported.

**Table 6.3.6.3.2-1: Simple data types**

| Type Name | Type Definition | Description |
|-----------|-----------------|-------------|
|           |                 |             |

### 6.3.6.3.3 Enumeration: LmfDataExposureCause

The enumeration LmfDataExposureCause represents the cause code for the LMF Data Exposure reports.

**Table 6.3.6.3.3-1: Enumeration LmfDataExposureCause**

| Enumeration value    | Description                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| "NOT_ENOUGH_SAMPLES" | Indicates that the requested number of sampling data cannot be met within the Area of Interest during the requested time window. |

#### 6.3.6.3.4 Enumeration: DataSourceType

The enumeration DataSourceType represents the data source from where the location measurement data to be collected for AI/ML positioning model training or performance monitoring.

**Table 6.3.6.3.4-1: Enumeration DataSourceType**

| Enumeration value | Description                                                         |
|-------------------|---------------------------------------------------------------------|
| "NG_RAN"          | Indicates that the measurement data to be collected from the NG-RAN |

#### 6.3.6.4 Binary data types

In this release of this specification, no binary data types are defined.

### 6.3.7 Error Handling

#### 6.3.7.1 General

HTTP error handling shall be supported as specified in clause 5.2.4 of 3GPP TS 29.500 [4].

#### 6.3.7.2 Protocol Errors

Protocol Error Handling shall be supported as specified in clause 5.2.7 of 3GPP TS 29.500 [4].

#### 6.3.7.3 Application Errors

The common application errors defined in Table 5.2.7.2-1 in 3GPP TS 29.500 [4] may be used for the Nlmf\_DataExposure service. The application errors defined for the Nlmf\_DataExposure service are listed in Table 6.3.7.3-1.

**Table 6.3.7.3-1: Application errors**

| Application Error    | HTTP status code | Description                                                             |
|----------------------|------------------|-------------------------------------------------------------------------|
| UNSUPPORTED_ML_MODEL | 403 Forbidden    | The requested ML Model for data collection is not supported by the LMF. |

### 6.3.8 Feature negotiation

Table 6.1.8-1 defines the features supported by the Nlmf\_DataExposure API. They shall be negotiated using the extensibility mechanism defined in clause 6.6 of 3GPP TS 29.500 [4].

No feature is defined in this release of the specification.

**Table 6.3.8-1: Supported Features**

| Feature number | Feature Name | M/O | Description |
|----------------|--------------|-----|-------------|
|                |              |     |             |

## 6.3.9 Security

As indicated in 3GPP TS 33.501 [9], the access to the Nlmf\_DataExposure API may be authorized by means of the OAuth2 protocol (see IETF RFC 6749 [10]), using the "Client Credentials" authorization grant, where the NRF (see 3GPP TS 29.510 [11]) plays the role of the authorization server.

If OAuth2 authorization is used, an NF Service Consumer, prior to consuming services offered by the Nlmf\_DataExposure API, shall obtain a "token" from the authorization server, by invoking the Access Token Request service, as described in 3GPP TS 29.510 [11], clause 5.4.2.2.

**NOTE:** When multiple NRFs are deployed in a network, the NRF used as authorization server is the same NRF that the NF Service Consumer used for discovering the Nlmf\_DataExposure service.

The Nlmf\_DataExposure API defines scopes for OAuth2 authorization as specified in 3GPP TS 33.501 [9];

**Table 6.3.9-1: OAuth2 scopes defined in Nlmf\_DataExposure API**

| Scope               | Description                          |
|---------------------|--------------------------------------|
| "nlmf-dataexposure" | Access to the Nlmf_DataExposure API. |

## 6.3.10 HTTP redirection

An HTTP request may be redirected to a different LMF service instance, within the same LMF or a different LMF of an LMF set, e.g. when an LMF service instance is part of an LMF (service) set or when using indirect communications (see 3GPP TS 29.500 [4]).

An SCP that reselects a different LMF producer instance will return the NF Instance ID of the new LMF producer instance in the 3gpp-Sbi-Producer-Id header, as specified in clause 6.10.3.4 of 3GPP TS 29.500 [4].

If an LMF within an LMF set redirects a service request to a different LMF of the set using an 307 Temporary Redirect or 308 Permanent Redirect status code, the identity of the new LMF towards which the service request is redirected shall be indicated in the 3gpp-Sbi-Target-Nf-Id header of the 307 Temporary Redirect or 308 Permanent Redirect response as specified in clause 6.10.9.1 of 3GPP TS 29.500 [4].

---

# Annex A (normative): OpenAPI specification

## A.1 General

This Annex specifies the formal definition of the Nlmf Service APIs. It consists of an OpenAPI 3.0.0 specification, in YAML format.

This Annex takes precedence when being discrepant to other parts of the specification with respect to the encoding of information elements and methods within the API(s).

**NOTE:** The semantics and procedures, as well as conditions, e.g. for the applicability and allowed combinations of attributes or values, not expressed in the OpenAPI definitions but defined in other parts of the specification also apply.

Informative copies of the OpenAPI specification files contained in this 3GPP Technical Specification are available on a Git-based repository, that uses the GitLab software version control system (see 3GPP TS 29.501 [5] clause 5.3.1 and 3GPP TR 21.900 [16] clause 5B).

## A.2 Nlmf\_Location API

openapi: 3.0.0

info:

```
version: '1.4.1'
title: 'LMF Location'
description: |
  LMF Location Service.
  © 2026, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
  All rights reserved.
```

externalDocs:

```
description: 3GPP TS 29.572 V19.6.0; 5G System; Location Management Services; Stage 3
url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.572/'
```

servers:

```
- url: '{apiRoot}/nlmf-loc/v1'
  variables:
    apiRoot:
      default: https://example.com
      description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501
```

security:

```
- {}
- OAuth2ClientCredentials:
  - nlmf-loc
```

paths:

```
/determine-location:
  post:
    summary: Determine Location of an UE
    operationId: DetermineLocation
    tags:
      - Determine Location
    security:
      - {}
      - OAuth2ClientCredentials:
        - nlmf-loc
      - OAuth2ClientCredentials:
        - nlmf-loc
      - nlmf-loc:determine-location:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/InputData'
        multipart/related: # message with binary body part(s)
          schema:
            type: object
            properties: # Request parts
              jsonData:
                $ref: '#/components/schemas/InputData'
              binaryDataLppMessage:
                type: string
                format: binary
            encoding:
              jsonData:
                contentType: application/json
              binaryDataLppMessage:
                contentType: application/vnd.3gpp.lpp
            headers:
              Content-Id:
                schema:
                  type: string
              binaryDataLppMessageExt1:
                contentType: application/vnd.3gpp.lpp
            headers:
              Content-Id:
                schema:
                  type: string
              binaryDataLppMessageExt2:
                contentType: application/vnd.3gpp.lpp
            headers:
              Content-Id:
                schema:
                  type: string
```

```
    required: true
  responses:
    '200':
      description: Expected response to a valid request
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/LocationDataExt'
    '204':
      description: Expected response for MO-LR requesting location assistance data.
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '501':
      $ref: 'TS29571_CommonData.yaml#/components/responses/501'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    '504':
      $ref: 'TS29571_CommonData.yaml#/components/responses/504'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'
  callbacks:
    EventNotify:
      '{$request.body#/hgmlcCallBackURI}':
        post:
          requestBody:
            description: UE Event Notification
            content:
              application/json:
                schema:
                  $ref: '#/components/schemas/EventNotifyDataExt'
  responses:
    '204':
      description: Expected response to a valid notification
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
```

```

    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  '504':
    $ref: 'TS29571_CommonData.yaml#/components/responses/504'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'
/up-subscriptions:
  post:
    summary: subscribe about status of a secure LCS-UP connection for a target UE
    operationId: UpSubscriptions
    tags:
      - UP Subscribe
    security:
      - {}
      - oAuth2ClientCredentials:
          - nlmf-loc
      - oAuth2ClientCredentials:
          - nlmf-loc
          - nlmf-loc:up-subscription:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/UpSubscription'
      required: true
    responses:
      '201':
        description: Expected response to successful UP Subscription
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/UpSubscription'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      UPNotify:
        '{$request.body#/upNotifyCallBackUri}':
          post:
            requestBody:
              description: UP Connection Status Notification
              content:
                application/json:
                  schema:
                    $ref: '#/components/schemas/UpNotifyData'
            responses:
              '204':
                description: Expected response to a valid notification
              '307':
                $ref: 'TS29571_CommonData.yaml#/components/responses/307'

```



```

    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':
      $ref: 'TS29571_CommonData.yaml#/components/responses/411'
    '413':
      $ref: 'TS29571_CommonData.yaml#/components/responses/413'
    '415':
      $ref: 'TS29571_CommonData.yaml#/components/responses/415'
    '429':
      $ref: 'TS29571_CommonData.yaml#/components/responses/429'
    '500':
      $ref: 'TS29571_CommonData.yaml#/components/responses/500'
    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    '504':
      $ref: 'TS29571_CommonData.yaml#/components/responses/504'
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/up-subscriptions/{subscriptionId}:
  delete:
    summary: Deletes a subscription
    operationId: DeleteSubscription
    tags:
      - UP Unsubscribe
    security:
      - {}
      - oAuth2ClientCredentials:
          - nlmf-loc
      - oAuth2ClientCredentials:
          - nlmf-loc
          - nlmf-loc:up-subscriptions:invoke
    parameters:
      - name: subscriptionId
        in: path
        required: true
        description: Unique subscription Id
        schema:
          type: string
    responses:
      '204':
        description: Expected response to a successful subscription removal
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '501':
        $ref: 'TS29571_CommonData.yaml#/components/responses/501'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'

```

```
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/cancel-location:
  post:
    summary: request cancellation of periodic or triggered location
    operationId: CancelLocation
    tags:
      - Cancel Location
    security:
      - {}
      - OAuth2ClientCredentials:
          - nlmf-loc
      - OAuth2ClientCredentials:
          - nlmf-loc
          - nlmf-loc:cancel-location:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/CancelLocData'
      required: true
    responses:
      '204':
        description: Expected response to a successful cancellation
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/location-context-transfer:
  post:
    summary: transfer context information for periodic or triggered location
    operationId: LocationContextTransfer
    tags:
      - Location Context Transfer
    security:
      - {}
      - OAuth2ClientCredentials:
          - nlmf-loc
      - OAuth2ClientCredentials:
          - nlmf-loc
          - nlmf-loc:location-context-transfer:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: 'TS29571_CommonData.yaml#/components/schemas/LocContextData'
      required: true
    responses:
      '204':
        description: Expected response to successful location context transfer
```

```
'307':
  $ref: 'TS29571_CommonData.yaml#/components/responses/307'
'308':
  $ref: 'TS29571_CommonData.yaml#/components/responses/308'
'400':
  $ref: 'TS29571_CommonData.yaml#/components/responses/400'
'401':
  $ref: 'TS29571_CommonData.yaml#/components/responses/401'
'403':
  $ref: 'TS29571_CommonData.yaml#/components/responses/403'
'404':
  $ref: 'TS29571_CommonData.yaml#/components/responses/404'
'411':
  $ref: 'TS29571_CommonData.yaml#/components/responses/411'
'413':
  $ref: 'TS29571_CommonData.yaml#/components/responses/413'
'415':
  $ref: 'TS29571_CommonData.yaml#/components/responses/415'
'429':
  $ref: 'TS29571_CommonData.yaml#/components/responses/429'
'500':
  $ref: 'TS29571_CommonData.yaml#/components/responses/500'
'502':
  $ref: 'TS29571_CommonData.yaml#/components/responses/502'
'503':
  $ref: 'TS29571_CommonData.yaml#/components/responses/503'
'504':
  $ref: 'TS29571_CommonData.yaml#/components/responses/504'
default:
  $ref: 'TS29571_CommonData.yaml#/components/responses/default'
/measure-location:
  post:
    summary: PRU location measurement
    operationId: LocationMeasure
    tags:
      - Location Measure
    security:
      - {}
      - OAuth2ClientCredentials:
          - nlmf-loc
      - OAuth2ClientCredentials:
          - nlmf-loc
      - nlmf-loc:measure-location:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/LocMeasurementReq'
      required: true
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/LocMeasurementResp'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
```

```

    '502':
      $ref: 'TS29571_CommonData.yaml#/components/responses/502'
    '503':
      $ref: 'TS29571_CommonData.yaml#/components/responses/503'
    '504':
      $ref: 'TS29571_CommonData.yaml#/components/responses/504'
    default:
      $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/configuration-up:
  post:
    summary: set up, modify or terminate a secure LCS-UP connection for a target UE
    operationId: UpConfig
    tags:
      - UP Configure
    security:
      - {}
      - oAuth2ClientCredentials:
          - nlmf-loc
      - oAuth2ClientCredentials:
          - nlmf-loc
          - nlmf-loc:configure-up:invoke
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/UpConfig'
      required: true
    responses:
      '204':
        description: >
          Expected response to successful set up, modify or terminate LCS-UP connection.
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nlmf-loc: Access to the Nlmf Location API
            nlmf-loc:determine-location:invoke: Access to invoke Determine Location
            nlmf-loc:cancel-location:invoke: Access to invoke Cancel Location
            nlmf-loc:location-context-transfer:invoke: Access to invoke Location Context
            nlmf-loc:measure-location:invoke: Access to invoke Location Measurement
            nlmf-loc:up-subscriptions:invoke: Access to invoke UP Subscription
            nlmf-loc:configure-up:invoke: Access to invoke UP Config

```

```
schemas:
#
# STRUCTURED TYPES
#
InputData:
  description: Information within Determine Location Request.
  type: object
  not:
    required: [ ecgi, ncgi ]
  properties:
    externalClientType:
      $ref: '#/components/schemas/ExternalClientType'
    correlationID:
      $ref: '#/components/schemas/CorrelationID'
    amfId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
    locationQoS:
      $ref: '#/components/schemas/LocationQoS'
    supportedGADShapes:
      type: array
      items:
        $ref: '#/components/schemas/SupportedGADShapes'
      minItems: 1
    supi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    pei:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Pei'
    gpsi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
    requestedRangingSlResult:
      type: array
      items:
        $ref: '#/components/schemas/RangingSlResult'
      minItems: 1
    relatedUes:
      type: array
      items:
        $ref: '#/components/schemas/RelatedUe'
      minItems: 1
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    ecgiOnSecondNode:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    ncgiOnSecondNode:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    priority:
      $ref: '#/components/schemas/LcsPriority'
    velocityRequested:
      $ref: '#/components/schemas/VelocityRequested'
    ueLcsCap:
      $ref: '#/components/schemas/UeLcsCapability'
    lcsServiceType:
      $ref: '#/components/schemas/LcsServiceType'
    ldrType:
      $ref: '#/components/schemas/LdrType'
    hgmlcCallBackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    lirGmlcCallBackUri:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    vgmlcAddress:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: '#/components/schemas/LdrReference'
    lirReference:
      $ref: '#/components/schemas/LirReference'
    periodicEventInfo:
      $ref: '#/components/schemas/PeriodicEventInfo'
    areaEventInfo:
      $ref: '#/components/schemas/AreaEventInfo'
    motionEventInfo:
      $ref: '#/components/schemas/MotionEventInfo'
    reportingAccessTypes:
      type: array
      items:
        $ref: '#/components/schemas/ReportingAccessType'
```

```
minItems: 1
ueConnectivityStates:
  $ref: '#/components/schemas/UeConnectivityState'
ueLocationServiceInd:
  $ref: '#/components/schemas/UeLocationServiceInd'
moAssistanceDataTypes:
  $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/LcsBroadcastAssistanceTypesData'
lppMessage:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
lppMessageExt:
  description: Indicates the lpp message extension.
  type: array
  items:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'
minItems: 1
supportedFeatures:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
uePositioningCap:
  $ref: '#/components/schemas/UePositioningCapabilities'
tnapId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/TnapId'
twapId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/TwapId'
ueCountryDetInd:
  type: boolean
scheduledLocTime:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
reliableLocReq:
  type: boolean
  default: false
evtRptAllowedAreas:
  type: array
  items:
    $ref: '#/components/schemas/ReportingArea'
minItems: 1
maxItems: 250
ueUnawareInd:
  type: boolean
  enum:
    - true
intermediateLocationInd:
  type: boolean
  default: false
maxRespTime:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/DurationSec'
lpHapType:
  $ref: 'TS29518_Namf_Location.yaml#/components/schemas/LpHapType'
ueUpPosCaps:
  type: array
  items:
    $ref: '#/components/schemas/UeUpPositioningCapabilities'
minItems: 1
reportingInd:
  allOf:
    - $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/ReportingInd'
  default: INSIDE_REPORTING
mbsrInfo:
  $ref: '#/components/schemas/MbsrInfo'
mwabSupportInd:
  type: boolean
  enum:
    - true
integrityRequirements:
  $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityRequirements'
upLocRepAddrAf:
  $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UpLocRepAddrAf'
upCumEvtRptCriteria:
  $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UpCumEvtRptCriteria'
mappedQoSeps:
  $ref: '#/components/schemas/MappedLocationQoSeps'
additionalUeInfo:
  $ref: '#/components/schemas/AdditionalUeInfo'
coordinateID:
  type: integer
rangingSlCapability:
  $ref: '#/components/schemas/RangingSlCapability'
slPositioningCapabilities:
  type: array
```

```

      items:
        $ref: '#/components/schemas/SlPositioningCapability'
      minItems: 1
    extendedFacilityInd:
      type: boolean
      enum:
        - true

LocationDataExt:
  description: Extended Location Data for UEs
  allOf:
    - $ref: '#/components/schemas/LocationData'
    - $ref: '#/components/schemas/AddLocationDatas'

LocationData:
  description: Information within Determine Location Response.
  type: object
  required:
    - locationEstimate
  properties:
    locationEstimate:
      $ref: '#/components/schemas/GeographicArea'
    accuracyFulfilmentIndicator:
      $ref: '#/components/schemas/AccuracyFulfilmentIndicator'
    ageOfLocationEstimate:
      $ref: '#/components/schemas/AgeOfLocationEstimate'
    timestampOfLocationEstimate:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    velocityEstimate:
      $ref: '#/components/schemas/VelocityEstimate'
    civicAddress:
      $ref: '#/components/schemas/CivicAddress'
    localLocationEstimate:
      $ref: '#/components/schemas/LocalArea'
    positioningDataList:
      type: array
      items:
        $ref: '#/components/schemas/PositioningMethodAndUsage'
      minItems: 1
    gnssPositioningDataList:
      type: array
      items:
        $ref: '#/components/schemas/GnssPositioningMethodAndUsage'
      minItems: 1
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    remoteUeInd:
      type: boolean
      enum:
        - true
    altitude:
      $ref: '#/components/schemas/Altitude'
    barometricPressure:
      $ref: '#/components/schemas/BarometricPressure'
    servingLMFIdentification:
      $ref: '#/components/schemas/LMFIdentification'
    uePositioningCap:
      $ref: '#/components/schemas/UePositioningCapabilities'
    ueAreaInd:
      $ref: '#/components/schemas/UeAreaIndication'
    supportedFeatures:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    achievedQos:
      $ref: '#/components/schemas/MinorLocationQoS'
    directReportInd:
      type: boolean
      default: false
    indoorOutdoorInd:
      $ref: '#/components/schemas/IndoorOutdoorInd'
    acceptedPeriodicEventInfo:
      $ref: '#/components/schemas/PeriodicEventInfo'
    haGnssMetrics:
      $ref: '#/components/schemas/HighAccuracyGnssMetrics'
    losNlosMeasureInd:
      $ref: '#/components/schemas/LosNlosMeasureInd'
    relatedApplicationlayerId:

```

```

    $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationlayerId'
  distanceDirection:
    $ref: '#/components/schemas/RangeDirection'
  2dRelativeLocation:
    $ref: '#/components/schemas/2DRelativeLocation'
  3dRelativeLocation:
    $ref: '#/components/schemas/3DRelativeLocation'
  relativeVelocity:
    $ref: '#/components/schemas/VelocityEstimate'
  rangingSlCapability:
    $ref: '#/components/schemas/RangingSlCapability'
  integrityResult:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityResult'
  nrppaPeriodicInd:
    type: boolean
    enum:
      - true

```

```

GeographicArea:
  description: Geographic area specified by different shape.
  anyOf:
    - $ref: '#/components/schemas/Point'
    - $ref: '#/components/schemas/PointUncertaintyCircle'
    - $ref: '#/components/schemas/PointUncertaintyEllipse'
    - $ref: '#/components/schemas/Polygon'
    - $ref: '#/components/schemas/PointAltitude'
    - $ref: '#/components/schemas/PointAltitudeUncertainty'
    - $ref: '#/components/schemas/EllipsoidArc'

```

```

GADShape:
  description: Common base type for GAD shapes.
  type: object
  required:
    - shape
  properties:
    shape:
      $ref: '#/components/schemas/SupportedGADShapes'
  discriminator:
    propertyName: shape
  mapping:
    POINT: '#/components/schemas/Point'
    POINT_UNCERTAINTY_CIRCLE: '#/components/schemas/PointUncertaintyCircle'
    POINT_UNCERTAINTY_ELLIPSE: '#/components/schemas/PointUncertaintyEllipse'
    POLYGON: '#/components/schemas/Polygon'
    POINT_ALTITUDE: '#/components/schemas/PointAltitude'
    POINT_ALTITUDE_UNCERTAINTY: '#/components/schemas/PointAltitudeUncertainty'
    ELLIPSOID_ARC: '#/components/schemas/EllipsoidArc'
    LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE: '#/components/schemas/Local2dPointUncertaintyEllipse'
    LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID:
      '#/components/schemas/Local3dPointUncertaintyEllipsoid'

```

```

Point:
  description: Ellipsoid Point.
  allOf:
    - $ref: '#/components/schemas/GADShape'
    - type: object
      required:
        - point
      properties:
        point:
          $ref: '#/components/schemas/GeographicalCoordinates'

```

```

PointUncertaintyCircle:
  description: Ellipsoid point with uncertainty circle.
  allOf:
    - $ref: '#/components/schemas/GADShape'
    - type: object
      required:
        - point
        - uncertainty
      properties:
        point:
          $ref: '#/components/schemas/GeographicalCoordinates'
        uncertainty:
          $ref: '#/components/schemas/Uncertainty'

```

```

PointUncertaintyEllipse:

```



```
description: Ellipsoid point with uncertainty ellipse.
allOf:
- $ref: '#/components/schemas/GADShape'
- type: object
  required:
  - point
  - uncertaintyEllipse
  - confidence
  properties:
    point:
      $ref: '#/components/schemas/GeographicalCoordinates'
    uncertaintyEllipse:
      $ref: '#/components/schemas/UncertaintyEllipse'
    confidence:
      $ref: '#/components/schemas/Confidence'

Polygon:
description: Polygon.
allOf:
- $ref: '#/components/schemas/GADShape'
- type: object
  required:
  - pointList
  properties:
    pointList:
      $ref: '#/components/schemas/PointList'

PointAltitude:
description: Ellipsoid point with altitude.
allOf:
- $ref: '#/components/schemas/GADShape'
- type: object
  required:
  - point
  - altitude
  properties:
    point:
      $ref: '#/components/schemas/GeographicalCoordinates'
    altitude:
      $ref: '#/components/schemas/Altitude'

PointAltitudeUncertainty:
description: Ellipsoid point with altitude and uncertainty ellipsoid.
allOf:
- $ref: '#/components/schemas/GADShape'
- type: object
  required:
  - point
  - altitude
  - uncertaintyEllipse
  - uncertaintyAltitude
  - confidence
  properties:
    point:
      $ref: '#/components/schemas/GeographicalCoordinates'
    altitude:
      $ref: '#/components/schemas/Altitude'
    uncertaintyEllipse:
      $ref: '#/components/schemas/UncertaintyEllipse'
    uncertaintyAltitude:
      $ref: '#/components/schemas/Uncertainty'
    confidence:
      $ref: '#/components/schemas/Confidence'
    vConfidence:
      $ref: '#/components/schemas/Confidence'

EllipsoidArc:
description: Ellipsoid Arc.
allOf:
- $ref: '#/components/schemas/GADShape'
- type: object
  required:
  - point
  - innerRadius
  - uncertaintyRadius
  - offsetAngle
  - includedAngle
  - confidence
```

```

    properties:
      point:
        $ref: '#/components/schemas/GeographicalCoordinates'
      innerRadius:
        $ref: '#/components/schemas/InnerRadius'
      uncertaintyRadius:
        $ref: '#/components/schemas/Uncertainty'
      offsetAngle:
        $ref: '#/components/schemas/Angle'
      includedAngle:
        $ref: '#/components/schemas/Angle'
      confidence:
        $ref: '#/components/schemas/Confidence'

LocalOrigin:
  description: Indicates a Local origin in a reference system
  type: object
  required:
    - coordinateId
  properties:
    coordinateId:
      type: string
    point:
      $ref: '#/components/schemas/GeographicalCoordinates'
    area:
      $ref: '#/components/schemas/GeographicArea'
    horizAxesOrientation:
      $ref: '#/components/schemas/HorizAxesOrientation'

RelativeCartesianLocation:
  description: Relative Cartesian Location
  type: object
  required:
    - x
    - y
  properties:
    x:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
    y:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'
    z:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Float'

LocalArea:
  description: Local area specified by different shape
  oneOf:
    - $ref: '#/components/schemas/Local2dPointUncertaintyEllipse'
    - $ref: '#/components/schemas/Local3dPointUncertaintyEllipsoid'

UeAreaIndication:
  description: >
    Indicates area (country, area in a country or international area) where UE is located
  type: object
  oneOf:
    - required:
        - country
        - internationalAreaInd
  properties:
    country:
      description: Indicates country or area in a country where UE is located
      type: string
    internationalAreaInd:
      description: >
        Indicates international area indication if UE is located in international area
      type: boolean
      default: false

Local2dPointUncertaintyEllipse:
  description: Local 2D point with uncertainty ellipse
  allOf:
    - $ref: '#/components/schemas/GADShape'
    - type: object
      required:
        - localOrigin
        - point
        - uncertaintyEllipse

```

```

    - confidence
  properties:
    localOrigin:
      $ref: '#/components/schemas/LocalOrigin'
    point:
      $ref: '#/components/schemas/RelativeCartesianLocation'
    uncertaintyEllipse:
      $ref: '#/components/schemas/UncertaintyEllipse'
    confidence:
      $ref: '#/components/schemas/Confidence'

Local3dPointUncertaintyEllipsoid:
  description: Local 3D point with uncertainty ellipsoid
  allOf:
    - $ref: '#/components/schemas/GADShape'
    - type: object
      required:
        - localOrigin
        - point
        - uncertaintyEllipsoid
        - confidence
      properties:
        localOrigin:
          $ref: '#/components/schemas/LocalOrigin'
        point:
          $ref: '#/components/schemas/RelativeCartesianLocation'
        uncertaintyEllipsoid:
          $ref: '#/components/schemas/UncertaintyEllipsoid'
        confidence:
          $ref: '#/components/schemas/Confidence'
        vConfidence:
          $ref: '#/components/schemas/Confidence'

GeographicalCoordinates:
  description: Geographical coordinates.
  type: object
  required:
    - lon
    - lat
  properties:
    lon:
      type: number
      format: double
      minimum: -180
      maximum: 180
    lat:
      type: number
      format: double
      minimum: -90
      maximum: 90

UncertaintyEllipse:
  description: Ellipse with uncertainty.
  type: object
  required:
    - semiMajor
    - semiMinor
    - orientationMajor
  properties:
    semiMajor:
      $ref: '#/components/schemas/Uncertainty'
    semiMinor:
      $ref: '#/components/schemas/Uncertainty'
    orientationMajor:
      $ref: '#/components/schemas/Orientation'

UncertaintyEllipsoid:
  description: Ellipsoid with uncertainty
  type: object
  required:
    - semiMajor
    - semiMinor
    - vertical
    - orientationMajor
  properties:
    semiMajor:
      $ref: '#/components/schemas/Uncertainty'
    semiMinor:

```

```
    $ref: '#/components/schemas/Uncertainty'
  vertical:
    $ref: '#/components/schemas/Uncertainty'
  orientationMajor:
    $ref: '#/components/schemas/Orientation'

PointList:
  description: List of points.
  type: array
  items:
    $ref: '#/components/schemas/GeographicalCoordinates'
  minItems: 3
  maxItems: 15

LocationQoS:
  description: QoS of Location request.
  type: object
  properties:
    hAccuracy:
      $ref: '#/components/schemas/Accuracy'
    vAccuracy:
      $ref: '#/components/schemas/Accuracy'
    verticalRequested:
      type: boolean
    responseTime:
      $ref: '#/components/schemas/ResponseTime'
    minorLocQoses:
      type: array
      items:
        $ref: '#/components/schemas/MinorLocationQoS'
      minItems: 1
      maxItems: 2
    lcsQosClass:
      $ref: '#/components/schemas/LcsQosClass'

PositioningMethodAndUsage:
  description: Indicates the usage of a positioning method.
  type: object
  required:
    - method
    - mode
    - usage
  properties:
    method:
      $ref: '#/components/schemas/PositioningMethod'
    mode:
      $ref: '#/components/schemas/PositioningMode'
    usage:
      $ref: '#/components/schemas/Usage'
    methodCode:
      type: integer
      minimum: 16
      maximum: 31

GnssPositioningMethodAndUsage:
  description: >
    Indicates the usage of a Global Navigation Satellite System (GNSS) positioning method.
  type: object
  required:
    - mode
    - gnss
    - usage
  properties:
    mode:
      $ref: '#/components/schemas/PositioningMode'
    gnss:
      $ref: '#/components/schemas/GnssId'
    usage:
      $ref: '#/components/schemas/Usage'

CivicAddress:
  description: Indicates a Civic address.
  type: object
  properties:
    country:
      type: string
    A1:
      type: string
```

```

A2:
  type: string
A3:
  type: string
A4:
  type: string
A5:
  type: string
A6:
  type: string
PRD:
  type: string
POD:
  type: string
STS:
  type: string
HNO:
  type: string
HNS:
  type: string
LMK:
  type: string
LOC:
  type: string
NAM:
  type: string
PC:
  type: string
BLD:
  type: string
UNIT:
  type: string
FLR:
  type: string
ROOM:
  type: string
PLC:
  type: string
PCN:
  type: string
POBOX:
  type: string
ADDCODE:
  type: string
SEAT:
  type: string
RD:
  type: string
RDSEC:
  type: string
RDBR:
  type: string
RDSUBBR:
  type: string
PRM:
  type: string
POM:
  type: string
usageRules:
  type: string
method:
  type: string
providedBy:
  type: string

VelocityEstimate:
  description: Velocity estimate.
  anyOf:
    - $ref: '#/components/schemas/HorizontalVelocity'
    - $ref: '#/components/schemas/HorizontalWithVerticalVelocity'
    - $ref: '#/components/schemas/HorizontalVelocityWithUncertainty'
    - $ref: '#/components/schemas/HorizontalWithVerticalVelocityAndUncertainty'
    - $ref: '#/components/schemas/RelativeVelocityWithUncertainty'

HorizontalVelocity:
  description: Horizontal velocity.
  type: object
  required:

```

```
- hSpeed
- bearing
properties:
  hSpeed:
    $ref: '#/components/schemas/HorizontalSpeed'
  bearing:
    $ref: '#/components/schemas/Angle'

HorizontalWithVerticalVelocity:
  description: Horizontal and vertical velocity.
  type: object
  required:
    - hSpeed
    - bearing
    - vSpeed
    - vDirection
  properties:
    hSpeed:
      $ref: '#/components/schemas/HorizontalSpeed'
    bearing:
      $ref: '#/components/schemas/Angle'
    vSpeed:
      $ref: '#/components/schemas/VerticalSpeed'
    vDirection:
      $ref: '#/components/schemas/VerticalDirection'

HorizontalVelocityWithUncertainty:
  description: Horizontal velocity with speed uncertainty.
  type: object
  required:
    - hSpeed
    - bearing
    - hUncertainty
  properties:
    hSpeed:
      $ref: '#/components/schemas/HorizontalSpeed'
    bearing:
      $ref: '#/components/schemas/Angle'
    hUncertainty:
      $ref: '#/components/schemas/SpeedUncertainty'

HorizontalWithVerticalVelocityAndUncertainty:
  description: Horizontal and vertical velocity with speed uncertainty.
  type: object
  required:
    - hSpeed
    - bearing
    - vSpeed
    - vDirection
    - hUncertainty
    - vUncertainty
  properties:
    hSpeed:
      $ref: '#/components/schemas/HorizontalSpeed'
    bearing:
      $ref: '#/components/schemas/Angle'
    vSpeed:
      $ref: '#/components/schemas/VerticalSpeed'
    vDirection:
      $ref: '#/components/schemas/VerticalDirection'
    hUncertainty:
      $ref: '#/components/schemas/SpeedUncertainty'
    vUncertainty:
      $ref: '#/components/schemas/SpeedUncertainty'

UeLcsCapability:
  description: Indicates the LCS capability supported by the UE..
  type: object
  properties:
    lppSupport:
      type: boolean
      default: true
    ciotOptimisation:
      type: boolean
      default: false

PeriodicEventInfo:
  description: Indicates the information of periodic event reporting.
```

```
type: object
required:
  - reportingAmount
  - reportingInterval
properties:
  reportingAmount:
    $ref: '#/components/schemas/ReportingAmount'
  reportingInterval:
    $ref: '#/components/schemas/ReportingInterval'
  reportingInfiniteInd:
    type: boolean
    enum:
      - true
  reportingIntervalMs:
    $ref: '#/components/schemas/ReportingIntervalMs'

AreaEventInfo:
  description: Indicates the information of area based event reporting.
  type: object
  required:
    - areaDefinition
  properties:
    areaDefinition:
      type: array
      items:
        $ref: '#/components/schemas/ReportingArea'
      minItems: 1
      maxItems: 250
    occurrenceInfo:
      $ref: '#/components/schemas/OccurrenceInfo'
    minimumInterval:
      $ref: '#/components/schemas/MinimumInterval'
    maximumInterval:
      $ref: '#/components/schemas/MaximumInterval'
    samplingInterval:
      $ref: '#/components/schemas/SamplingInterval'
    reportingDuration:
      $ref: '#/components/schemas/ReportingDuration'
    reportingLocationReq:
      type: boolean
      default: true

ReportingArea:
  description: Indicates an area for event reporting.
  type: object
  required:
    - areaType
  properties:
    areaType:
      $ref: '#/components/schemas/ReportingAreaType'
    tai:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'

MotionEventInfo:
  description: Indicates the information of motion based event reporting.
  type: object
  required:
    - linearDistance
  properties:
    linearDistance:
      $ref: '#/components/schemas/LinearDistance'
    occurrenceInfo:
      $ref: '#/components/schemas/OccurrenceInfo'
    minimumInterval:
      $ref: '#/components/schemas/MinimumInterval'
    maximumInterval:
      $ref: '#/components/schemas/MaximumInterval'
    samplingInterval:
      $ref: '#/components/schemas/SamplingInterval'
    reportingDuration:
      $ref: '#/components/schemas/ReportingDuration'
    reportingLocationReq:
      type: boolean
      default: true
```

```

CancelLocData:
  description: Information within Cancel Location Request.
  type: object
  required:
    - hgmlcCallBackURI
    - ldrReference
  properties:
    hgmlcCallBackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: '#/components/schemas/LdrReference'
    lcsCorrelationID:
      $ref: '#/components/schemas/CorrelationID'
    supportedFeatures:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

LocContextData:
  description: Information within Transfer Location Context Request.
  type: object
  required:
    - amfId
    - ldrType
    - hgmlcCallBackURI
    - ldrReference
    - eventReportMessage
  properties:
    amfId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/NfInstanceId'
    locationQoS:
      $ref: '#/components/schemas/LocationQoS'
    supportedGADShapes:
      type: array
      items:
        $ref: '#/components/schemas/SupportedGADShapes'
      minItems: 1
    supi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    gpsi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
    addLocContextList:
      type: array
      items:
        $ref: '#/components/schemas/AddLocationContext'
      minItems: 1
    ldrType:
      $ref: '#/components/schemas/LdrType'
    hgmlcCallBackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: '#/components/schemas/LdrReference'
    periodicEventInfo:
      $ref: '#/components/schemas/PeriodicEventInfo'
    areaEventInfo:
      $ref: '#/components/schemas/AreaEventInfo'
    motionEventInfo:
      $ref: '#/components/schemas/MotionEventInfo'
    eventReportMessage:
      $ref: '#/components/schemas/EventReportMessage'
    eventReportingStatus:
      $ref: '#/components/schemas/EventReportingStatus'
    ueLocationInfo:
      $ref: '#/components/schemas/UELocationInfo'
    cIoT5GSOptimisation:
      type: boolean
      default: false
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    guami:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Guami'
    supportedFeatures:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
    uePositioningCap:
      $ref: '#/components/schemas/UePositioningCapabilities'
    ueUpPosCaps:
      type: array

```



```

    items:
      $ref: '#/components/schemas/UeUpPositioningCapabilities'
    minItems: 1
  scheduledLocTime:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
  indoorOutdoorInd:
    $ref: '#/components/schemas/IndoorOutdoorInd'
  losNlosMeasureInd:
    $ref: '#/components/schemas/LosNlosMeasureInd'
  upCumEvtRptCriteria:
    $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UpCumEvtRptCriteria'
  lcsUppExistInd:
    type: boolean
    enum:
      - true
  srcLmfRoutingInfo:
    $ref: '#/components/schemas/LMFIdentification'
  extendedFacilityInd:
    type: boolean
    enum:
      - true

AddLocationContext:
  description: Information within Location context(s).
  type: object
  required:
    - ldrType
    - hgmlcCallbackURI
    - ldrReference
  properties:
    locationQoS:
      $ref: '#/components/schemas/LocationQoS'
    ldrType:
      $ref: '#/components/schemas/LdrType'
    hgmlcCallbackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: '#/components/schemas/LdrReference'
    periodicEventInfo:
      $ref: '#/components/schemas/PeriodicEventInfo'
    areaEventInfo:
      $ref: '#/components/schemas/AreaEventInfo'
    motionEventInfo:
      $ref: '#/components/schemas/MotionEventInfo'
    eventReportMessage:
      $ref: '#/components/schemas/EventReportMessage'
    eventReportingStatus:
      $ref: '#/components/schemas/EventReportingStatus'
    ueLocationInfo:
      $ref: '#/components/schemas/UELocationInfo'
    scheduledLocTime:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    indoorOutdoorInd:
      $ref: '#/components/schemas/IndoorOutdoorInd'
    losNlosMeasureInd:
      $ref: '#/components/schemas/LosNlosMeasureInd'
    upCumEvtRptCriteria:
      $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/UpCumEvtRptCriteria'
    lcsUppExistInd:
      type: boolean
      enum:
        - true

EventReportMessage:
  description: Indicates an event report message.
  type: object
  required:
    - eventClass
    - eventContent
  properties:
    eventClass:
      $ref: '#/components/schemas/EventClass'
    eventContent:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/RefToBinaryData'

EventReportingStatus:
  description: Indicates the status of event reporting.
  type: object

```

```

    properties:
      eventReportCounter:
        $ref: '#/components/schemas/EventReportCounter'
      eventReportDuration:
        $ref: '#/components/schemas/EventReportDuration'

UELocationInfo:
  description: Indicates location information of a UE.
  type: object
  properties:
    locationEstimate:
      $ref: '#/components/schemas/GeographicArea'
    ageOfLocationEstimate:
      $ref: '#/components/schemas/AgeOfLocationEstimate'
    timestampOfLocationEstimate:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    velocityEstimate:
      $ref: '#/components/schemas/VelocityEstimate'
    ageOfVelocityEstimate:
      $ref: '#/components/schemas/AgeOfLocationEstimate'
    timestampOfVelocityEstimate:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'

EventNotifyDataExt:
  description: Extended Event Notify Data for UEs
  allOf:
    - $ref: '#/components/schemas/EventNotifyData'
    - $ref: '#/components/schemas/AddEventNotifyDatas'

EventNotifyData:
  description: Information within Event Notify Request.
  type: object
  required:
    - reportedEventType
    - ldrReference
  properties:
    reportedEventType:
      $ref: '#/components/schemas/ReportedEventType'
    supi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    gpsi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
    hgmLcCallBackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    ldrReference:
      $ref: '#/components/schemas/LdrReference'
    lirReference:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LirReference'
    locationEstimate:
      $ref: '#/components/schemas/GeographicArea'
    ageOfLocationEstimate:
      $ref: '#/components/schemas/AgeOfLocationEstimate'
    timestampOfLocationEstimate:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    civicAddress:
      $ref: '#/components/schemas/CivicAddress'
    localLocationEstimate:
      $ref: '#/components/schemas/LocalArea'
    positioningDataList:
      type: array
      items:
        $ref: '#/components/schemas/PositioningMethodAndUsage'
      minItems: 1
    gnssPositioningDataList:
      type: array
      items:
        $ref: '#/components/schemas/GnssPositioningMethodAndUsage'
      minItems: 1
    servingLMFidentification:
      $ref: '#/components/schemas/LMFIdentification'
    terminationCause:
      $ref: '#/components/schemas/TerminationCause'
    velocityEstimate:
      $ref: '#/components/schemas/VelocityEstimate'
    altitude:
      $ref: '#/components/schemas/Altitude'
    achievedQos:
      $ref: '#/components/schemas/MinorLocationQoS'

```

```

supportedFeatures:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'
indoorOutdoorInd:
  $ref: '#/components/schemas/IndoorOutdoorInd'
haGnssMetrics:
  $ref: '#/components/schemas/HighAccuracyGnssMetrics'
losNlosMeasureInd:
  $ref: '#/components/schemas/LosNlosMeasureInd'
upLocRepStatAf:
  type: integer
relatedApplicationlayerId:
  $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationlayerId'
distanceDirection:
  $ref: '#/components/schemas/RangeDirection'
2dRelativeLocation:
  $ref: '#/components/schemas/2DRelativeLocation'
3dRelativeLocation:
  $ref: '#/components/schemas/3DRelativeLocation'
relativeVelocity:
  $ref: '#/components/schemas/VelocityEstimate'
integrityResult:
  $ref: 'TS29515_Ngmlc_Location.yaml#/components/schemas/IntegrityResult'

UeConnectivityState:
  description: Indicates the connectivity state of a UE.
  type: object
  required:
    - accessType
  properties:
    accessType:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
    connectivitystate:
      $ref: 'TS29518_Namf_EventExposure.yaml#/components/schemas/CmState'

MinorLocationQoS:
  description: Contain Minor Location QoS.
  type: object
  properties:
    hAccuracy:
      $ref: '#/components/schemas/Accuracy'
    vAccuracy:
      $ref: '#/components/schemas/Accuracy'

MbsrInfo:
  description: MBSR Information
  type: object
  properties:
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'

LocMeasurementReq:
  description: Location Measurement Request.
  type: object
  properties:
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    ecgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ecgi'
    preCalcuLocEstimate:
      $ref: '#/components/schemas/GeographicArea'
    timestampOfPreCalcuLocEstimate:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    timeWindows:
      type: array
      items:
        $ref: '#/components/schemas/TimeWindow'
      minItems: 1
    timeWindowsNrppa:
      $ref: '#/components/schemas/TimeWindowsNrppa'
    method:
      $ref: '#/components/schemas/PositioningMethod'
    trpParameters:
      type: array
      items:

```

```
    $ref: '#/components/schemas/TrpParameter'
    minItems: 1

LocMeasurementResp:
  description: Location Measurement Response.
  type: object
  required:
    - locMeasurements
  properties:
    locMeasurements:
      type: array
      items:
        $ref: '#/components/schemas/LocMeasurements'
      minItems: 1

LocMeasurements:
  description: PRU Location Measurements.
  type: object
  properties:
    locInfo:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'

HighAccuracyGnssMetrics:
  description: High Accuracy GNSS Positioning Metrics.
  type: object
  properties:
    nrOfUsedSatellites:
      type: integer
      minimum: 0
      maximum: 64
    hdopi:
      type: integer
      minimum: 1
      maximum: 256
    pdopi:
      type: integer
      minimum: 1
      maximum: 256
    age:
      type: integer
      minimum: 0
      maximum: 99
    fixType:
      $ref: '#/components/schemas/FixType'

UpNotifyData:
  description: UP Subscription
  type: object
  required:
    - notifCorrelationId
    - upConnectionStatus
  properties:
    notifCorrelationId:
      $ref: '#/components/schemas/CorrelationID'
    upConnectionStatus:
      $ref: '#/components/schemas/UpConnectionStatus'
    targetLMFId:
      $ref: '#/components/schemas/LMFIdentification'

UpSubscription:
  description: UP Subscription
  type: object
  required:
    - upNotifyCallBackUri
    - notifCorrelationId
    - supi
  properties:
    upNotifyCallBackUri:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    notifCorrelationId:
      $ref: '#/components/schemas/CorrelationID'
    supi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
    gpsi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'

RelatedUe:
  description: Related UE Information
```

```
type: object
required:
- applicationlayerId
- relatedUeType
properties:
  applicationlayerId:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/ApplicationlayerId'
  relatedUeType:
    $ref: '#/components/schemas/RelatedUeType'

UpConfig:
description: UP Config
type: object
required:
- upNotifyCallBackUri
- notifCorrelationId
anyOf:
- required: [supi]
- required: [gpsi]
properties:
  upNotifyCallBackUri:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
  notifCorrelationId:
    $ref: '#/components/schemas/CorrelationID'
  supi:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Supi'
  gpsi:
    $ref: 'TS29571_CommonData.yaml#/components/schemas/Gpsi'
  amfReallocationInd:
    type: boolean
    default: false
  lcsUpConnectionInd:
    $ref: '#/components/schemas/LcsUpConnectionInd'
  targetLMFId:
    $ref: '#/components/schemas/LMFIdentification'
  ueUpPosCaps:
    type: array
    items:
      $ref: '#/components/schemas/UeUpPositioningCapabilities'
    minItems: 1

RangeDirection:
description: Represents a distance and direction from a point A to a point B.
type: object
properties:
  distance:
    type: number
  azimuthDirection:
    $ref: '#/components/schemas/Angle'
  elevationDirection:
    $ref: '#/components/schemas/Angle'

2DRelativeLocation:
description: Represents a relative 2D location with uncertainty ellipse.
type: object
properties:
  semiMinor:
    $ref: '#/components/schemas/Uncertainty'
  semiMajor:
    $ref: '#/components/schemas/Uncertainty'
  orientationAngle:
    $ref: '#/components/schemas/Angle'

3DRelativeLocation:
description: Represents a relative 3D location with uncertainty ellipsoid.
type: object
properties:
  semiMinor:
    $ref: '#/components/schemas/Uncertainty'
  semiMajor:
    $ref: '#/components/schemas/Uncertainty'
  verticalUncertainty:
    $ref: '#/components/schemas/Uncertainty'
  orientationAngle:
    $ref: '#/components/schemas/Angle'

AddLocationDatas:
description: Additional Location Data.
```

```
    type: object
    properties:
      addLocationDatas:
        type: array
        items:
          $ref: '#/components/schemas/LocationData'
        minItems: 1

AddEventNotifyDatas:
  description: Additional Event Notify Data.
  type: object
  properties:
    addEventNotifyDatas:
      type: array
      items:
        $ref: '#/components/schemas/EventNotifyData'
      minItems: 1

MappedLocationQoSeps:
  description: Mapped Location QoS for EPS.
  type: object
  required:
    - hAccuracy
  properties:
    hAccuracy:
      $ref: '#/components/schemas/Accuracy'
    vAccuracy:
      $ref: '#/components/schemas/Accuracy'

AdditionalUeInfo:
  description: MBSR UE or MWAB UE Information
  type: object
  properties:
    ncgi:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Ncgi'
    tai:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Tai'

TimeWindowsNrppa:
  description: Time windows when network assisted positioning is used.
  type: object
  properties:
    measurementList:
      $ref: '#/components/schemas/TimeWindowInfoMeasurementList'
    srsList:
      $ref: '#/components/schemas/TimeWindowInfoSrsList'
  anyOf:
    - required: [measurementList]
    - required: [srsList]

RelativeVelocityWithUncertainty:
  description: Relative velocity with radial velocity and transverse velocity.
  type: object
  properties:
    rVelocity:
      $ref: '#/components/schemas/RadialVelocity'
    aTransverseVelocity:
      $ref: '#/components/schemas/AngularVelocity'
    eTransverseVelocity:
      $ref: '#/components/schemas/AngularVelocity'

RadialVelocity:
  description: Rate of change of a range.
  type: object
  required:
    - unitsRadialVelocity
    - radialVelocity
    - rVelocityUncertainty
  properties:
    unitsRadialVelocity:
      $ref: '#/components/schemas/UnitsLinearVelocity'
    radialVelocity:
      $ref: '#/components/schemas/RadialVelocityValue'
    rVelocityUncertainty:
      $ref: '#/components/schemas/RadialVelocityUncertainty'
```

```
AngularVelocity:
  description: Rate of change of an angle.
  type: object
  required:
    - unitsAngularVelocity
    - angularVelocity
    - aVelocityUncertainty
  properties:
    unitsAngularVelocity:
      $ref: '#/components/schemas/UnitsAngularVelocity'
    angularVelocity:
      $ref: '#/components/schemas/AngularVelocityValue'
    aVelocityUncertainty:
      $ref: '#/components/schemas/AngularVelocityUncertainty'

TrpParameter:
  description: The identifiers of TRPs if the network assisted positioning is used
  type: object
  properties:
    trpIds:
      type: array
      items:
        $ref: '#/components/schemas/TrpId'
      minItems: 1
    globalGnbId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/GlobalRanNodeId'
```

```
#
#
# SIMPLE TYPES
#
```

```
Altitude:
  description: Indicates value of altitude.
  type: number
  format: double
  minimum: -32767
  maximum: 32767
```

```
Angle:
  description: Indicates value of angle.
  type: integer
  minimum: 0
  maximum: 360
```

```
Uncertainty:
  description: Indicates value of uncertainty.
  type: number
  format: float
  minimum: 0
```

```
Orientation:
  description: Indicates value of orientation angle.
  type: integer
  minimum: 0
  maximum: 180
```

```
HorizAxesOrientation:
  description: Horizontal axes orientation angle clockwise from northing in 0.1 degrees.
  type: integer
  minimum: 0
  maximum: 3600
```

```
Confidence:
  description: Indicates value of confidence.
  type: integer
  minimum: 0
  maximum: 100
```

```
Accuracy:
  description: Indicates value of accuracy.
  type: number
  format: float
  minimum: 0
```

```
InnerRadius:
  description: Indicates value of the inner radius.
  type: integer
  format: int32
```

minimum: 0  
maximum: 327675

CorrelationID:  
description: LCS Correlation ID.  
type: string  
minLength: 1  
maxLength: 255

AgeOfLocationEstimate:  
description: Indicates value of the age of the location estimate.  
type: integer  
minimum: 0  
maximum: 32767

HorizontalSpeed:  
description: Indicates value of horizontal speed.  
type: number  
format: float  
minimum: 0  
maximum: 2047

VerticalSpeed:  
description: Indicates value of vertical speed.  
type: number  
format: float  
minimum: 0  
maximum: 255

SpeedUncertainty:  
description: Indicates value of speed uncertainty.  
type: number  
format: float  
minimum: 0  
maximum: 255

BarometricPressure:  
description: Specifies the measured uncompensated atmospheric pressure.  
type: integer  
minimum: 30000  
maximum: 115000

LcsServiceType:  
description: LCS service type.  
type: integer  
minimum: 0  
maximum: 127

LdrReference:  
description: LDR Reference.  
type: string  
pattern: "^[0-9a-fA-F]{2}){1,255}\$"

LirReference:  
description: LIR Reference.  
type: string  
pattern: "^[0-9a-fA-F]{2,510}\$"

ReportingAmount:  
description: Number of required periodic event reports.  
type: integer  
minimum: 1  
maximum: 8639999

ReportingInterval:  
description: Event reporting periodic interval in seconds.  
type: integer  
minimum: 1  
maximum: 8639999

ReportingIntervalMs:  
description: Event reporting periodic interval in milliseconds.  
type: integer  
minimum: 1  
maximum: 999

MinimumInterval:  
description: Minimum interval between event reports.



```
    type: integer
    minimum: 1
    maximum: 32767

MaximumInterval:
  description: Maximum interval between event reports.
  type: integer
  minimum: 1
  maximum: 86400

SamplingInterval:
  description: Maximum time interval between consecutive evaluations by a UE of a trigger event.
  type: integer
  minimum: 1
  maximum: 3600

ReportingDuration:
  description: Maximum duration of event reporting.
  type: integer
  minimum: 1
  maximum: 8640000

LinearDistance:
  description: Minimum straight line distance moved by a UE to trigger a motion event report.
  type: integer
  minimum: 1
  maximum: 10000

LMFIdentification:
  description: LMF identification.
  type: string

EventReportCounter:
  description: Number of event reports received from the target UE.
  type: integer
  minimum: 1
  maximum: 8640000

EventReportDuration:
  description: Duration of event reporting.
  type: integer
  minimum: 1
  maximum: 8640000

UePositioningCapabilities:
  description: >
    Positioning capabilities supported by the UE. A string encoding the
    "ProvideCapabilities-r9-IEs" IE as specified in clause 6.3 of 3GPP
    TS 37.355 (start from octet 1).
  type: string
  format: byte

TimeWindow:
  description: Time Window when UE assisted positioning is used.
  type: string
  format: byte

TimeWindowInfoMeasurementList:
  description: Contains the Time Window Information Measurement List.
  type: string
  format: byte

TimeWindowInfoSrsList:
  description: Contains the Time Window Information SRS List.
  type: string
  format: byte

RangingSlCapability:
  description: Ranging/Sidelink Positioning capabilities supported by the UE.
  type: string
  format: byte

RadialVelocityValue:
  description: Indicates value of rate of change of a range between the device A and device B.
  type: integer
  minimum: -2048
  maximum: 2047
```

```
AngularVelocityValue:
  description: Indicates rate of change of an angle.
  type: integer
  minimum: -1024
  maximum: 1023

AngularVelocityUncertainty:
  description: Indicates uncertainty for rate of change of an angle.
  type: integer
  minimum: 0
  maximum: 255

RadialVelocityUncertainty:
  description: Indicates uncertainty for rate of change of an range.
  type: integer
  minimum: 0
  maximum: 255

TrpId:
  description: Identifiers of TRP.
  type: integer
  minimum: 1
  maximum: 65535

#
# ENUMS
#

ExternalClientType:
  description: Indicates types of External Clients.
  anyOf:
    - type: string
      enum:
        - EMERGENCY_SERVICES
        - VALUE_ADDED_SERVICES
        - PLMN_OPERATOR_SERVICES
        - LAWFUL_INTERCEPT_SERVICES
        - PLMN_OPERATOR_BROADCAST_SERVICES
        - PLMN_OPERATOR_OM
        - PLMN_OPERATOR_ANONYMOUS_STATISTICS
        - PLMN_OPERATOR_TARGET_MS_SERVICE_SUPPORT
        - SL_POS
    - type: string

SupportedGADShapes:
  description: Indicates supported GAD shapes.
  anyOf:
    - type: string
      enum:
        - POINT
        - POINT_UNCERTAINTY_CIRCLE
        - POINT_UNCERTAINTY_ELLIPSE
        - POLYGON
        - POINT_ALTITUDE
        - POINT_ALTITUDE_UNCERTAINTY
        - ELLIPSOID_ARC
        - LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE
        - LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID
        - DISTANCE_DIRECTION
        - RELATIVE_2D_LOCATION_UNCERTAINTY_ELLIPSE
        - RELATIVE_3D_LOCATION_UNCERTAINTY_ELLIPSOID
    - type: string

ResponseTime:
  description: Indicates acceptable delay of location request.
  anyOf:
    - type: string
      enum:
        - LOW_DELAY
        - DELAY_TOLERANT
        - NO_DELAY
    - type: string

PositioningMethod:
  description: Indicates supported positioning methods.
  anyOf:
    - type: string
```

```

enum:
  - CELLID
  - ECID
  - OTDOA
  - BAROMETRIC_PRESSURE
  - WLAN
  - BLUETOOTH
  - MBS
  - MOTION_SENSOR
  - DL_TDOA
  - DL_AOD
  - MULTI-RTT
  - NR_ECID
  - UL_TDOA
  - UL_AOA
  - NETWORK_SPECIFIC
  - SL_TDOA
  - SL_TOA
  - SL_AoA
  - SL_RT
  - AIML

```

```

# Some of the values in the enumeration deviate from the naming conventions indicated in clause
# 5.1.4 of 3GPP TS 29.501 (i.e. to use UPPER_WITH_UNDERSCORE); however, it is kept as currently
# defined in this specification to maintain backwards compatibility
- type: string

```

```

PositioningMode:
  description: Indicates supported modes used for positioning method.
  anyOf:
    - type: string
    enum:
      - UE_BASED
      - UE_ASSISTED
      - CONVENTIONAL
    - type: string

```

```

GnssId:
  description: Global Navigation Satellite System (GNSS) ID.
  anyOf:
    - type: string
    enum:
      - GPS
      - GALILEO
      - SBAS
      - MODERNIZED_GPS
      - QZSS
      - GLONASS
      - BDS
      - NAVIC
    - type: string

```

```

Usage:
  description: Indicates usage made of the location measurement.
  anyOf:
    - type: string
    enum:
      - UNSUCCESS
      - SUCCESS_RESULTS_NOT_USED
      - SUCCESS_RESULTS_USED_TO_VERIFY_LOCATION
      - SUCCESS_RESULTS_USED_TO_GENERATE_LOCATION
      - SUCCESS_METHOD_NOT_DETERMINED
    - type: string

```

```

LcsPriority:
  description: Indicates priority of the LCS client.
  anyOf:
    - type: string
    enum:
      - HIGHEST_PRIORITY
      - NORMAL_PRIORITY
    - type: string

```

```

VelocityRequested:
  description: Indicates velocity requirement.
  anyOf:
    - type: string
    enum:
      - VELOCITY_IS_NOT_REQUESTED

```

```
- VELOCITY_IS_REQUESTED
- type: string

AccuracyFulfilmentIndicator:
  description: Indicates fulfilment of requested accuracy.
  anyOf:
    - type: string
    enum:
      - REQUESTED_ACCURACY_FULFILLED
      - REQUESTED_ACCURACY_NOT_FULFILLED
    - type: string

VerticalDirection:
  description: Indicates direction of vertical speed.
  type: string
  enum:
    - UPWARD
    - DOWNWARD

LdrType:
  description: Indicates LDR types.
  anyOf:
    - type: string
    enum:
      - UE_AVAILABLE
      - PERIODIC
      - ENTERING_INTO_AREA
      - LEAVING_FROM_AREA
      - BEING_INSIDE_AREA
      - MOTION
    - type: string

ReportingAreaType:
  description: Indicates type of event reporting area.
  anyOf:
    - type: string
    enum:
      - EPS_TRACKING_AREA_IDENTITY
      - E-UTRAN_CELL_GLOBAL_IDENTIFICATION
      - 5GS_TRACKING_AREA_IDENTITY
      - NR_CELL_GLOBAL_IDENTITY
    - type: string

OccurrenceInfo:
  description: Specifies occurrence of event reporting.
  anyOf:
    - type: string
    enum:
      - ONE_TIME_EVENT
      - MULTIPLE_TIME_EVENT
    - type: string

ReportingAccessType:
  description: Specifies access types of event reporting.
  anyOf:
    - type: string
    enum:
      - NR
      - EUTRA_CONNECTED_TO_5GC
      - NON_3GPP_CONNECTED_TO_5GC
      - NR_LEO
      - NR_MEO
      - NR_GEO
      - NR_OTHER_SAT
      - EUTRA_CONNECTED_TO_EPC
    - type: string

EventClass:
  description: Specifies event classes.
  anyOf:
    - type: string
    enum:
      - SUPPLEMENTARY_SERVICES
      - DUMMY
    - type: string

ReportedEventType:
  description: Specifies type of event reporting.
```

```
anyOf:
- type: string
  enum:
    - PERIODIC_EVENT
    - ENTERING_AREA_EVENT
    - LEAVING_AREA_EVENT
    - BEING_INSIDE_AREA_EVENT
    - MOTION_EVENT
    - MAXIMUM_INTERVAL_EXPIRATION_EVENT
    - LOCATION_CANCELLATION_EVENT
    - INTERMEDIATE_EVENT
    - DIRECT_REPORT_EVENT
    - CUMULATIVE_EVENT_REPORT
- type: string

TerminationCause:
description: Specifies causes of event reporting termination.
anyOf:
- type: string
  enum:
    - TERMINATION_BY_UE
    - TERMINATION_BY_NETWORK
    - NORMAL_TERMINATION
- type: string

LcsQosClass:
description: Specifies LCS QoS class.
anyOf:
- type: string
  enum:
    - BEST_EFFORT
    - ASSURED
    - MULTIPLE_QOS
- type: string

UeLocationServiceInd:
description: Specifies location service types requested by UE.
anyOf:
- type: string
  enum:
    - LOCATION_ESTIMATE
    - LOCATION_ASSISTANCE_DATA
- type: string

IndoorOutdoorInd:
description: Specifies UE location indoor or outdoor.
anyOf:
- type: string
  enum:
    - INDOOR
    - OUTDOOR
- type: string

FixType:
description: Specifies the positioning fix type.
anyOf:
- type: string
  enum:
    - CARRIER_PHASE_FLOAT
    - CARRIER_PHASE_FIX
- type: string

LosNlosMeasureInd:
description: Specifies LOS measurement or NLOS measurement.
anyOf:
- type: string
  enum:
    - LOS
    - NLOS
- type: string

UpConnectionStatus:
description: UP Connection Status.
anyOf:
- type: string
  enum:
    - ESTABLISHED
    - RELEASED
```

```
- MOVE
- type: string

RangingSlResult:
  description: Specifies the type of result requested for ranging and sidelink positioning.
  anyOf:
    - type: string
      enum:
        - ABSOLUTE_LOCATION
        - RELATIVE_LOCATION
        - RANGING_DIRECTION
        - RANGING_DIRECTION
        - DIRECTION
        - VELOCITY
        - RELATIVE_VELOCITY
    - type: string

RelatedUeType:
  description: Specifies the different roles of UE for ranging and sidelink positioning service.
  anyOf:
    - type: string
      enum:
        - LOCATED_UE
        - REFERENCE_UE
    - type: string

LcsUpConnectionInd:
  description: LCS UP Connection Indication.
  anyOf:
    - type: string
      enum:
        - TERMINATION
        - SETUP
    - type: string

UeUpPositioningCapabilities:
  description: User plane positioning capabilities supported by the UE.
  anyOf:
    - type: string
      enum:
        - LCS_UPP
        - SUPL
        - MULTIPLE_LCS_UPP
    - type: string

SlPositioningCapability:
  description: The Ranging and sidelink positioning capabilities supported by the UE.
  anyOf:
    - type: string
      enum:
        - V2X
        - 5G_PROSE
    - type: string

UnitsLinearVelocity:
  description: The the units of linear velocity.
  anyOf:
    - type: string
      enum:
        - MPERS
        - CMPERS
    - type: string

UnitsAngularVelocity:
  description: The units of angular velocity.
  anyOf:
    - type: string
      enum:
        - DEGPERS1
        - DEGPERS01
    - type: string
```

## A.3 Nlmf\_Broadcast API

openapi: 3.0.0

info:

```
version: '1.3.0'
title: 'LMF Broadcast'
description: |
  LMF Broadcast Service.
  © 2025, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
  All rights reserved.
```

externalDocs:

```
description: 3GPP TS 29.572 V19.5.0; 5G System; Location Management Services; Stage 3
url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.572/'
```

servers:

```
- url: '{apiRoot}/nlmf-broadcast/v1'
  variables:
    apiRoot:
      default: https://example.com
      description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501
```

paths:

```
/cipher-key-data:
  post:
    summary: Request ciphering key data
    operationId: CipheringKeyData
    tags:
      - Request Ciphering Key Data
    requestBody:
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/CipherRequestData'
      required: true
    responses:
      '200':
        description: Expected response to a valid request
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/CipherResponseData'
      '307':
        $ref: 'TS29571_CommonData.yaml#/components/responses/307'
      '308':
        $ref: 'TS29571_CommonData.yaml#/components/responses/308'
      '400':
        $ref: 'TS29571_CommonData.yaml#/components/responses/400'
      '401':
        $ref: 'TS29571_CommonData.yaml#/components/responses/401'
      '403':
        $ref: 'TS29571_CommonData.yaml#/components/responses/403'
      '404':
        $ref: 'TS29571_CommonData.yaml#/components/responses/404'
      '411':
        $ref: 'TS29571_CommonData.yaml#/components/responses/411'
      '413':
        $ref: 'TS29571_CommonData.yaml#/components/responses/413'
      '415':
        $ref: 'TS29571_CommonData.yaml#/components/responses/415'
      '429':
        $ref: 'TS29571_CommonData.yaml#/components/responses/429'
      '500':
        $ref: 'TS29571_CommonData.yaml#/components/responses/500'
      '502':
        $ref: 'TS29571_CommonData.yaml#/components/responses/502'
      '503':
        $ref: 'TS29571_CommonData.yaml#/components/responses/503'
      '504':
        $ref: 'TS29571_CommonData.yaml#/components/responses/504'
      default:
        $ref: 'TS29571_CommonData.yaml#/components/responses/default'
    callbacks:
      CipheringKeyData:
        '{$request.body#/amfCallbackURI}':
          post:
```

```

requestBody:
  description: Ciphering Key Data Notification
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/CipheringKeyInfo'
responses:
  '200':
    description: Expected response to a valid request
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/CipheringKeyResponse'
  '307':
    $ref: 'TS29571_CommonData.yaml#/components/responses/307'
  '308':
    $ref: 'TS29571_CommonData.yaml#/components/responses/308'
  '400':
    $ref: 'TS29571_CommonData.yaml#/components/responses/400'
  '401':
    $ref: 'TS29571_CommonData.yaml#/components/responses/401'
  '403':
    $ref: 'TS29571_CommonData.yaml#/components/responses/403'
  '404':
    $ref: 'TS29571_CommonData.yaml#/components/responses/404'
  '411':
    $ref: 'TS29571_CommonData.yaml#/components/responses/411'
  '413':
    $ref: 'TS29571_CommonData.yaml#/components/responses/413'
  '415':
    $ref: 'TS29571_CommonData.yaml#/components/responses/415'
  '429':
    $ref: 'TS29571_CommonData.yaml#/components/responses/429'
  '500':
    $ref: 'TS29571_CommonData.yaml#/components/responses/500'
  '502':
    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  '504':
    $ref: 'TS29571_CommonData.yaml#/components/responses/504'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'

```

```

components:
  schemas:
    #
    # STRUCTURED TYPES
    #
    CipheringKeyInfo:
      description: Information within Ciphering Key Data Notification request.
      type: object
      required:
        - cipheringData
      properties:
        cipheringData:
          type: array
          items:
            $ref: '#/components/schemas/CipheringDataSet'
          minItems: 1
      supportedFeatures:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

    CipheringKeyResponse:
      description: Information within Ciphering Key Data Notification Response.
      type: object
      properties:
        cipheringDataReport:
          type: array
          items:
            $ref: '#/components/schemas/CipheringSetReport'
          minItems: 1

    CipheringDataSet:
      description: Represents a Ciphering Data Set.
      type: object
      required:
        - cipheringSetID

```



```

    - cipheringKey
    - c0
    - validityStartTime
    - validityDuration
  properties:
    cipheringSetID:
      $ref: '#/components/schemas/CipheringSetID'
    cipheringKey:
      $ref: '#/components/schemas/CipheringKey'
    c0:
      $ref: '#/components/schemas/C0'
    ltePosSibTypes:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    nrPosSibTypes:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'
    validityStartTime:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    validityDuration:
      $ref: '#/components/schemas/ValidityDuration'
    tailList:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Bytes'

CipheringSetReport:
  description: Represents a report of Ciphering Data Set storage.
  type: object
  required:
    - cipheringSetID
    - storageOutcome
  properties:
    cipheringSetID:
      $ref: '#/components/schemas/CipheringSetID'
    storageOutcome:
      $ref: '#/components/schemas/StorageOutcome'

CipherRequestData:
  description: Information within Ciphering Key Data request.
  type: object
  required:
    - amfCallbackURI
  properties:
    amfCallbackURI:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
    supportedFeatures:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/SupportedFeatures'

CipherResponseData:
  description: Information within Ciphering Key Data Response.
  type: object
  required:
    - dataAvailability
  properties:
    dataAvailability:
      $ref: '#/components/schemas/DataAvailability'
#
#
# SIMPLE TYPES
#
CipheringSetID:
  description: Ciphering Data Set Identifier.
  type: integer
  minimum: 0
  maximum: 65535

CipheringKey:
  description: Ciphering Key.
  format: byte
  type: string

C0:
  description: First component of the initial ciphering counter.
  format: byte
  type: string

ValidityDuration:
  description: Validity Duration of the Ciphering Data Set.
  type: integer
  minimum: 1
  maximum: 65535

```

```
#
# ENUMS
#
StorageOutcome:
  description: Indicates the result of Ciphering Data Set storage.
  anyOf:
    - type: string
      enum:
        - STORAGE_SUCCESSFUL
        - STORAGE_FAILED

DataAvailability:
  description: Indicates availability of ciphering key data at an LMF.
  anyOf:
    - type: string
      enum:
        - CIPHERING_KEY_DATA_AVAILABLE
        - CIPHERING_KEY_DATA_NOT_AVAILABLE
```

---

## A.4 Nlmf\_DataExposure API

openapi: 3.0.0

```
info:
  version: 1.0.0
  title: Nlmf_DataExposure
  description: |
    LMF Data Exposure Service.
    © 2025, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
```

```
security:
  - {}
  - oAuth2ClientCredentials:
      - nlmf-dataexposure
```

```
externalDocs:
  description: 3GPP TS 29.572 V19.5.0; 5G System; Location Management Services; Stage 3
  url: 'https://www.3gpp.org/ftp/Specs/archive/29_series/29.572/'
```

```
servers:
  - url: '{apiRoot}/nlmf-dataexposure/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 4.4 of 3GPP TS 29.501
```

```
paths:
  /subscriptions:
    post:
      summary: Nlmf_Data Exposure Subscribe Service Operation
      tags:
        - Subscriptions(Collection)
      operationId: CreateSubscription
      requestBody:
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/LmfDataExposureSubscription'
            required: true
      responses:
        '201':
          description: Subscription Created
          headers:
            Location:
              description: >
                Contains the URI of the newly created resource, according to the structure
                {apiRoot}/nlmf-dataexposure/<apiVersion>/subscriptions/{subscriptionId}
              required: true
              schema:
                type: string
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/LmfDataExposureSubscription'
        '307':
```

```

    $ref: 'TS29571_CommonData.yaml#/components/responses/307'
  '308':
    $ref: 'TS29571_CommonData.yaml#/components/responses/308'
  '400':
    $ref: 'TS29571_CommonData.yaml#/components/responses/400'
  '401':
    $ref: 'TS29571_CommonData.yaml#/components/responses/401'
  '403':
    $ref: 'TS29571_CommonData.yaml#/components/responses/403'
  '404':
    $ref: 'TS29571_CommonData.yaml#/components/responses/404'
  '411':
    $ref: 'TS29571_CommonData.yaml#/components/responses/411'
  '413':
    $ref: 'TS29571_CommonData.yaml#/components/responses/413'
  '415':
    $ref: 'TS29571_CommonData.yaml#/components/responses/415'
  '429':
    $ref: 'TS29571_CommonData.yaml#/components/responses/429'
  '500':
    $ref: 'TS29571_CommonData.yaml#/components/responses/500'
  '501':
    $ref: 'TS29571_CommonData.yaml#/components/responses/501'
  '502':
    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  default:
    $ref: 'TS29571_CommonData.yaml#/components/responses/default'
callbacks:
  LmfDataExposureNotification:
    '{$request.body#/subscription/notificationUri}':
      post:
        summary: LMF Data Exposure Notification
        requestBody:
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/LmfDataExposureNotification'
        required: true
      responses:
        '204':
          description: Successful acknowledgement
        '307':
          $ref: 'TS29571_CommonData.yaml#/components/responses/307'
        '308':
          $ref: 'TS29571_CommonData.yaml#/components/responses/308'
        '400':
          $ref: 'TS29571_CommonData.yaml#/components/responses/400'
        '401':
          $ref: 'TS29571_CommonData.yaml#/components/responses/401'
        '403':
          $ref: 'TS29571_CommonData.yaml#/components/responses/403'
        '404':
          $ref: 'TS29571_CommonData.yaml#/components/responses/404'
        '411':
          $ref: 'TS29571_CommonData.yaml#/components/responses/411'
        '413':
          $ref: 'TS29571_CommonData.yaml#/components/responses/413'
        '415':
          $ref: 'TS29571_CommonData.yaml#/components/responses/415'
        '429':
          $ref: 'TS29571_CommonData.yaml#/components/responses/429'
        '500':
          $ref: 'TS29571_CommonData.yaml#/components/responses/500'
        '502':
          $ref: 'TS29571_CommonData.yaml#/components/responses/502'
        '503':
          $ref: 'TS29571_CommonData.yaml#/components/responses/503'
        default:
          $ref: 'TS29571_CommonData.yaml#/components/responses/default'

/subscriptions/{subscriptionId}:
  patch:
    summary: Nlmf_DataExposure Subscribe Modify service Operation
    tags:
      - Individual subscription (Document)
    operationId: ModifySubscription

```

```

parameters:
  - name: subscriptionId
    in: path
    required: true
    description: Unique ID of the subscription to be modified
    schema:
      type: string
requestBody:
  content:
    application/json-patch+json:
      schema:
        type: array
        items:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/PatchItem'
        minItems: 1
      required: true
responses:
  '204':
    description: Expected response to a successful subscription modification
  '307':
    $ref: 'TS29571_CommonData.yaml#/components/responses/307'
  '308':
    $ref: 'TS29571_CommonData.yaml#/components/responses/308'
  '400':
    $ref: 'TS29571_CommonData.yaml#/components/responses/400'
  '401':
    $ref: 'TS29571_CommonData.yaml#/components/responses/401'
  '403':
    $ref: 'TS29571_CommonData.yaml#/components/responses/403'
  '404':
    $ref: 'TS29571_CommonData.yaml#/components/responses/404'
  '411':
    $ref: 'TS29571_CommonData.yaml#/components/responses/411'
  '413':
    $ref: 'TS29571_CommonData.yaml#/components/responses/413'
  '415':
    $ref: 'TS29571_CommonData.yaml#/components/responses/415'
  '429':
    $ref: 'TS29571_CommonData.yaml#/components/responses/429'
  '500':
    $ref: 'TS29571_CommonData.yaml#/components/responses/500'
  '501':
    $ref: 'TS29571_CommonData.yaml#/components/responses/501'
  '502':
    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  default:
    description: Unexpected error

delete:
  summary: Nlmf_DataExposure Unsubscribe service Operation
  tags:
    - Individual subscription (Document)
  operationId: DeleteSubscription
  parameters:
    - name: subscriptionId
      in: path
      required: true
      description: Deletion of Subscription
      schema:
        type: string
  responses:
    '204':
      description: Subscription deleted successfully
    '307':
      $ref: 'TS29571_CommonData.yaml#/components/responses/307'
    '308':
      $ref: 'TS29571_CommonData.yaml#/components/responses/308'
    '400':
      $ref: 'TS29571_CommonData.yaml#/components/responses/400'
    '401':
      $ref: 'TS29571_CommonData.yaml#/components/responses/401'
    '403':
      $ref: 'TS29571_CommonData.yaml#/components/responses/403'
    '404':
      $ref: 'TS29571_CommonData.yaml#/components/responses/404'
    '411':

```

```

    $ref: 'TS29571_CommonData.yaml#/components/responses/411'
  '413':
    $ref: 'TS29571_CommonData.yaml#/components/responses/413'
  '415':
    $ref: 'TS29571_CommonData.yaml#/components/responses/415'
  '429':
    $ref: 'TS29571_CommonData.yaml#/components/responses/429'
  '500':
    $ref: 'TS29571_CommonData.yaml#/components/responses/500'
  '502':
    $ref: 'TS29571_CommonData.yaml#/components/responses/502'
  '503':
    $ref: 'TS29571_CommonData.yaml#/components/responses/503'
  default:
    description: Unexpected error

components:
  securitySchemes:
    oAuth2ClientCredentials:
      type: oauth2
      flows:
        clientCredentials:
          tokenUrl: '{nrfApiRoot}/oauth2/token'
          scopes:
            nlmf-dataexposure: Access to the Nlmf_DataExposure API

  schemas:
#
# STRUCTURED TYPES
#
  LmfDataExposureSubscription:
    description: Describes subscription operation of LMF data exposure
    type: object
    properties:
      notificationUri:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
      notifyCorrelationId:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CorrelationID'
      aoi:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PresenceInfo'
      recurTime:
        $ref: 'TS29503_Nudm_SDM.yaml#/components/schemas/RecurTime'
      numOfSamples:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      dataSources:
        type: array
        items:
          $ref: '#/components/schemas/DataSourceType'
        minItems: 1
      qualityThreshold:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocationQoS'
      mlModelId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
      expiryTime:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
      pruIndication:
        type: boolean
        enum:
          - true
      required:
        - notificationUri
        - notifyCorrelationId
        - aoi

  LmfDataExposureNotification:
    description: Describes Notification for LMF data collection
    properties:
      notifyCorrelationId:
        $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/CorrelationID'
      reports:
        type: array
        items:
          $ref: '#/components/schemas/LmfDataExposureReport'
        minItems: 1
      required:
        - notifyCorrelationId
        - reports

```

```
LmfDataExposureReport:
  description: Describes an LMF data collection exposure report
  properties:
    samplingDataList:
      type: array
      items:
        $ref: '#/components/schemas/LmfDataExposureSamplingData'
      minItems: 1
    mlModelId:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/Uinteger'
    cause:
      $ref: '#/components/schemas/LmfDataExposureCause'

LmfDataExposureSamplingData:
  description: Describes an LMF data collection exposure Sampling Data
  properties:
    timestamp:
      $ref: 'TS29571_CommonData.yaml#/components/schemas/DateTime'
    locMeasureData:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocMeasurements'
    groundTruth:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocationData'
    qualityIndicator:
      type: number
      format: float
      minimum: 0
      maximum: 1
    mlLocEstimate:
      $ref: 'TS29572_Nlmf_Location.yaml#/components/schemas/LocationData'
  required:
    - timestamp
    - locMeasureData
    - groundTruth

#
# SIMPLE TYPES
#

#
# ENUMERATIONS
#

DataSourceType:
  description: The data source for location measurement data collection.
  anyOf:
    - type: string
      enum:
        - NG_RAN
    - type: string

LmfDataExposureCause:
  description: The cause code for LMF Data Exposure Report.
  anyOf:
    - type: string
      enum:
        - NOT_ENOUGH_SAMPLES
    - type: string
```

---

## Annex B (informative): Change history

| Date    | Meeting | TDoc      | CR   | Rev | Cat | Subject/Comment                                                                                                                  | New version |
|---------|---------|-----------|------|-----|-----|----------------------------------------------------------------------------------------------------------------------------------|-------------|
| 2018-01 | CT4#82  |           |      |     |     | TS Skeleton agreed in CT4#82                                                                                                     | 0.0.0       |
| 2018-01 | CT4#82  | C4-181398 |      |     |     | Initial draft (C4-181119)<br>Incorporation of agreed pCRs from CT4#82: C4-181121, C4-181233, C4-181234                           | 0.1.0       |
| 2018-03 | CT4#83  | C4-182444 |      |     |     | Incorporation of agreed pCRs from CT4#83: C4-182181, C4-182427                                                                   | 0.2.0       |
| 2018-03 | CT#79   | CP-180034 |      |     |     | Presented for information                                                                                                        | 1.0.0       |
| 2018-04 | CT4#84  | C4-183524 |      |     |     | Incorporation of agreed pCRs from CT4#84: C4-183184, C4-183363, C4-183510                                                        | 1.1.0       |
| 2018-05 | CT4#85  | C4-184640 |      |     |     | Incorporation of agreed pCRs from CT4#85: C4-184195, C4-184197, C4-184198, C4-184199, C4-184202, C4-184443, C4-184446, C4-184547 | 1.2.0       |
| 2018-06 | CT#80   | CP-181111 |      |     |     | Presented for approval                                                                                                           | 2.0.0       |
| 2018-06 | CT#80   |           |      |     |     | Approved in CT#80                                                                                                                | 15.0.0      |
| 2018-09 | CT#81   | CP-182066 | 0002 | 2   |     | Error Cases                                                                                                                      | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0003 | -   |     | Custom Headers                                                                                                                   | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0004 | -   |     | Overall Clean-up                                                                                                                 | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0005 | -   |     | Description of Structured data types                                                                                             | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0006 | 1   |     | Resource structure presentation                                                                                                  | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0007 | 1   |     | LMF servers clause in OpenAPI                                                                                                    | 15.1.0      |
| 2018-09 | CT#81   | CP-182066 | 0008 | -   |     | API Version Update                                                                                                               | 15.1.0      |
| 2018-12 | CT#82   | CP-183025 | 0010 | 1   | F   | Cardinality                                                                                                                      | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0011 | -   | F   | APIRoot Clarification                                                                                                            | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0012 | -   | F   | AMF Id                                                                                                                           | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0013 | -   | F   | Barometric Pressure in Location Data                                                                                             | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0014 | 1   | F   | Clarify Serving Cell in Input Data                                                                                               | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0015 | 1   | F   | Oauth2 Corrections                                                                                                               | 15.2.0      |
| 2018-12 | CT#82   | CP-183025 | 0016 | -   | F   | API Version                                                                                                                      | 15.2.0      |
| 2018-12 | CT#82   | CP-183179 | 0017 | -   | F   | ExternalDocs Update                                                                                                              | 15.2.0      |
| 2019-03 | CT#83   | CP-190030 | 0018 | 1   | F   | OpenAPI Corrections                                                                                                              | 15.3.0      |
| 2019-03 | CT#83   | CP-190030 | 0019 | 1   | F   | Application Errors                                                                                                               | 15.3.0      |
| 2019-03 | CT#83   | CP-190030 | 0020 | 1   | F   | Essential Correction to InnerRadius                                                                                              | 15.3.0      |
| 2019-03 | CT#83   | CP-190030 | 0021 | 1   | F   | Mandatory Response Codes                                                                                                         | 15.3.0      |
| 2019-03 | CT#83   | CP-190030 | 0022 | 1   | F   | Essential correction to OpenAPI definition of GeographicArea                                                                     | 15.3.0      |
| 2019-03 | CT#83   | CP-190030 | 0023 | -   | F   | API version update                                                                                                               | 15.3.0      |
| 2019-06 | CT#84   | CP-191042 | 0024 | 2   | F   | UE Capabilities                                                                                                                  | 15.4.0      |
| 2019-06 | CT#84   | CP-191042 | 0025 | 2   | F   | Storage of OpenAPI specification files                                                                                           | 15.4.0      |
| 2019-06 | CT#84   | CP-191042 | 0027 | 1   | F   | Copyright Note in OpenAPI Spec                                                                                                   | 15.4.0      |
| 2019-06 | CT#84   | CP-191042 | 0028 | 1   | F   | Major API version                                                                                                                | 15.4.0      |
| 2019-06 | CT#84   | CP-191042 | 0030 | -   | F   | Open API Version                                                                                                                 | 15.4.0      |
| 2019-09 | CT#85   | CP-192113 | 0031 | 1   | F   | Missing attribute FLR in Civic Address                                                                                           | 16.0.0      |
| 2019-09 | CT#85   | CP-192192 | 0033 | 2   | B   | LMF service operations for a deferred 5GC-MT-LR                                                                                  | 16.0.0      |
| 2019-09 | CT#85   | CP-192192 | 0034 | 1   | B   | LMF service operations for a commercial 5GC-MT-LR                                                                                | 16.0.0      |
| 2019-09 | CT#85   | CP-192192 | 0035 | -   | F   | High Accuracy Support                                                                                                            | 16.0.0      |
| 2019-09 | CT#85   | CP-192113 | 0037 | 1   | D   | Correct type Polygon                                                                                                             | 16.0.0      |
| 2019-09 | CT#85   | CP-192120 | 0039 | -   | F   | 3GPP TS 29.572 API version update                                                                                                | 16.0.0      |
| 2019-12 | CT#86   | CP-193033 | 0041 | 1   | A   | Motion Sensor Position Method                                                                                                    | 16.1.0      |
| 2019-12 | CT#86   | CP-193165 | 0042 | 3   | B   | Addition of the LMF Broadcast Service Operations                                                                                 | 16.1.0      |
| 2019-12 | CT#86   | CP-193055 | 0043 | 1   | B   | LCS QoS Class                                                                                                                    | 16.1.0      |
| 2019-12 | CT#86   | CP-193036 | 0045 | 1   | F   | ExternalDoc Clause                                                                                                               | 16.1.0      |
| 2019-12 | CT#86   | CP-193036 | 0046 | 1   | F   | ProblemDetails Optional in Error Response                                                                                        | 16.1.0      |
| 2019-12 | CT#86   | CP-193044 | 0048 | -   | F   | 3GPP TS 29.572 API version update                                                                                                | 16.1.0      |
| 2020-03 | CT#87   | CP-200039 | 0049 | 2   | F   | Add Corresponding API descriptions in clause 5.1                                                                                 | 16.2.0      |
| 2020-03 | CT#87   | CP-200039 | 0050 | 2   | D   | Editorial corrections                                                                                                            | 16.2.0      |
| 2020-03 | CT#87   | CP-200039 | 0051 | 1   | F   | Correction - formatting consistency                                                                                              | 16.2.0      |
| 2020-03 | CT#87   | CP-200018 | 0052 |     | B   | Connectivity state per access type                                                                                               | 16.2.0      |
| 2020-03 | CT#87   | CP-200018 | 0053 |     | B   | Primary Cell in the Secondary RAN node                                                                                           | 16.2.0      |
| 2020-03 | CT#87   | CP-200052 | 0055 |     | F   | 3GPP TS 29.572 Rel16 API External doc update                                                                                     | 16.2.0      |
| 2020-03 | CT#87   | CP-200180 | 0054 | 4   | B   | Request Type and embedded LPP message                                                                                            | 16.2.0      |
| 2020-06 | CT#88e  | CP-201060 | 0056 | 1   | F   | Add a new Notifications Overview Table                                                                                           | 16.3.0      |
| 2020-06 | CT#88e  | CP-201060 | 0057 | 1   | F   | Add custom operation Name                                                                                                        | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0058 |     | F   | Location Context Transfer                                                                                                        | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0059 | 1   | B   | Network Specific Positioning Methods                                                                                             | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0060 |     | B   | Positioning Methods Support                                                                                                      | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0061 | 2   | F   | Storage of YAML files in ETSI Forge                                                                                              | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0062 | 1   | F   | Resolve Editor Notes                                                                                                             | 16.3.0      |
| 2020-06 | CT#88e  | CP-201032 | 0063 | 1   | F   | LDRreference                                                                                                                     | 16.3.0      |



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| 2020-06 | CT#88e | CP-201032 | 0065 | 1 | F | Resolution of EN on NR positioning SIBs                                                  | 16.3.0 |
| 2020-06 | CT#88e | CP-201032 | 0068 | 1 | F | Adding ResponseTime enumeration value                                                    | 16.3.0 |
| 2020-06 | CT#88e | CP-201060 | 0069 |   | F | Missing Descriptions                                                                     | 16.3.0 |
| 2020-06 | CT#88e | CP-201073 | 0070 |   | F | 29.572 Rel-16 API version and External doc update                                        | 16.3.0 |
| 2020-09 | CT#89e | CP-202112 | 0071 | 1 | F | Optionality of ProblemDetails in TS29.572 cleanup                                        | 16.4.0 |
| 2020-09 | CT#89e | CP-202112 | 0073 | 1 | F | Adding missing navigation satellite systems for positioning                              | 16.4.0 |
| 2020-09 | CT#89e | CP-202112 | 0074 | 1 | F | Including VGMLC address towards LMF when requesting LMF's Location service               | 16.4.0 |
| 2020-09 | CT#89e | CP-202112 | 0075 | 1 | F | Corrections on EventNotify service operation                                             | 16.4.0 |
| 2020-09 | CT#89e | CP-202043 | 0077 | 1 | F | Correct mismatch on GeographicArea between table and yaml                                | 16.4.0 |
| 2020-09 | CT#89e | CP-202096 | 0078 | - | F | 29.572 Rel-16 API version and External doc update                                        | 16.4.0 |
| 2020-12 | CT#90e | CP-203050 | 0080 | 1 | F | Essential corrections in clause 5.2.2.4 CancelLocation                                   | 16.5.0 |
| 2020-12 | CT#90e | CP-203050 | 0081 | 1 | F | Indication of control plane CIoT 5GS optimization in LocationContextTransfer             | 16.5.0 |
| 2020-12 | CT#90e | CP-203035 | 0082 | 1 | F | YAML files in 3GPP Forge                                                                 | 16.5.0 |
| 2020-12 | CT#90e | CP-203036 | 0085 | 1 | F | 29.572 Rel-16 API version and External doc update                                        | 16.5.0 |
| 2021-03 | CT#91e | CP-210041 | 0087 | - | F | Missing PIDL-LO elements in Location Information                                         | 17.0.0 |
| 2021-03 | CT#91e | CP-210037 | 0088 | 1 | F | HTTP 3xx redirection                                                                     | 17.0.0 |
| 2021-03 | CT#91e | CP-210034 | 0089 | - | F | Reference change                                                                         | 17.0.0 |
| 2021-03 | CT#91e | CP-210054 | 0091 | - | F | 29.572 Rel-16 API version and External doc update                                        | 17.0.0 |
| 2021-06 | CT#92e | CP-211055 | 0092 | 1 | F | Resolving Warning in Nlmf_BroadCast API                                                  | 17.1.0 |
| 2021-06 | CT#92e | CP-211055 | 0093 |   | F | Resolving Warning in Nlmf_Location API                                                   | 17.1.0 |
| 2021-06 | CT#92e | CP-211026 | 0094 | 1 | B | Add List of Assistance Data Types for MO-LR                                              | 17.1.0 |
| 2021-06 | CT#92e | CP-211026 | 0095 | 3 | B | Add Local Coordinates                                                                    | 17.1.0 |
| 2021-06 | CT#92e | CP-211051 | 0096 |   | F | OpenAPI Reference                                                                        | 17.1.0 |
| 2021-06 | CT#92e | CP-211059 | 0098 |   | A | 3xx description correction for SCP                                                       | 17.1.0 |
| 2021-06 | CT#92e | CP-211059 | 0102 | 1 | A | Redirect Response                                                                        | 17.1.0 |
| 2021-06 | CT#92e | CP-211050 | 0103 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.1.0 |
| 2021-09 | CT#93e | CP-212034 | 0105 | 1 | B | Add UE Positioning Capabilities                                                          | 17.2.0 |
| 2021-09 | CT#93e | CP-212034 | 0108 | 2 | B | Support for Area Decision of Satellite Access                                            | 17.2.0 |
| 2021-09 | CT#93e | CP-212034 | 0109 | 1 | B | LMF Parameters Support for non-3GPP Access                                               | 17.2.0 |
| 2021-09 | CT#93e | CP-212064 | 0111 |   | A | Encoding of binary attributes in JSON objects                                            | 17.2.0 |
| 2021-09 | CT#93e | CP-212034 | 0114 | 1 | B | Multiple QoS Class                                                                       | 17.2.0 |
| 2021-09 | CT#93e | CP-212059 | 0115 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.2.0 |
| 2021-12 | CT#94e | CP-213096 | 0118 | 1 | B | Multiple LPP messages                                                                    | 17.3.0 |
| 2021-12 | CT#94e | CP-213096 | 0119 | 1 | B | Higher Resolution Timestamp for Location Estimates                                       | 17.3.0 |
| 2021-12 | CT#94e | CP-213096 | 0120 |   | F | UE Positioning Capabilities Data Type                                                    | 17.3.0 |
| 2021-12 | CT#94e | CP-213096 | 0121 | 1 | B | Update allowed access type for event report                                              | 17.3.0 |
| 2021-12 | CT#94e | CP-213118 | 0122 |   | F | Resolve OpenAPI Warnings                                                                 | 17.3.0 |
| 2021-12 | CT#94e | CP-213120 | 0123 |   | F | 307/308 redirection                                                                      | 17.3.0 |
| 2021-12 | CT#94e | CP-213121 | 0124 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.3.0 |
| 2022-03 | CT#95e | CP-220034 | 0128 |   | F | Corrections on the RelativeCartesianLocation                                             | 17.4.0 |
| 2022-03 | CT#95e | CP-220227 | 0130 | 1 | A | Corrections on attributes                                                                | 17.4.0 |
| 2022-03 | CT#95e | CP-220090 | 0131 |   | F | Update incorrect table number                                                            | 17.4.0 |
| 2022-03 | CT#95e | CP-220217 | 0133 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.4.0 |
| 2022-06 | CT#96  | CP-221022 | 0135 | 1 | B | Schedule location time for LMF                                                           | 17.5.0 |
| 2022-06 | CT#96  | CP-221022 | 0136 |   | F | Update the title of RelativeCartesianLocation                                            | 17.5.0 |
| 2022-06 | CT#96  | CP-221051 | 0138 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.5.0 |
| 2022-09 | CT#97  | CP-222030 | 0139 | 1 | F | Alignment on the service name used with template                                         | 17.6.0 |
| 2022-09 | CT#97  | CP-222036 | 0140 | 1 | F | Indication of Network Assisted Positioning method                                        | 17.6.0 |
| 2022-09 | CT#97  | CP-222058 | 0141 |   | F | 29.572 Rel-17 API version and External doc update                                        | 17.6.0 |
| 2022-12 | CT#98  | CP-223028 | 0142 | 1 | F | Missing mandatory status codes in OpenAPI                                                | 18.0.0 |
| 2022-12 | CT#98  | CP-223055 | 0143 | 1 | F | Correction on country verification for satellite access                                  | 18.0.0 |
| 2022-12 | CT#98  | CP-223064 | 0146 | 1 | F | Boolean type IEs definition correction                                                   | 18.0.0 |
| 2022-12 | CT#98  | CP-223033 | 0147 |   | F | 29.572 Rel-18 API version and External doc update                                        | 18.0.0 |
| 2023-03 | CT#99  | CP-230032 | 0153 | 2 | B | Support of Event Report Allowed Area                                                     | 18.1.0 |
| 2023-03 | CT#99  | CP-230032 | 0154 | 2 | B | Support of UE Unaware Positioning                                                        | 18.1.0 |
| 2023-03 | CT#99  | CP-230324 | 0155 | 3 | B | Multiple location report for MT-LR Immediate Location Request for the regulatory service | 18.1.0 |
| 2023-03 | CT#99  | CP-230032 | 0158 | 2 | B | Location service in PNI-NPN with signalling optimisation                                 | 18.1.0 |
| 2023-03 | CT#99  | CP-230032 | 0159 | 2 | B | Support of low power and high accuracy positioning                                       | 18.1.0 |
| 2023-03 | CT#99  | CP-230071 | 0161 |   | F | 29.572 Rel-18 API version and External doc update                                        | 18.1.0 |
| 2023-06 | CT#100 | CP-231028 | 0160 | 4 | F | Location header description                                                              | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0164 | 2 | B | NWDAF assisted LMF positioning method determination                                      | 18.2.0 |
| 2023-06 | CT#100 | CP-231075 | 0166 | 1 | A | Missing finer periodicities than 1s and an infinite reporting amount                     | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0168 | 1 | B | Support of UE User Plane Positioning Capability                                          | 18.2.0 |
| 2023-06 | CT#100 | CP-231026 | 0172 |   | B | OAuth2 scopes in the Nlmf_Location API                                                   | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0175 | 3 | B | Clarification of the location QoS mapping                                                | 18.2.0 |
| 2023-06 | CT#100 | CP-231031 | 0176 | 2 | B | Add reporting indication                                                                 | 18.2.0 |
| 2023-06 | CT#100 | CP-231055 | 0177 | 1 | B | Support of location service involving Mobile Base Station Relay                          | 18.2.0 |
| 2023-06 | CT#100 | CP-231070 | 0180 |   | F | 29.572 Rel-18 API version and External doc update                                        | 18.2.0 |

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| 2023-09 | CT#101 | CP-232035 | 0183 | 1 | F | Correction on Reporting Indication                                            | 18.3.0 |
| 2023-09 | CT#101 | CP-232063 | 0185 | 1 | A | Missed HA GNSS Metrics Support over SBI                                       | 18.3.0 |
| 2023-09 | CT#101 | CP-232154 | 0186 | 2 | B | The service operation of Nlmf_Location_MeasurementData                        | 18.3.0 |
| 2023-09 | CT#101 | CP-232035 | 0188 | 2 | B | Support on NLOS/LOS measurement indication                                    | 18.3.0 |
| 2023-09 | CT#101 | CP-232155 | 0189 | 3 | B | Nlmf_Location_UPNotify service operation                                      | 18.3.0 |
| 2023-09 | CT#101 | CP-232035 | 0190 | 2 | B | Nlmf_Location_UPSubscribe service operation                                   | 18.3.0 |
| 2023-09 | CT#101 | CP-232062 | 0192 | 1 | A | Add GNSS integrity requirement                                                | 18.3.0 |
| 2023-09 | CT#101 | CP-232178 | 0193 | 3 | B | Update on LMF service for MT procedures for ranging_SL                        | 18.3.0 |
| 2023-09 | CT#101 | CP-232060 | 0194 |   | F | 29.572 Rel-18 API version and External doc update                             | 18.3.0 |
| 2023-12 | CT#102 | CP-233037 | 0182 | 3 | B | Periodic or triggered location events via user plane to an LCS Client or AF   | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0196 | 1 | F | Addition of missing interface between LMF and GMLC                            | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0197 | 1 | B | Add the Nlmf_Location_UPConfig                                                | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0198 |   | F | Align with the SBI template                                                   | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0199 |   | F | Correction on the description of scheduledLocTime                             | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0201 |   | B | Update on UpNotifyData                                                        | 18.4.0 |
| 2023-12 | CT#102 | CP-233029 | 0203 | 1 | F | HTTP RFCs obsoleted by IETF RFC 9110, 9111 and 9113                           | 18.4.0 |
| 2023-12 | CT#102 | CP-233294 | 0204 | 4 | B | Update on LMF service for ranging_SL                                          | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0205 | 1 | F | Reporting Indication Definition Alignment to Stage 2                          | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0206 | 2 | B | Multiple QoS for Deferred Location Service Continuation from 5GS to EPS       | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0207 | 1 | B | Allowed Reporting Access Type for EUTRAN Connected to EPC                     | 18.4.0 |
| 2023-12 | CT#102 | CP-233030 | 0208 |   | F | ProblemDetails RFC 7807 obsoleted by 9457                                     | 18.4.0 |
| 2023-12 | CT#102 | CP-233052 | 0209 | 1 | B | Addition of additional ULI of the MBSR UE                                     | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0210 |   | F | Correction on PRU Location Measurements                                       | 18.4.0 |
| 2023-12 | CT#102 | CP-233037 | 0211 | 2 | B | Update user plane positioning capabilities                                    | 18.4.0 |
| 2023-12 | CT#102 | CP-233031 | 0214 |   | F | Correction on OAuth Scopes Names                                              | 18.4.0 |
| 2023-12 | CT#102 | CP-233060 | 0215 | 3 | F | 29.572 Rel-18 API version and External doc update                             | 18.4.0 |
| 2024-03 | CT#103 | CP-240053 | 0216 |   | F | Correct supportedGADShapes Cardinality in LocContextData                      | 18.5.0 |
| 2024-03 | CT#103 | CP-240140 | 0217 | 2 | F | Add correlation ID to location UP service operations                          | 18.5.0 |
| 2024-03 | CT#103 | CP-240054 | 0222 |   | A | Missed Vertical Confidence                                                    | 18.5.0 |
| 2024-03 | CT#103 | CP-240155 | 0225 | 2 | B | Add UPUnSubscribe and update UPSubscribe service operations                   | 18.5.0 |
| 2024-03 | CT#103 | CP-240030 | 0226 | 1 | F | Update the description of UPConfig service operation                          | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0229 | 1 | B | coordinate id in case of absolute locations                                   | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0230 | 1 | B | Data type extension to support Ngmlc_Location_ProvideRanging                  | 18.5.0 |
| 2024-03 | CT#103 | CP-240030 | 0232 | 1 | B | Add time window                                                               | 18.5.0 |
| 2024-03 | CT#103 | CP-240030 | 0233 |   | F | Corrections on the URI of location-measure                                    | 18.5.0 |
| 2024-03 | CT#103 | CP-240030 | 0234 |   | F | Corrections on the URI of up-configure                                        | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0239 | 1 | F | Aligning distance and location terminology with stage 2                       | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0240 | 1 | F | Corrections on Application Layer ID                                           | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0241 | 1 | B | Add Ranging/Sidelink Positioning Capability                                   | 18.5.0 |
| 2024-03 | CT#103 | CP-240045 | 0242 |   | F | Update the incorrect implementation                                           | 18.5.0 |
| 2024-03 | CT#103 | CP-240053 | 0245 | 2 | F | Clarification on URI Path Segment Naming Conventions for Custom operations    | 18.5.0 |
| 2024-03 | CT#103 | CP-240056 | 0246 |   | F | 29.572 Rel-18 API version and External doc update                             | 18.5.0 |
| 2024-06 | CT#104 | CP-241030 | 0224 | 2 | F | Resolve the EN and update attributes of MeasurementData service operation     | 18.6.0 |
| 2024-06 | CT#104 | CP-241056 | 0250 | 1 | A | Integrity Result                                                              | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0251 | 2 | F | Corrections on Local Origin                                                   | 18.6.0 |
| 2024-06 | CT#104 | CP-241056 | 0253 | 1 | A | Missed Vertical Confidence in Local Location                                  | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0254 | 1 | F | Correction on Missed Description Fields in OpenAPI and Enum Naming Convention | 18.6.0 |
| 2024-06 | CT#104 | CP-241030 | 0255 | 1 | F | Update the security configuration for UPUnSubscribe                           | 18.6.0 |
| 2024-06 | CT#104 | CP-241030 | 0256 | 5 | F | Support of LCS user plane connection binding to the UE                        | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 0257 | 1 | F | Updates on feature description                                                | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0258 | 1 | D | Editorial corrections                                                         | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0259 |   | F | Misalignment on AddEventNotifyDatas                                           | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0260 | 1 | F | Style Corrections of Nlmf_Location API                                        | 18.6.0 |
| 2024-06 | CT#104 | CP-241065 | 0264 |   | A | Definition of VelocityEstimate                                                | 18.6.0 |
| 2024-06 | CT#104 | CP-241038 | 0266 | 1 | B | Remote UE Indication in Location Information                                  | 18.6.0 |

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| 2024-06 | CT#104 | CP-241028 | 0267 |   | B | Returning UNSUPPORTED_EVENT_TYPE                                                | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0268 | 1 | F | Add missing condition for conditional parameters                                | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0269 | 1 | F | Feature negotiation correction                                                  | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0270 |   | F | Update on reference number                                                      | 18.6.0 |
| 2024-06 | CT#104 | CP-241050 | 0271 | 1 | F | Updates on naming convention for enumeration                                    | 18.6.0 |
| 2024-06 | CT#104 | CP-241045 | 0272 | 1 | F | Alignment with naming conventions.                                              | 18.6.0 |
| 2024-06 | CT#104 | CP-241052 | 0274 |   | F | 29.572 Rel-18 API version and External doc update                               | 18.6.0 |
| 2024-09 | CT#105 | CP-242049 | 0278 | 3 | F | Supported Ranging and Sidelink Positioning Methods for Located UE Selection     | 19.0.0 |
| 2024-09 | CT#105 | CP-242053 | 0279 | 2 | F | Usage of the common application errors by Nlmf_Broadcast service                | 19.0.0 |
| 2024-09 | CT#105 | CP-242030 | 0281 | 2 | F | The term Payload is replaced with Content due to RFC 9113                       | 19.0.0 |
| 2024-09 | CT#105 | CP-242053 | 0282 | 2 | F | Miscellaneous corrections                                                       | 19.0.0 |
| 2024-09 | CT#105 | CP-242054 | 0283 | 1 | F | 29.572 Rel-19 API version and External doc update                               | 19.0.0 |
| 2024-12 | CT#106 | CP-243272 | 0285 | 4 | F | Support multiple LCS contexts transfer while modifying the UP connection        | 19.1.0 |
| 2024-12 | CT#106 | CP-243025 | 0287 | 1 | A | LCS Correlation ID in Nlmf_Location_CancelLocation Request                      | 19.1.0 |
| 2024-12 | CT#106 | CP-243026 | 0293 | 1 | A | Clarification on the RangeDirection parameter                                   | 19.1.0 |
| 2024-12 | CT#106 | CP-243046 | 0294 | 3 | B | Support for MWAB involved positioning                                           | 19.1.0 |
| 2024-12 | CT#106 | CP-243046 | 0297 | 1 | B | Support of additional ULI in MWAB                                               | 19.1.0 |
| 2024-12 | CT#106 | CP-243060 | 0299 | 1 | B | NRPPa Periodic Indication                                                       | 19.1.0 |
| 2024-12 | CT#106 | CP-243314 | 0301 | 3 | A | Update to support NRPPa related measurements                                    | 19.1.0 |
| 2024-12 | CT#106 | CP-243044 | 0304 | 1 | B | Updates on the Reference model                                                  | 19.1.0 |
| 2024-12 | CT#106 | CP-243273 | 0307 | 6 | B | LMF Enhancements for LMF based AI/ML Positioning – Stage 3 aspects              | 19.1.0 |
| 2024-12 | CT#106 | CP-243059 | 0310 |   | F | Incorrect resource URI definition                                               | 19.1.0 |
| 2024-12 | CT#106 | CP-243059 | 0311 |   | F | Incorrect usage of HTTP for Nlmf_Broadcast service                              | 19.1.0 |
| 2024-12 | CT#106 | CP-243059 | 0312 | 1 | F | Eliminate the duplicate reference                                               | 19.1.0 |
| 2024-12 | CT#106 | CP-243069 | 0315 |   | F | API version and External doc update                                             | 19.1.0 |
| 2025-03 | CT#107 | CP-250027 | 0296 | 4 | A | Updates on relative velocity                                                    | 19.2.0 |
| 2025-03 | CT#107 | CP-250045 | 0317 | 1 | B | Updates on Nlmf_DataExposure                                                    | 19.2.0 |
| 2025-03 | CT#107 | CP-250032 | 0318 | 1 | F | Miscellaneous corrections                                                       | 19.2.0 |
| 2025-03 | CT#107 | CP-250046 | 0319 |   | F | Remove Nullvalue for UP Location Reporting AF Address                           | 19.2.0 |
| 2025-03 | CT#107 | CP-250054 | 0320 | 1 | B | Support of PRU Usage Extension                                                  | 19.2.0 |
| 2025-03 | CT#107 | CP-250045 | 0321 |   | F | Corrections to Nlmf_DataExposure service                                        | 19.2.0 |
| 2025-03 | CT#107 | CP-250045 | 0322 | 1 | B | LMF functionalities and consumers                                               | 19.2.0 |
| 2025-03 | CT#107 | CP-250045 | 0323 |   | F | Removal of subscriptionCorrelationId from LmfDataExposureNotification data type | 19.2.0 |
| 2025-03 | CT#107 | CP-250048 | 0324 | 1 | B | Enhancement of Additional ULI                                                   | 19.2.0 |
| 2025-03 | CT#107 | CP-250137 | 0326 | 4 | F | API version and External doc update                                             | 19.2.0 |
| 2025-06 | CT#108 | CP-251067 | 0329 |   | F | Corrections for User Plane Connection                                           | 19.3.0 |
| 2025-06 | CT#108 | CP-251067 | 0330 |   | F | Correct the references                                                          | 19.3.0 |
| 2025-06 | CT#108 | CP-251050 | 0332 | 1 | A | Correction to the additional ULI                                                | 19.3.0 |
| 2025-06 | CT#108 | CP-251069 | 0333 | 1 | F | Correction to MWAB                                                              | 19.3.0 |
| 2025-06 | CT#108 | CP-251066 | 0334 | 2 | B | Abnormal Handling for LMF Data Exposure Service                                 | 19.3.0 |
| 2025-06 | CT#108 | CP-251066 | 0335 | 4 | B | Support of Performance Monitoring for LMF Based AIML Positioning                | 19.3.0 |
| 2025-06 | CT#108 | CP-251055 | 0337 |   | F | Correct the descriptions of the semiMinor and the semiMajor                     | 19.3.0 |
| 2025-06 | CT#108 | CP-251045 | 0342 | 2 | A | Add missing Rel-17 posSibType                                                   | 19.3.0 |
| 2025-06 | CT#108 | CP-251051 | 0344 | 2 | A | Add missing Rel-18 posSibType                                                   | 19.3.0 |
| 2025-06 | CT#108 | CP-251055 | 0345 |   | F | Update the reference number and editorial issue                                 | 19.3.0 |
| 2025-06 | CT#108 | CP-251067 | 0346 |   | F | Range concept usage instead of undefined distance                               | 19.3.0 |
| 2025-06 | CT#108 | CP-251066 | 0348 | 2 | B | Enhancements to Nlmf_DataExposure API                                           | 19.3.0 |
| 2025-06 | CT#108 | CP-251048 | 0352 | 3 | A | Correction on Location Context Transfer with LCS-UPP Connection                 | 19.3.0 |
| 2025-06 | CT#108 | CP-251067 | 0357 |   | D | Editorial corrections                                                           | 19.3.0 |
| 2025-06 | CT#108 | CP-251079 | 0359 |   | F | API version and External doc update                                             | 19.3.0 |
| 2025-09 | CT#109 | CP-252049 | 0364 | 1 | F | Support Multiple LCSUPP Connections per UE                                      | 19.4.0 |
| 2025-09 | CT#109 | CP-252049 | 0365 |   | F | Correction to the references                                                    | 19.4.0 |
| 2025-09 | CT#109 | CP-252048 | 0366 | 1 | B | AIML Positioning Method                                                         | 19.4.0 |
| 2025-09 | CT#109 | CP-252192 | 0367 | 2 | F | Resolve Editor's Notes                                                          | 19.4.0 |

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|---------|--------|-----------|------|---|---|-------------------------------------------------------------------------------|--------|
| 2025-09 | CT#109 | CP-252037 | 0368 |   | F | Essential Clarification for Bits Order for SIB Types Bitmap                   | 19.4.0 |
| 2025-09 | CT#109 | CP-252176 | 0374 | 1 | F | API version and External doc update                                           | 19.4.0 |
| 2025-12 | CT#110 | CP-253152 | 0377 | 2 | F | Correction on the enum values                                                 | 19.5.0 |
| 2025-12 | CT#110 | CP-253150 | 0378 | 2 | F | Adding missing application error code                                         | 19.5.0 |
| 2025-12 | CT#110 | CP-253143 | 0379 | 2 | B | UP Configuration for existing LCS-UPP connection after AMF Change             | 19.5.0 |
| 2025-12 | CT#110 | CP-253141 | 0381 | 5 | A | UE ID with LCS User Plane Connection                                          | 19.5.0 |
| 2025-12 | CT#110 | CP-253153 | 0382 |   | F | Correction of figure name for Location Measurements procedure                 | 19.5.0 |
| 2025-12 | CT#110 | CP-253153 | 0383 |   | F | Correction of reference and table numbering errors                            | 19.5.0 |
| 2025-12 | CT#110 | CP-253154 | 0384 | 1 | F | Update the UE User Plane Positioning Capabilities                             | 19.5.0 |
| 2025-12 | CT#110 | CP-253154 | 0386 | 1 | F | Transfer UE UP Positioning capabilities in LocContextData                     | 19.5.0 |
| 2025-12 | CT#110 | CP-253154 | 0388 | 1 | F | Convey the updated UE user plane positioning capabilities in UP Configuration | 19.5.0 |
| 2025-12 | CT#110 | CP-253153 | 0391 | 1 | F | Correction on the Condition for Conditional IE                                | 19.5.0 |
| 2025-12 | CT#110 | CP-253143 | 0392 | 1 | F | Correction on LDR Reference                                                   | 19.5.0 |
| 2025-12 | CT#110 | CP-253154 | 0393 | 4 | F | Support of UE with Multiple LCS-UPP connections Capability                    | 19.5.0 |
| 2025-12 | CT#110 | CP-253170 | 0394 | 3 | F | Support data collection for LMF based AIML positioning                        | 19.5.0 |
| 2025-12 | CT#110 | CP-253167 | 0395 |   | F | API version and External doc update                                           | 19.5.0 |
| 2026-03 | CT#111 | CP-260031 | 0397 | 1 | F | Inclusion of SUPI as Mandatory UE Identifier                                  | 19.6.0 |
| 2026-03 | CT#111 | CP-260031 | 0399 | 2 | F | Extended Facility IE Support Indication                                       | 19.6.0 |
| 2026-03 | CT#111 | CP-260040 | 0401 |   | F | API version and External doc update                                           | 19.6.0 |